



**University of
Zurich** ^{UZH}

Flexible Techniques for Automatic Text Recognition of Historical Documents

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by
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Abstract

English This thesis explores flexible techniques for automatically recognising text in historical documents. It delves into various approaches to *optical character recognition* for printed documents and *handwritten text recognition* for manuscripts. The effectiveness of these approaches is tested using both the traditional *convolutional neural network* and *recurrent neural network architecture*, as well as the newer *Transformer*-based architecture. Additionally, new datasets are introduced. Results show that the Transformer-based models perform equally well or better than the traditional approaches. The thesis also introduces and compares methods for evaluating text generated by automatic text recognition methods without ground truth material.

Deutsch Diese Arbeit beschäftigt sich mit flexiblen Techniken für die *automatische Texterkennung* historischer Dokumente. Sie untersucht verschiedene Ansätze zur *optischen Zeichenerkennung* von gedruckten Dokumenten und zur *Handschriftenerkennung* von Manuskripten. Um die verschiedenen Ansätze zu testen, die einerseits auf der klassischen *Convolutional-Neural-Network* $\times$ *Recurrent-Neural-Network*-Architektur und andererseits auf *Transformer*-basierten Architekturen beruhen, präsentiert diese Arbeit auch neue Datensätze. Es zeigt sich, dass die neuen Transformer-basierten Modelle entweder gleich gut oder sogar besser abschneiden als die klassischen Ansätze. Ausserdem präsentiert die vorliegende Arbeit Methoden zur Bewertung von Texten, die mit automatischen Texterkennungsmethoden ohne manuell korrigiertem Testmaterial erzeugt wurden.

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List of Abbreviations

ATR	Automatic Text Recognition
BLSTM	Bi-directional Long Short-Term Memory
CA	Character Accuracy
CER	Character Error Rate
CNN	Convolutional Neural Network
CTC	Connectionist Temporal Classification
DE	German
EN	English
ENHG	Early New High German
GLAM	Galleries, Libraries, Archives, and Museums
GT	Ground Truth
HTR	Handwritten Text Recognition
HMM	Hidden Markov Model
LA	Latin
LR	Layout Recognition
LSTM	Long Short-Term Memory
NLP	Natural Language Processing
NZZ	Neue Zürcher Zeitung
OCR	Optical Character Recognition
PPL	Perplexity
PPPL	Pseudo-Perplexity
RNN	Recurrent Neural Network

Chapter 1

Introduction

This chapter highlights the importance of flexible techniques for automatically transcribing historical documents. In Section 1.1, we provide an overview of the current state of digitisation of historical documents in Switzerland, which serves as the motivation for this work. The second part identifies the various challenges encountered during the digitisation of printed and handwritten documents (1.2). The remaining sections describe the scope and contributions (1.3) and provide an outline (1.4) of this thesis. Finally, the last section lists previous publications (1.5), which serve as the foundation for this thesis.

1.1 Digitisation of Historical Documents in Switzerland

The landscape of electronically available and fully digitised historical documents in Switzerland is limited to a few platforms. The most readily available items are **written** documents. The platforms categorise the documents they provide into two types: printed texts and handwritten documents. The most important portals for accessing **printed** documents are *e-rara*¹ for old prints, *e-newspaperarchives*² for newspapers, and *e-periodica*³ for periodicals. Most of the **handwritten** documents,

¹<https://www.e-rara.ch>

²<https://www.e-newspaperarchives.ch>

³<https://www.e-periodica.ch>

i. e., manuscripts,⁴ are provided by *e-manuscripta*⁵ and *e-codices*⁶. Additionally, there is the *swisscollections*⁷ gateway, which provides a meta-catalogue of various data pools for historical and modern collections.

The platforms mentioned above are populated by **cultural heritage institutions** such as galleries, libraries, archives, and museums⁸ and their digitisation departments. Together, they contribute to the increasingly fast-paced **digital transformation** and convert analogue collections into digital ones [Jannidis et al., 2017]. Hence, they provide bigger and bigger amounts of data from different access points:⁹ a digitised object, e. g., the first printed copy of the Bible translation¹⁰ by Huldrych Zwingli (1484–1531) and his colleague Leo Jud (1482–1542), becomes available to a researcher from Bolivia – or from any other part of the world – via *WorldCat*¹¹ after it has been digitised and published on e-rara.

Most GLAMs providing digitised historical material in Switzerland are the digitisation services of university libraries, e. g., the *DigiCenter* of the ETH Library. However, other institutions like the *Zentralbibliothek Zürich Digitisation Centre* (DigiZ) from the Zurich Central library, the *Zurich State Archive* or the *Swiss National Library* also contribute to the online platforms. In light of their digitisation strategies, they have steadily increased the amount of electronically available items. This means that at least high-quality facsimiles,¹² i. e., faithful reproductions of the original documents in the form of scans, become more and more available. Alone the DigiZ has digitised over 6.8 million (text) pages or images over the last six years.¹³ Table 1.1 summarises the situation for available printed and handwritten documents on the most important Swiss access points mentioned above, to which various digitisation

⁴**manuscript**: a borrowing from Latin consisting of the words *manū*, ablative of *manus* hand + *scriptus*, past participle of *scribere* to write

⁵<https://www.e-manuscripta.ch>

⁶<https://www.e-codices.unifr.ch/>

⁷<https://www.swisscollections.ch>

⁸commonly known as **GLAM** institutions

⁹By **access points** we mean portals, catalogues, or other web services dedicated to the indexing of digitised material.

¹⁰Die gantze Bibel / der ursprünglichen ebraischen und griechischen Waarheyt nach auff's aller treüwlichst verteütschet, Zürich: Christoph Froschauer, 1531 (VD 16 B 2690); Grossmünster Zürich.

¹¹WorldCat is the global counterpart to swisscollections, linking thousands of libraries' collections in one place.

¹²**facsimile**: a borrowing from Latin consisting of the words *fac*, imperative of *facere* to make + *simile*, neuter of *similis* similar

¹³According to numbers the Zurich Central Library has published in its annual reports. The current digitisation rate lies at over 800k scans per year.

services have delivered scans.¹⁴

documents	source	specifications	# of titles	# of available transcriptions	in %
printed	e-rara	15c.	528	1	0.19
		16c.	7,282	42	0.58
		17c.	10,524	133	1.26
		18c.	9,687	7,647	78.94
		19c.	6,683	5,235	78.33
			# of issues	# of pages	# of articles
	e-newspaperarchives	newspapers	1,083,331	11,388,997	81,638,771
	e-periodica	periodicals	NA	9,720,826	945,694
handwritten	e-manuscripta	texts	5,242	198	3.78
		correspondence	52,034	529	1.02
	e-codices		2,700	NA	NA

Table 1.1: As of November 7, 2022, here is a rundown of the items available on various platforms. The numbers in the specifications for e-rara denote the century. The statistics for e-newspaperarchives and e-periodica are based on issue and page counts only. In the case of handwritten documents, the top column names are once again relevant.

When dealing with written documents, simply scanning the source does not amount to complete digitisation, unlike with objects such as antique vases, rocks, or photographs that lack textual content. As such, **metadata**, i. e., author, title, year, place of publication, and publisher of, e. g., a book, along with its scans only suffice to *identify* it but do not allow one to *search through* its contents. Thus, a written document, or an entire document collection, that is *available* online (via its metadata or even with its scans) is not automatically *accessible*. Although the provision of scans by an archive would make such a document accessible for humans (e. g., for close reading), its contents are still inaccessible for machines, distant reading approaches, and search engines.

To make written documents more accessible, it is important to provide their **full texts** and index them in a search system. As the digital transformation continues to progress, the number of full texts available will increase, which will directly impact

¹⁴In Section 2.3, we will refer to further collections where digitised documents were the main focus.

researchers in humanities and social sciences who rely on written primary sources for their work. With all data being available and accessible online, researchers can work remotely and explore new research questions by sifting through numerous documents using keyword searches and other advanced search methods offered by search engines. This shift towards digital research methods in the humanities results from the digital transformation and is a major driver of the *Digital Humanities*.

1.2 From Facsimiles to Electronic Texts

When working with a scan that contains text, it is important to ensure that it is of high quality.¹⁵ The first step is to extract the text regions via **page layout analysis** or **layout recognition**. Layout recognition first identifies text regions in the scan and second the corresponding text lines. It is important to note that the upstream step of layout recognition is crucial, as it heavily impacts the automatic text recognition that follows [Martínek et al., 2020]. Recent approaches, such as those introduced by Grüning et al. [2019], have achieved high $F1$ -scores¹⁶ of 0.9 for complex and 0.96 for simple layouts in handwritten and printed historical documents. Although the performance of layout recognition impacts the results of automatic text recognition, it will not be the main focus of this thesis.

To obtain text from images of lines that have been recognised by layout recognition, we use **optical character recognition** (OCR) models. Regarding text recognition in manuscripts, we apply **handwritten text recognition** (HTR) models. Both OCR and HTR models are machine learning models trained on pairs of line images and text (see Chapter 2 for further details). Figure 1.1 demonstrates the digitisation process from a physical object on the archive shelf to the extracted text. As an example, we include a letter from the [Bullinger correspondence](#)¹⁷ (see Section 4.2.2 for further details).

In terms of OCR of printed documents, the current state-of-the-art achieves average results ranging from 5% to 1% **character error rate** (CER),¹⁸ depending on factors

¹⁵We speak of “high-quality” if the scan has a resolution of at least 300 dots per inch (dpi).

¹⁶We will introduce the $F1$ -score in Section 3.5. Although we use it to measure the text quality, the calculations for measuring layout recognition are the same.

¹⁷Heinrich Bullinger Briefwechsel [HBBW], Bd. 13, Zürich 2008, 293–298. Here I link to the web service also hosting the facsimiles.

¹⁸CER is a measure which informs about the quality of OCR and HTR models. The lower the CER, the better the resulting text. See Section 3.5.2 for further details.

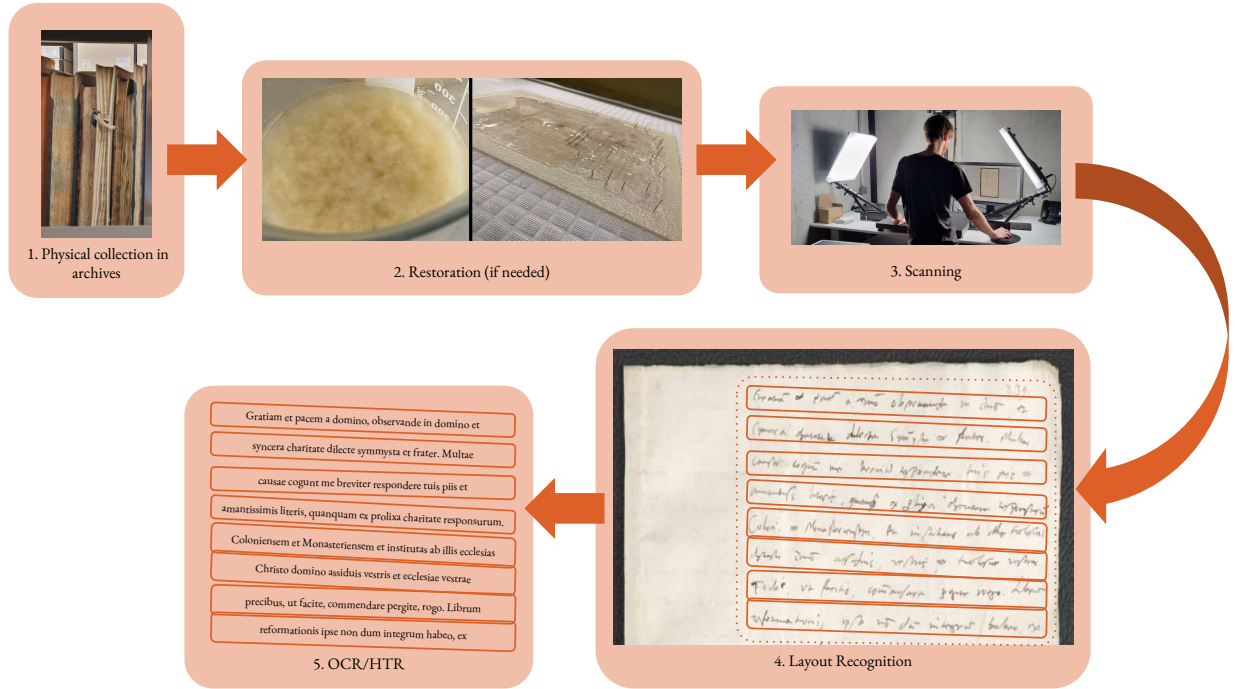


Figure 1.1: Digitisation process from the shelf in the archive, over restoration through scanning, layout recognition and OCR/HTR. Images are taken from Zurich States Archive’s Twitter account [@staatsarchiv_zh](https://twitter.com/staatsarchiv_zh)^a (1, 2, and 3) and [Bullinger Digital](https://www.bullinger-digital.ch/)^b (4). The dotted box in the fourth step represents a text zone, which contains several text lines (solid boxes).

^ahttps://twitter.com/staatsarchiv_zh

^b<https://www.bullinger-digital.ch/>

such as font, quality of the images, amount of training material, and heterogeneity of the data [Springmann and Lüdeling, 2017, Brandt Skelbye and Dannélls, 2021]. For the more challenging HTR, the most recent results range from over 10% to just above 1% [Hodel et al., 2021]. Generally, we consider automatically extracted full texts with a **character accuracy** (CA)¹⁹ of below 90% as *poor*, between 92% and 98% as *average*, and above 98% as *good* [Holley, 2009]. We will refer to Holley’s classification of OCR and HTR results throughout this thesis.

GLAMs actively support and accelerate the research processes by making the full text extracted via OCR and HTR and the scans available to researchers. These techniques have become catalysts in the Digital Humanities and are responsible for providing huge text collections that serve as research subjects for various inquiries.

¹⁹CA is defined as 1-CER. See Section 3.5.2 for further details.

However, there is an issue when comparing different portals and text types: although textbases are growing, there are still imbalances on digitisation platforms.

1.2.1 Old Prints

Starting with e-rara, we see in Table 1.1 that prints from the 18th and 19th centuries have a high availability of transcriptions,²⁰ reaching almost 80%. For earlier centuries, this percentage drops to around 1%. We can only speculate about the causes of this difference for printed documents on e-rara. The most likely explanation is a change in the digitisation routine. We also need to take into account that when GLAMs started with mass digitisation, the performance of OCR models for text extraction was not as advanced as it is today. If GLAMs prioritised quality, they would refrain from publishing OCRed texts when the performance delivered unsatisfactory results. With increasingly precise OCR models, however, we will see more electronic texts becoming available.

Let us consider a few examples to understand what kind of difficult prints GLAMs have to deal with. The first is a double page from a collection of Psalms²¹ printed in parallel in different languages (Latin, Greek, Arabic, and Aramaic) together with the so-called *scholia*, i. e., grammatical, critical, or explanatory comments of the Psalms (also in different languages, see Figure 1.2). Firstly, the layout is inherently complex. On the left-hand page, a layout recognition system would be required to recognise at least six different regions: the first is the red commentary on top which spans the double page, regions two to five consist of the four columns for the Psalms, and the sixth is the scholia, which is, to make matters more difficult, a continuation of the scholia that only starts in the right-most column on the next page. Secondly, the OCR model applied to these pages must be *multilingual* since we are dealing with texts in different languages. Thirdly, the OCR model would need to be able to deal with five different scripts: Latin, Hebrew, Greek, Arabic and Aramaic. Lastly, the ageing process of the paper, in this case so-called “foxing,” i. e., the process of the paper turning brown and developing spots, further complicates the OCR.

The example shown in Figure 1.3 is simpler regarding its typesetting. However, the printers chose to use a **blackletter** font,²² instead of the **Antiqua** font used for the

²⁰These transcriptions must have been produced automatically.

²¹Agostino *Giustiani*, *Psalterium Hebræum, Græcū, Arabicū & Chaldaëū cū tribus latinis interp[re]tat[i]o[n]ibus & glossis*, Genova: Petrus Paulus Porrus, 1516; Zentralbibliothek Zürich, III B 35 | G.

²²also known as **Gothic** font

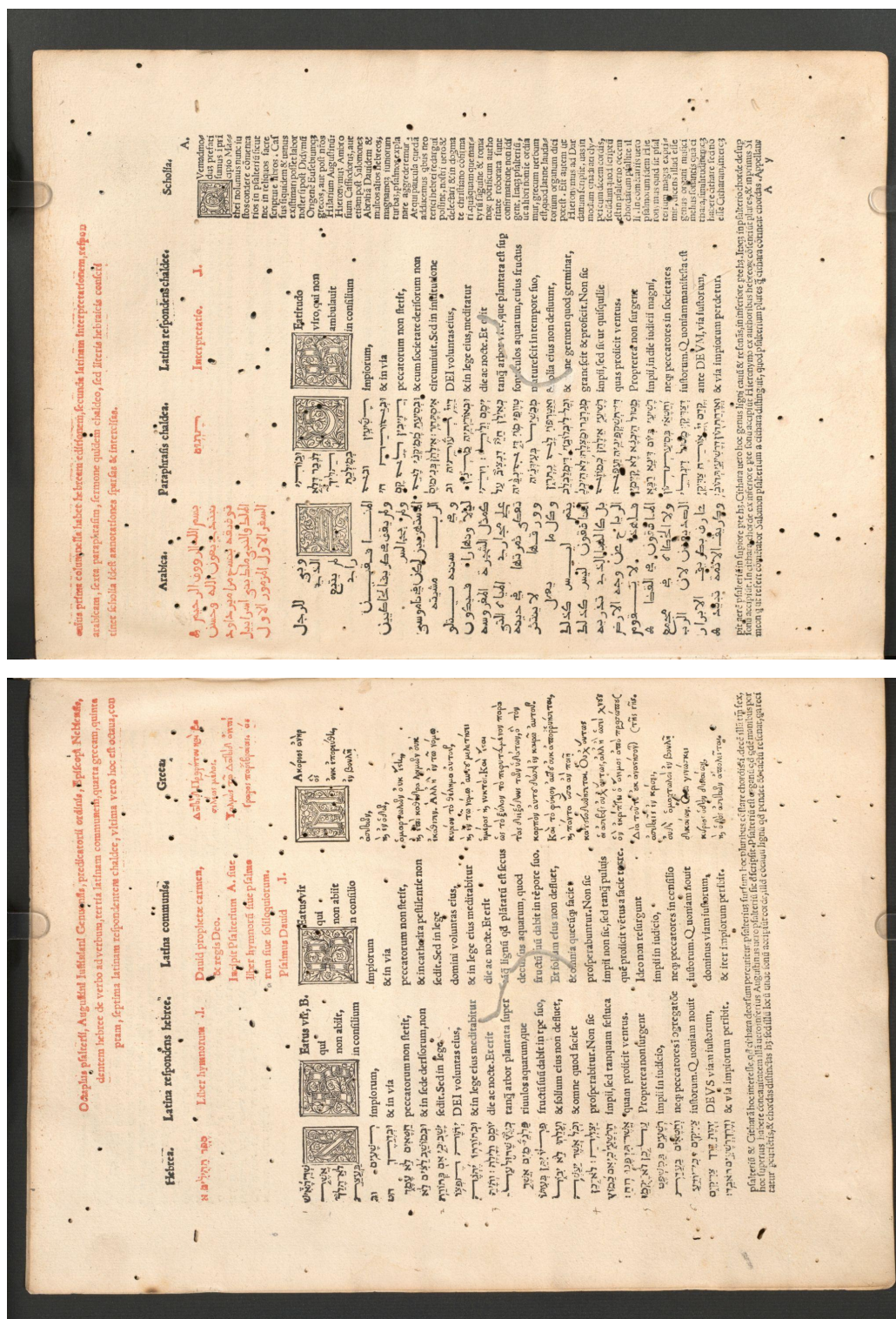
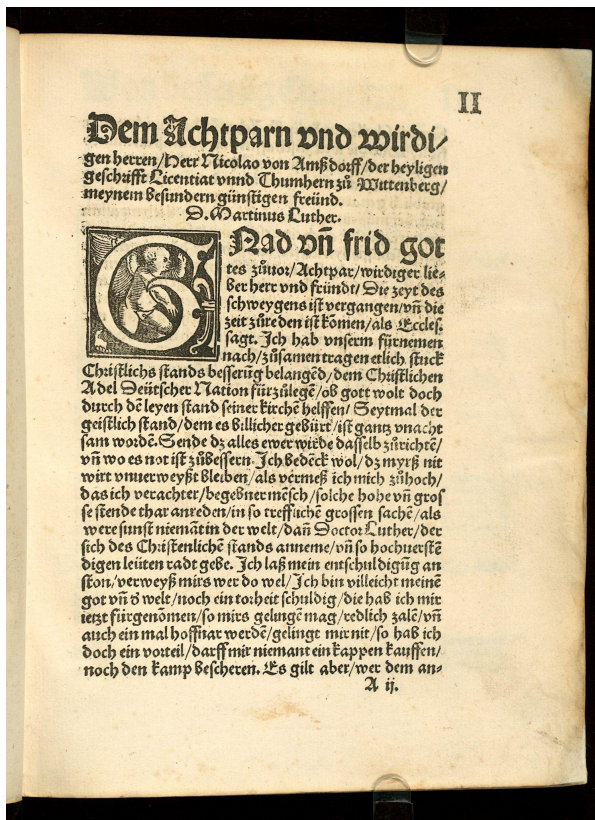


Figure 1.2: Example from a book of Psalms in several languages. From left to right, we find the Psalms in Hebrew, a Latin translation close to the Hebrew text, a common Latin translation, followed by Greek (all on the left page), Arabic, and an Aramaic translation, which is also accompanied by a Latin translation close to the Aramaic text (on the right page). The scholia is typeset around the Psalms (on both pages).



DemAchtpam vndxpfrty/
8ct*bemn/tottt Nicolao von Amßdorff/der heyiligen
Zescyrffr Licentiae vnnd Thumhern zu Wttenberg/
meynem besunderngimDgen freund.

i- V .Martinus Lucher.

fM Vfe.

Oadyn frib goe

tes znuor/Achtpar/wirdtger lie-

ber Herr vnd fcündr/ Sre zeyt des

fchweygens ist vergangen/vn die

zeck zureden ist kömen/als Ecclef.

fe&'Jcb hab vnferm furnemen

nach/znsamentragenerlich strck

A-del Bentscher Narion flirzulege/ob sfott wolt doch
durch deleyen stand seiner kirche helsten/ Seytmal der
geistlich stand/dem es brUichergebür/istgantz vnacht
sam worde.Sende dz alles ewer wiede dasselb zurichre/
vn wo es nor ist zubestern Ich bedeck wol/dz myrß nie
wirr vnuerwysße bletven/als vcrmeß ich mich zühoch/
das ich vecachter/bsgebnecmcsch/soiche hohevngros
sestenbecharanreben/insorreglkche grossen sache/als
weresunst niematin der welr/ban DocrorLuther/ber
sich des Christenliche stands anneme/vnso hochuerste
digenleütenrabt gebe.Ich laß mein emschndigng an
ston/verweyß mirs wer do wel/Ich bin villeichr meine
Aot vn 6 welkJnoch ein torhcik scbnldig/die hab ich mir
ttztstlirgenomen/so »mrs gekngcmag/redlich zalc/vn
auch ein malhoffnar wede/gelingc mnnik/so hab ich
doch ein vorreil/darffmir niemam einkappen kauffen/
noch den kamp bescheren.Eö gilt aber/wec%
dem an- ih

Figure 1.3: Simple layout of a print in blackletter with the OCR result taken from the e-rara platform on the right.

Latin parts of the Psalms. The image depicts the very first page of Martin Luther's *An den Christlichen Adel deutscher Nation von der christlichen Standesbesserung*,²³ and next to it is the text generated by an OCR system, which is available on e-rara via the PDF download.

The OCR'd text on the right-hand side of Figure 1.3 contains numerous errors that can make it challenging for readers to comprehend it without referring to the original. Moreover, it can be unreliable to index the text in a search system due to OCR errors (because many words would not be found due to these errors (like the occurrence of *Luther* himself, which has been mistranscribed as *Lucher*). It is likely that the OCR model used to extract the text from this facsimile was not trained

²³Martin *Luther* *An den Christlichen Adel deutscher Nation von der christlichen Standesbesserung*, Basel: Adam Petri, 1520 (VD 16 L 3752); Universitätsbibliothek Basel, UBH AN VII 10:2.

on the language of the text, which is Early New High German (ENHG), nor has it encountered this particular font before. The most common errors occur at very specific places:

1. Certain characters are easy to confuse, like the normal *f* and the *f* (so-called *long s*).
2. Characters like *v* can be interpreted as either *v* or *u*, and *j* as *j* or *i*, respectively, depending on the context. E. g., the occurrence of *vnnd* (EN *and*, DE *und*) should be recognised as either *vnnd* or *unnd* (depending on the transcription guidelines of the text data that served as training material for the OCR model). The same applies to *Jch* (EN *I*, DE, *Ich*), which should be realised as either *Jch* or *Ich*.
3. Abbreviations, which are much more common in older texts since they allow to save space and thus expensive paper, must be known to the OCR model. Otherwise, it struggles to either resolve them or correctly recognise the characters. In the above text, we find, among others, the tilde symbol “~” above certain characters. As such, the occurrence of *vñ* means *vnnd* (EN *and*), and *fachẽ* means *sachen* (EN *things*). This also shows that the tilde symbol causes different resolutions depending on the context.
4. There are special characters, like the letter *u* with an *o* above: *ũ*. It abbreviates the diphthong *uo*. The word *zũuor* (EN *before*, DE *zuvor*) should be transcribed as such or at least as *zũvor*.

Errors like these are common in OCRred material printed in blackletter. Post-OCR correction methods [Volk et al., 2011, Kissos and Dershowitz, 2016, Khirbat, 2017, Amrhein and Clematide, 2018, Schaefer and Neudecker, 2020] have proven effective in reducing the amount of erroneously recognised characters. Still, since different OCR or HTR models make different errors [Volk et al., 2011], it is preferable for a model to recognise the text correctly from the beginning.²⁴

1.2.2 Historical Newspapers

Achieving good results on blackletter script is still possible, as demonstrated by the left-hand side of Figure 1.4. This image displays an OCRred text from a historical

²⁴We are aware that post-OCR procedures improve OCR results, as does the combination of different OCR systems. These measures, however, are not part of this thesis.

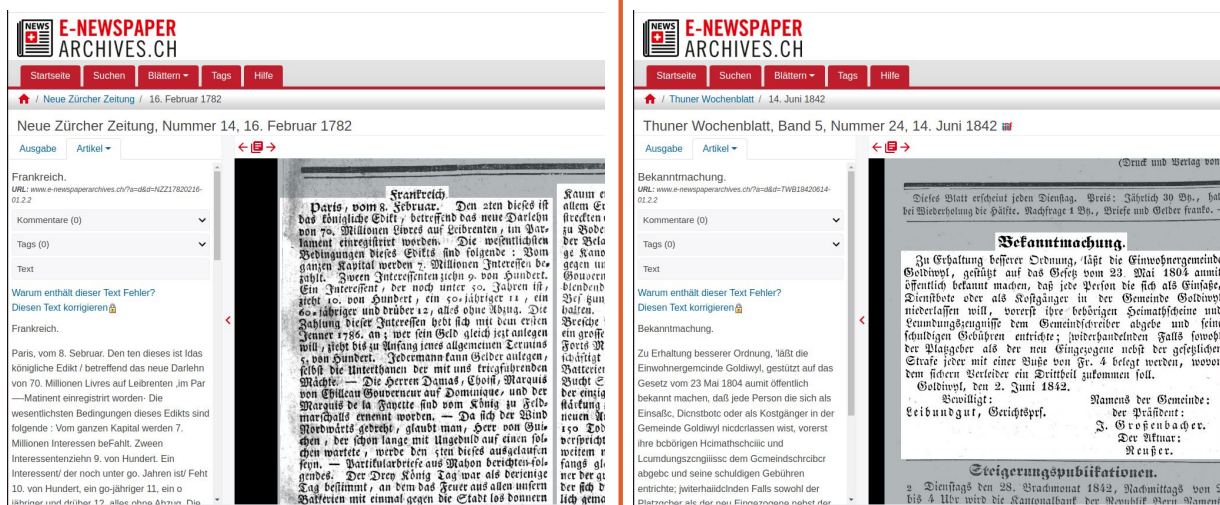


Figure 1.4: Two examples from the e-newspaperarchives platform, with the text on the left and the facsimile on the right.

newspaper, specifically an issue of the *Neue Zürcher Zeitung* (NZZ) dating back to [February 16, 1782](https://www.e-newspaperarchives.ch/?a=d&d=NZZ17820216-01.2.2).²⁵ We did not encounter any confusion between the normal *f* and the long *s*, or any other issues previously identified. However, there were still some challenges, such as mistranscribed word boundaries (e. g., we find *Interessenzenziehen* instead of *Interessenten ziehen*), incorrect recognition of numbers, and irregularities caused by line breaks. But contrary to the example from Luther’s publication, the text remains highly readable, and the ability to index such texts can benefit researchers in locating relevant documents via search interfaces. E. g., newspapers are a particularly valuable source for historians [Allen and Sieczkiewicz, 2010], as they offer insights into both past and present societies, covering significant regional and international events. Since keyword search is the preferred research method for newspaper collections, it is essential to ensure that OCR is accurate.

The accuracy of the OCR in the NZZ newspaper issue we examined may not be consistent across all newspapers, as seen in the right-hand side of Figure 1.4. An issue of the *Thuner Wochenblatt* dated June 14, 1842, had mediocre OCR, with many instances of *e* being mistaken for *c*, and *n* being confused with *ii*, among other errors. Therefore, the e-newspaperarchives platform presents a heterogeneous situation, with some items having excellent OCR and others containing multiple errors. This highlights the importance of quality control for services that collect and index material from GLAMs’ digitisation departments [Verheusen, 2008] and is one of

²⁵www.e-newspaperarchives.ch/?a=d&d=NZZ17820216-01.2.2

the main drivers for the development of unsupervised measures we put forward in this thesis.

Many documents published in blackletter have some of the aforementioned challenges in common. A general OCR model often does not work for all the documents GLAMs need to process without further tweaks and amendments. This makes OCR, in general, a costly, time-consuming, and error-prone process, which can be applied with sufficient quality control. Still, the availability of full texts presents itself differently for newspapers and periodicals, as Table 1.1 shows. On e-periodica, every digitised page comes with a text version. This means there is a different workflow in place than for old prints and handwritings and shows the confidence with which OCR is being applied by the different digitisation centres, both for Antiqua and blackletter fonts.

1.2.3 Handwritings

While automatic transcriptions for printed documents are fairly common, the same cannot be said for handwritten ones. This is evident on e-manuscripta, where only 1% of correspondences and 3% of text items have transcriptions available. It is worth noting that human transcribers manually added these transcriptions. To address this issue, e-manuscripta encourages its community to help transcribe manuscripts using their transcription tool.

The project *Transcribe Bentham* focuses on transcribing the collection of manuscripts by the English philosopher and reformer Jeremy Bentham (1748–1832). This initiative has successfully involved interested volunteers, as stated in a report by Causer et al. [2018]. Over a period of four years, the crowd transcribed nearly 11,000 documents, equivalent to approximately 1.2 million words. If the transcribers on e-manuscripta maintained a similar yearly rate of 2,750 documents, it would still take 20 years to transcribe the entire holdings as of November 2022.²⁶

Thus, manual transcription does not scale with the vast number of available manuscripts [Terras, 2022b]. HTR can, on the one hand, provide a first hypothesis which the crowd could then correct, and, on the other hand, profit from these transcriptions since they can be used as new training material. Moreover, it could also help to make the manuscripts more accessible by enabling indexing of the auto-

²⁶This assumes a comparable page layout and word density to that of the Bentham collection.

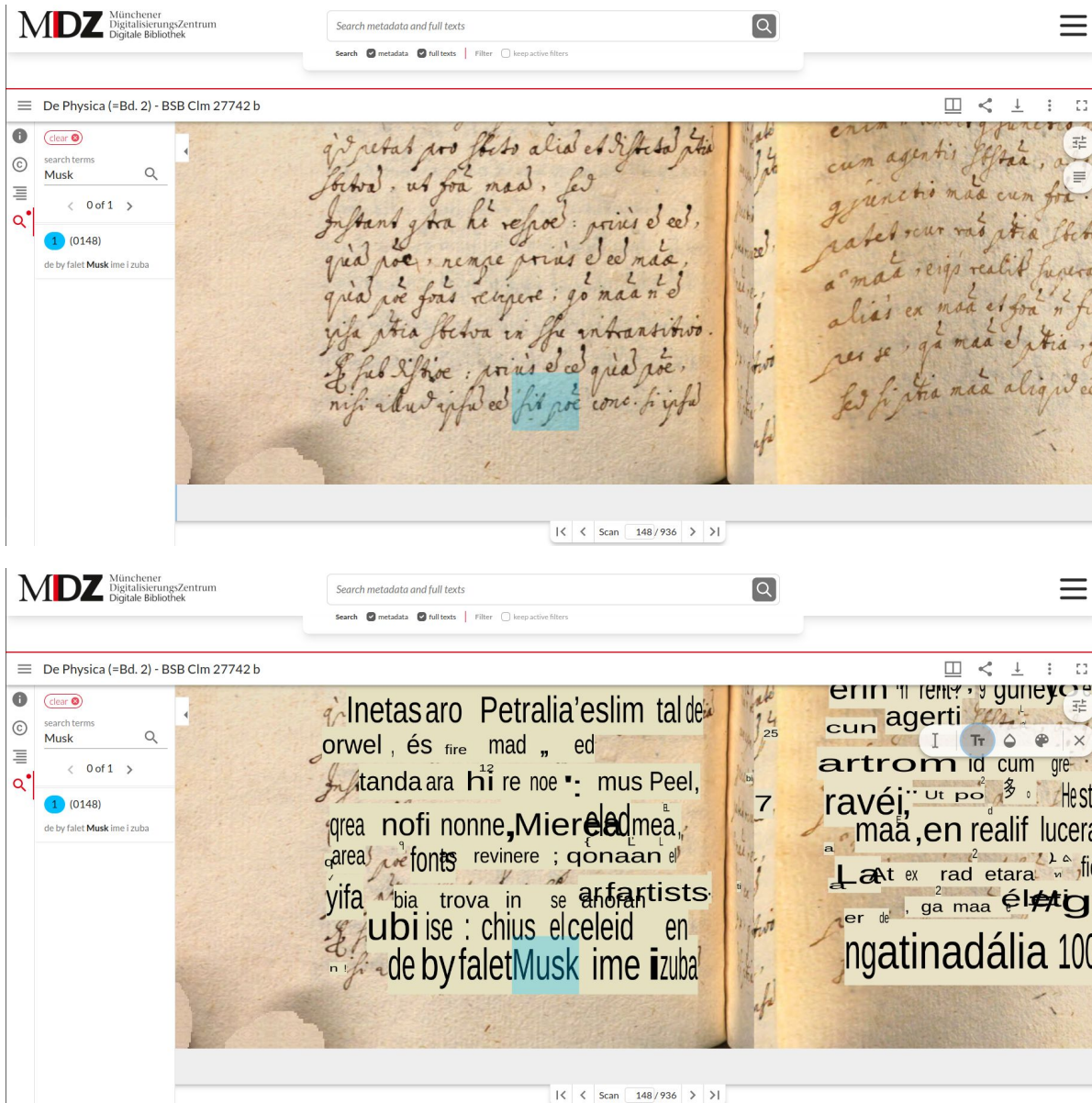


Figure 1.5: A result for the search of the keyword *Musk* in the Digital Collections by the Munich Digitization Center. The top screenshot shows where the search word is supposed to occur on the facsimile. The bottom screenshot shows the option of text overlay of the automatically recognised text in the facsimile.

matically generated transcription in a search system.

However, applying HTR models and using their results blindly is of no benefit to the users, as an example from the *Munich Digitization Center* (MDZ) shows in Figure 1.5. The MDZ offers 2,841,946 digitised items in its Digital Collections.²⁷ A search for the last name *Musk* returns exactly eight manuscripts, of which we randomly chose one. The blue box in the top screenshot of Figure 1.5 is supposed to mark where in the document the writing of *Musk* has been found. The MDZ interface allows overlaying the facsimile with the automatically recognised text. Doing this results in the bottom screenshot of Figure 1.5. The text recognition is random at best and does not correspond to the writing in the facsimile. Effects like these are the result of non-adapted layout recognition and the application of non-suitable HTR models.

But even if an HTR model had been adapted to the hand of a certain writer, it can not easily be used to decipher a different hand from another writer. The challenge of greater variability in characters is, after all, more pronounced in manuscripts than it is in printed documents. The digitisation of **correspondences**, i. e., letters, is thus especially demanding, in particular, if the correspondence network includes several hundred authors, each of which has a different hand [Hotson and Wallnig, 2019]. Figure 1.6 shows three examples from different authors who corresponded with Swiss reformer Heinrich Bullinger (1504–1575) and one example of his own handwriting. Adding to the complexity that we as humans rarely succeed in writing the same character in the same way is the fact that the writing style changed depending on the language Heinrich Bullinger and his contemporaries wrote in. For Latin, they mostly used a **humanist cursive**, while for Early New High German, a form of the **German Kurrent** script was more common.²⁸

The challenge HTR models face is, first and foremost, much greater variability of hands, i. e., different writing styles, than we find them for OCR. This means that efforts to train reliable HTR models which can cope with a great variety of hands are crucial. Thus, research has to focus on mainly three matters: first, it needs to develop universal models which can recognise multiple hands. Second, as follows from the previous paragraph, it must handle different scripts. Third, and as an extension of the second point,²⁹ it must be able to process input that had possibly been written in multiple languages.³⁰

²⁷as of November 21, 2022

²⁸see Marcos [2017] for more details about these scripts

²⁹We just mentioned that scripts differed according to the language an author wrote in.

³⁰After all, multilingualism is not only intricate to printed documents but also to manuscripts.



Figure 1.6: Different handwriting styles in the same correspondence. Not only do the hands of writers differ, but there is also variance within writers, namely when changing from Latin to Early New High German or vice versa. (a) Johannes Fabricius (1527–1566), (b) Johannes Haller (1523–1575), (c) Heinrich Bullinger (1504–575), (d) Johannes Pontisella (1510–1574).

1.3 Scope and Contributions

From the preceding discussion, we identify the following requirements that OCR and HTR models must possess to improve the quality of automatically recognised texts:

1. For **prints**: The ability to handle Antiqua and blackletter fonts.
2. For **manuscripts**: The ability to handle a large variety of handwriting styles.
3. The ability to handle multilingual texts.
4. Additionally: there is a need for quality control in mass digitisation efforts.

In what follows, we will focus on the OCR of text from historical newspapers and the HTR from correspondence data. Concerning the application of OCR to historical newspapers, we will limit ourselves to the more challenging recognition of blackletter fonts. The part concerning handwriting will mainly address the generalisability of HTR models in very heterogeneous settings. Concretely, this thesis makes the following contributions:

- It produces a ground truth (GT)³¹ for the training of OCR models for blackletter-typed newspapers.
- It provides a training set for HTR with a great variety of hands to study writer adaptation of HTR models. We worked in close collaboration with the University of Bern for creating this dataset.
- It examines the usability of neural networks for blackletter fonts in newspapers and compares different training regimes and network architectures (see Chapter 3).
- It compares traditional HTR models to the most recent Transformer-based architectures and suggests training procedures for their successful application (see Chapter 4).
- It suggests methods to assess the quality of the OCRed and HTRed text in the absence of GT (see Chapter 5).

³¹A GT consists of manually annotated data, in the case of OCR and HTR the transcription of historical documents per line, including possible corrections of the layout recognition.

1.4 Outline

Chapter 2 presents the background of different OCR and HTR models to lay the foundations for the architectures of automatic text recognition models. In Chapter 3, we focus on the procedures for the successful OCR of blackletter fonts in newspapers. Chapter 4 suggests methods for the recognition of handwriting in a diverse corpus. Chapter 5 then introduces different methods to assess the quality of automatically recognised texts without GT data. Finally, Chapter 6 draws the main conclusions from this work and hints at possible future extensions of this thesis.

1.5 Publications

This section mentions publications that are related to this thesis. Chapter 3 builds on the following papers and posters published at different conferences:

- Simon Clematide, **Phillip Benjamin Ströbel**. 2018. Improving OCR quality of Historical Newspapers with Handwritten Text Recognition Models. In *Workshop DARIAH-CH*, Neuchâtel, Switzerland.
- **Phillip Benjamin Ströbel** and Simon Clematide. 2019. Improving OCR of Black Letter in Historical Newspapers: The Unreasonable Effectiveness of HTR Models on Low-Resolution Images. In *Digital Humanities Conference 2019 (DH2019)*, Utrecht, Netherlands.
- Maud Ehrmann, Matteo Romanello, Simon Clematide, **Phillip Benjamin Ströbel**, and Raphaël Barman. 2020a. Language Resources for Historical Newspapers: the Impresso Collection. In *Proceedings of the 12th Language Resources and Evaluation Conference*, pages 958–968, Marseille, France.
- **Phillip Benjamin Ströbel**, Simon Clematide, and Martin Volk. How Much Data Do You Need? About the Creation of a Ground Truth for Black Letter and the Effectiveness of Neural OCR. 2020b. In *Proceedings of the 12th Language Resources and Evaluation Conference*, pages 3551–3559, Marseille, France.

The subsequent body of works touches upon topics we will encounter in Chapter 4:

- **Phillip Benjamin Ströbel**, Simon Clematide, Martin Volk, and Tobias Hodel. 2022a. Transformer-based HTR for Historical Documents. *arXiv preprint arXiv:2203.11008*.
- Martin Volk, Lukas Fischer, Patricia Scheurer, Raphael Schwitter, **Phillip Benjamin Ströbel**, Benjamin Suter. 2022c. Nunc profana tractemus. Detecting Code-Switching in a Large Corpus of 16th Century Letters. In *Proceedings of the 13th Language Resources and Evaluation Conference*, pages 2901–2908, Marseille, France.
- **Phillip Benjamin Ströbel**, Tobias Hodel, Andreas Fischer, Anna Scius-Bertrand, Anna Janka, Jonas Widmer, Beat Wolf, Patricia Scheurer, Martin Volk. 2023. Bullingers Briefwechsel zugänglich machen: Stand der Handschriftenerkennung. In *Abstracts zur 9. Jahrestagung des Verbands Digital Humanities im deutschsprachigen Raum e. V. “DHd2023: Open Humanities, Open Culture”*, Luxembourg/Trier, Luxembourg/Germany.
- Anna Scius-Bertrand, Andreas Fischer, Tobias Hodel, **Phillip Benjamin Ströbel**, Martin Volk. (in press). The Bullinger Dataset: A Writer Adaptation Challenge. In *International Conference on Document Analysis and Recognition: 17th International Conference*, San José, California, USA.

Lastly, Chapter 5 has its foundation in the following publication:

- **Phillip Benjamin Ströbel**, Simon Clematide, Martin Volk, Raphael Schwitter, Tobias Hodel, and David Schoch. 2022b. Evaluation of HTR models without Ground Truth Material. In *Proceedings of the 13th Language Resources and Evaluation Conference*, pages 4395–4404, Marseille, France.

Chapter 2

General Background

This chapter lays the technical foundations for most of the parts from Chapters 3 and 4 and examines the role of both OCR and HTR in the digitisation of GLAM collections. Thus, we will start with the general problem definition (Section 2.1), accompanied by a brief historical background (Section 2.2). We recommend reading at least Sections 2.2.3 and 2.2.4 to guarantee an easier understanding of what we present in the following chapters. In the end, we will proceed to the importance of OCR and HTR techniques in the context of the digital transformation (Section 2.3). To save some space and formulate problems more concisely, we will refer to OCR and HTR as **automatic text recognition** (ATR) in the following.

2.1 The Problem Definition of ATR

This section states the general problem definition of ATR and is not a mandatory read to be able to follow the remaining chapters. However, it gives a broad overview of the development of ATR techniques.

ATR combines the world of **computer vision** (CV) and **natural language processing** (NLP) by translating a visual signal into written human language. The general task of any ATR model formulated as a **supervised** machine learning problem is to extract **features** from an image of a word or a whole text line (i. e., the **input**) and transform them into a sequence of characters (i. e., the **output**). Depending on the overall architecture, the feature extraction from the images looks different,

which we detail in Sections 2.2.2, 2.2.3 and 2.2.4.

Hence, we want to find a function f which maps the input (generally an image of a line or a word) to its output (text), as in Equation 2.1:

$$\text{text} = f(\text{image}) \quad (2.1)$$

We formalise the output as a character sequence $\mathbf{y}$ and the input as a matrix $\mathbf{X}$ because we want to predict a **character sequence** from an image consisting of pixels. Moreover, the parameters θ of the function f need to be learnt to arrive at the optimal mapping of $\mathbf{X}$ to $\mathbf{y}$. This leaves us in the simplest case with the function in Equation 2.2.

$$\mathbf{y} = f(\mathbf{X}; \theta). \quad (2.2)$$

What most feature extraction procedures for ATR have in common is that they depend on some image segmentation.¹ We can then understand the input matrix $\mathbf{X}$ as consisting of a sequence of *feature vectors* (see Section 2.2 for more details) extracted from *input segments* (i. e., from partitions of the whole line or word image.) They are arranged as vectors and used as input to our function f . Assuming an optimal segmentation into the single characters in the image, an ATR model then learns to assign each segment to a character from the alphabet. Thus, it solves a **sequential problem** of mapping an input of the form $\mathbf{X}$ to an output $\mathbf{y}$, with $\mathbf{X}$ re-defined as a matrix containing of image segment feature vectors $\mathbf{x}$, i. e., $\mathbf{X} = \langle \mathbf{x}_{s_1}, \mathbf{x}_{s_2}, \dots, \mathbf{x}_{s_T} \rangle$, where each $\mathbf{x}$ consists of N features, and where T denotes the total number of segments, thus $\mathbf{X} \in \mathbb{R}^{N \times T}$. The sequence $\mathbf{y} = \langle y_{s_1}, y_{s_2}, \dots, y_{s_T} \rangle$ contains the corresponding character labels. Naturally, the output sequence consisting of single characters needs to make sense as a whole. Ideally, the model considers the history of already-produced characters to determine the next one. As such, it simultaneously learns and considers language information to decode the information from the image. In the easiest case, we aim to maximise the probability of a character y_t of $\mathbf{y}$ given the feature values $\mathbf{x}_t$ of an input image segment $t \in \{s_1 \dots s_T\}$, i. e., $Pr(y_t | \mathbf{x}_t)$.

We can now summarise the threefold task of an ATR model and formulate it as a probabilistic model:

1. It needs to adequately segment an image of a word or text line image into its segments $\mathbf{s}$, resulting in T segments.
2. At each step $t \in \{s_1, \dots, s_T\}$, it must map each segment (or aggregation of segments) to a character c (or a character sequence). We can express this as

¹Taking all the pixels as features, possibly over three channels for coloured images, to predict the character sequence is computationally too expensive.

maximising the probability of a character from a known alphabet $\mathcal{C}$ given a segment s_t : $\operatorname{argmax}_{c \in \mathcal{C}} Pr(c_t | s_t)$. c_t denotes the most likely character given a segment (i. e., the character the segment has been mapped to).²

3. It must ensure to choose the most likely character sequence. This requires the ATR model to take the history of choices it has already made into account. One way to model this is via the probability of **character n -grams** $Pr(c_t | c_{t-1})$, where we make the decision about the character c_t dependent on the $t - 1$ preceding characters. E. g., by setting n to 2, i. e., modelling character **bi-grams**, we would make the choice about which character to put in the e -slot in the word *glitter* depending on the preceding t . We can also directly bring n -grams into the equation by modelling an n -gram g at timestep t depending on its final character and the preceding n -grams g_{t-1} . E. g., with n again being 2, at the e -slot of *glitter* we would model the bi-gram te based on the bi-gram tt .

We can also formulate the problem as a **generative model** of a word or text line image. This model assumes the image segments are generated from the character sequence. Equation 2.3, which we adapted from Edwards III [2007], illustrates this as follows:

$$Pr(\text{line}) = \overbrace{Pr(g_1)Pr(s_1|c_1)}^{\text{language modeling}} \underbrace{\prod_{t=2}^N Pr(g_t|g_{t-1})Pr(s_t|c_t)}_{\text{visual segment-to-character mapping}} \quad (2.3)$$

This problem formulation assumes an *a priori* character segmentation on which earlier approaches to ATR were built. We will see in Section 2.2.1 how to use this for ATR. This is, of course, not the only way to formulate the problem and we will employ and experiment with first and foremost two different kinds of segmentation-free methods (see Sections 2.2.3 and 2.2.4) in this thesis. However, it nicely highlights the interaction between visual components and (at least here) character-level language modelling.³ In the following, we focus on how this problem has been solved and (re-)formulated in the past.

²This corresponds to $Pr(y_t | \mathbf{x}_t)$ for the t -th character from the previous paragraph.

³as opposed to sub-word language models of Transformers (see Section 2.2.4)

2.2 Approaches to ATR

There are several approaches to solving the problem mentioned in Equation 2.3. In the following, we highlight the most important ones. Section 2.2.1 starts with a brief overview of the character segmentation problem.⁴ Segmentation still played a major role in approaches that used **Hidden Markov Models** (HMM), as detailed in Section 2.2.2.⁵ With the advent of **Convolutional Neural Networks** and **Recurrent Neural Networks** (see Section 2.2.3, which provides more insights about character segmentation and which we recommend reading) HMMs lost importance in ATR. Lastly, Section 2.2.4, which also should get the reader’s attention, introduces the concept of the **Transformer**, altering a lot on the side of image segmentation and language generation for ATR. We encounter the Transformer again in Chapter 4, where we experiment with how it solves the ATR problem for historical documents.

2.2.1 Character Segmentation

Finding the correct character segmentation of a printed or handwritten text line is a challenging undertaking. Ideally, we segment the line image according to the respective characters and map each image segment to a character. However, Sayre [1973] formulated a paradox which tells us that “the individual [characters] must be recognized as such before it can be determined where one inscription leaves off and another begins.” As such, we are confronted with a chicken or egg problem: we cannot segment a line image without recognising characters, and we cannot recognise characters without segmentation.

In printed texts, most characters are separable by white spaces, as shown in Figure 2.1 a) and b). An exception are so-called **ligatures**⁶ like *fi* in Figure 2.1 a), where the top of the *f* is connected with the dot of the *i* and the horizontal stroke of the *f* connects with the downstroke of the *i*, making segmentation more difficult.

For handwriting, on the other hand, segmentation becomes much more challenging. Consider the handwritten Latin word *litteras* (EN *letters*) in Figure 2.1 c). As humans, we have little difficulty segmenting this word into its single characters, al-

⁴For a more detailed overview of research until the early 1990s, please see Mori et al. [1992].

⁵Here, we recommend Steinhilber et al. [1999] for an in-depth survey and El-Yacoubi et al. [1999] for an introduction to HMMs for ATR.

⁶**ligature**: from Latin *ligāre* to bind

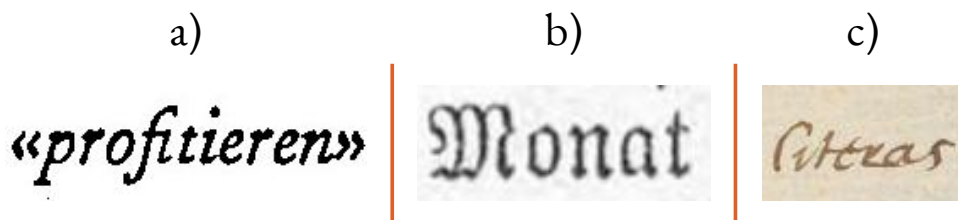


Figure 2.1: Example of (a) the German word *profitieren* (EN *to profit*) in Antiqua and (b) the German word *Monat* (EN *month*) printed in blackletter, both taken from a newspaper, (c) the handwritten Latin word *litteras* (EN *letters*) taken from a letter by Matthieu Coignet (1514–1586).

beit this also depends on our familiarity with (1) the handwriting and (2) the language. If we know that the word in Figure 2.1 c) is a Latin word we can deduce that, given the context, the character between the *t* and the *r* must be an *e*, although it is likely to be confused with another *t*. The wider the context gets, the easier it is to guess a missing character: having the left context of *lit* and the right context of *ras*, a person knowledgeable of Latin easily infers that the missing character must be an *e*.

An ATR model, however, must either be provided with the segmentation beforehand or learn the segmentation from the data. Comparing the different segmentations of *litteras* in Figure 2.2, only 2.2 a) is correct and is most likely to result in the transcription of *l-i-t-e-r-a-s*. Segmentations 2.2 b) and 2.2 c) could be other possibilities an ATR model could infer from the visual signals, and 2.2 d) represents a segmentation into equal-sized image blocks. As such, segmentation 2.2 b) could result in *A-t-e-r-c-u-r*, the one in 2.2 c) in *l-i-t-t-c-a_-s*, and the one in 2.2 d) in *l-i-b-n-c-u-r*, none of which are correct.

In reality, the segmentation does not happen at random but is guided by visual cues. Using **pixel densities** along the columns of an image is one way to identify where to draw boundaries because a low pixel density at a specific place would indicate character separations. We illustrate this with the handwritten word *litteras* and the word *Monat* in blackletter print in Figure 2.3. The segments follow either the minima (left-hand side) or the maxima (right-hand side) of the pixel densities (orange curves below the image) in the image.

The difference between the handwritten and the printed word is the smoother curve for the handwritten one due to the connection between the single characters. We see that the minima and maxima for *Monat* are much more pronounced and therefore

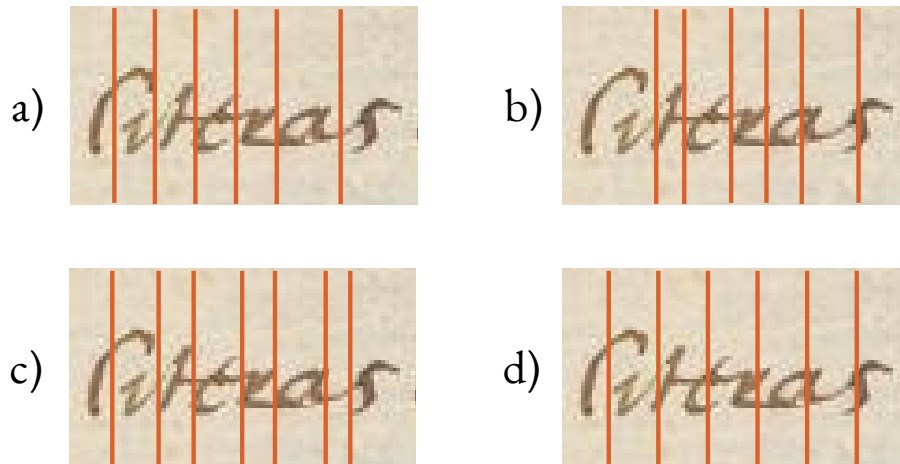


Figure 2.2: Different segmentation possibilities. The one portrayed in a) would be the correct one, b) and c) represent random but not unlikely segmentations and d) a segmentation into equally-sized chunks.

facilitate the segmentation into characters. However, it is possible in both cases to map a character to either one segment or to an aggregation of segments (as the blue brackets below the pixel densities show).

Some of the earliest approaches were, in fact, dedicated to the recognition of handwritten words with data obtained from a handwriting transmitter (telautograph) where information like pen velocity, stops, minima and maxima were used as features to decipher words participants wrote [Eden, 1962, Mermelstein and Eden, 1964a,b].⁷ In more detail, they used these features to segment the written words into strokes and mapped single strokes or a combination of strokes to prototype characters. For example, the *a* in *literas* is composed of four different prototype strokes (defined as the stroke alphabet by Eden [1962]). Figure 2.4 illustrates this process.

Sayre [1973]’s contribution then extended the above-mentioned techniques to already-written words, i. e., it did not rely on stroke sequences, as shown in Figure 2.4. Rather, Sayre modelled the recognition of words as a three-step approach

⁷These features are nowadays still used in the so-called *online* recognition of handwriting, which corresponds to the instantaneous deciphering of a person writing on a phone, computer, or tablet. We are only concerned with *offline* handwriting here.

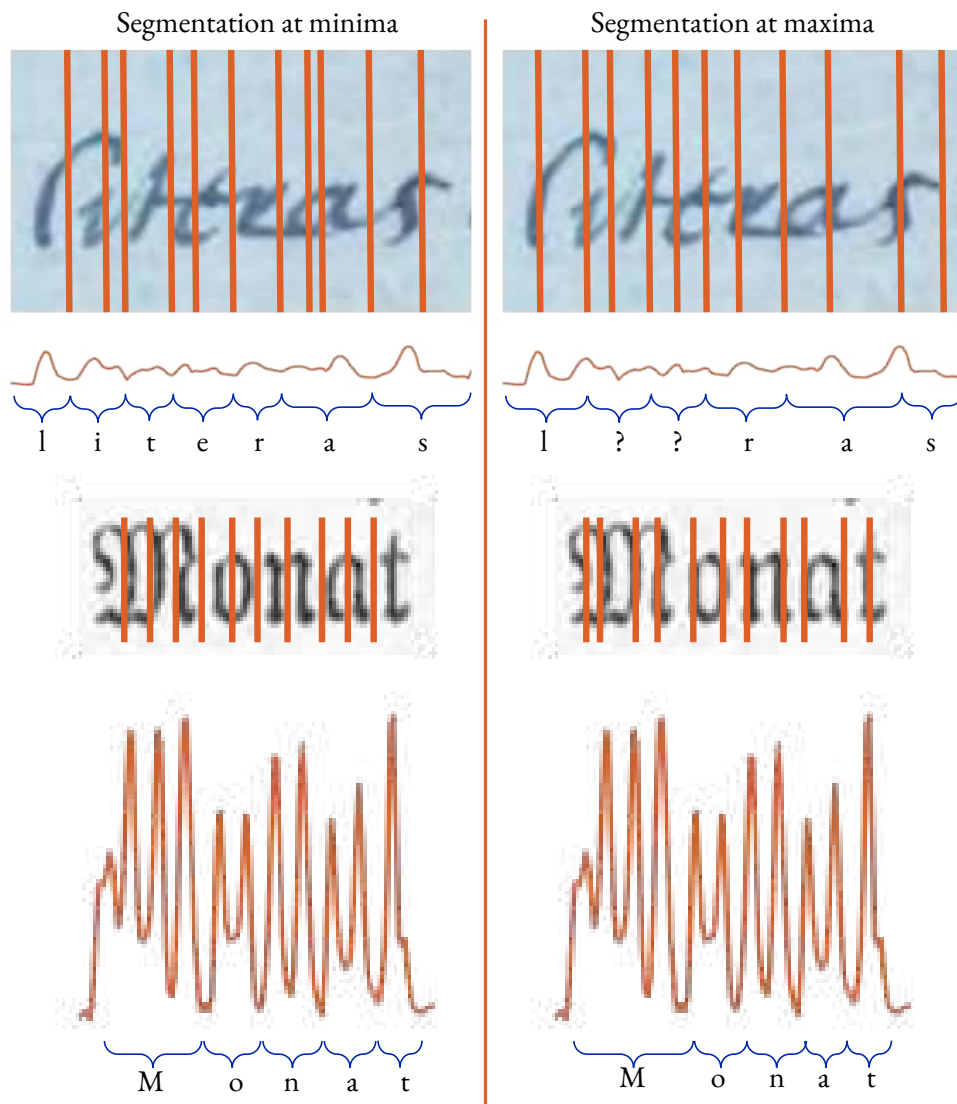


Figure 2.3: Different segmentation strategies based on minima and maxima of pixel densities (orange curve) of the handwritten word *literas* and the blackletter print *Monat*.



Figure 2.4: The prototype *a* composed into its stroke sequence and the *a* from *litteras* split up accordingly. The left-hand prototype *a* is taken from Eden [1962]

where a first routine segmented the image based on visual cues, a second added additional information about the presence of other indicative features of characters (e. g., the presence of a cross hinting at a *t*), and the third consisted of a contextual interpretation which was supposed to map the segments to characters based on bi- and tri-gram information. As such, Sayre’s approach already included the language modelling information formulated in Equation 2.3.

In subsequent work, Ehrich and Koehler [1975] highlighted the importance of context, especially for segmentation and the mapping to characters. Segmentation occurred with the help of defining pre-segmentation marks based on downstrokes. The *pre-recogniser* then attributed a set of possible characters to each pre-segment based on 15 invariant features. The authors used this information to build a binary tree, for which a *correlator* would output the most probable path and hence the most probable character sequence.

2.2.2 Hidden Markov Models

As a result of Sayre [1973]’s paradox mentioned in the previous section, research split into two branches: one branch examined the possibilities of segmentation-based methods, using some of the segmentation techniques explained in Section 2.2.1, while the other branch developed segmentation-free approaches. For both of them, HMMs proved essential to solving the ATR problem. Early uses of HMMs for ATR go back to the late 1980s [Kundu et al., 1988] and gained momentum, especially for handwriting recognition [Bunke et al., 1995, Cho et al., 1995, Procter et al., 2000, Gilloux et al., 1995], but also for font-invariant and multilingual OCR models [Bazzi et al., 1999]. And although the deep learning wave had its impact on the research of HMMs for ATR (see Sections 2.2.3 and 2.2.4), they are still being researched, albeit often combined with neural approaches [Arora et al., 2019].

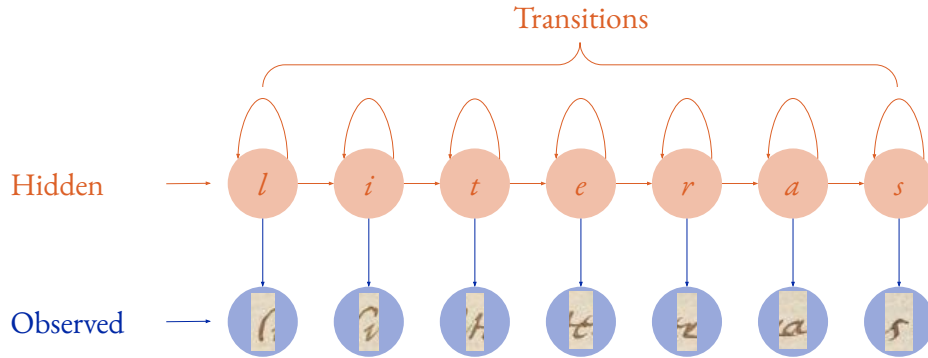


Figure 2.5: A simplified illustration of a Markov chain with the help of the *literas* example. We assume an already segmented input as the observed image segments, while a model should infer the hidden characters from the observed values, i. e., the visual information. As in Equation 2.3, the orange part refers to the language model (and the orange arrows thus to the transition probabilities), while the blue arrows represent the emission probabilities.

Model Architecture

First, we explain the HMM and its use for ATR. HMMs were first employed in the 1960s [Baum and Petrie, 1966, Baum and Eagon, 1967, Baum et al., 1970, Baum, 1972] to achieve inference on *Markov processes*. A Markov process, also known as a *Markov chain*, is a tool to model stochastic processes over time during which an outcome is predicted based on the current and previous states [Markov, 1906]. In ATR, we have such a process if we look at each character in a sequence as being dependent on the preceding one. As such, a string of characters can be modelled as a Markov process and constitutes a character-based language model, as hinted at in Equation 2.3. For our *literas* example, the Markov chain looks depicted in Figure 2.5.

Assuming that the image of the word *literas* is already split into segments, Figure 2.5 shows that these segments correspond to the observed part of the HMM model.⁸ Hidden, and what we want to infer from the image segments, are the single characters. Different probabilities characterise the HMM. The **transition**

⁸It is common to pre-process the images before segmentation, to reduce variation. Frequently applied techniques include binarisation, baseline slant normalisation, character skew correction, etc. We will not detail all of these procedures here but mention essential pre-processing steps we took in Section 4.3.2.

probability⁹ is defined as the likelihood of a character c_t if preceded by c_{t-1} , i. e., $Pr(c_t|c_{t-1})$. In the above example, this could be the probability of an i preceded by l , or $Pr(c_t = i|c_{t-1} = l)$. HMMs normally learn these probabilities from dictionaries or other text sources. The transition probabilities correspond to the character-level language modeling part in Equation 2.3. Up until the emergence of HMMs in ATR, n -gram statistics have mainly been used to correct or identify errors.¹⁰ The **emission probability** of an image segment is given by $Pr(s_t|c_t)$, i. e., the features of an image segment given a character. The features are usually handcrafted and comprise image information such as loops, ascenders, descenders, height, positions respective to helping lines like top and bottom lines, baselines, middle lines, and pixel densities, to name a few.

Model Training and Recognition

During the training process,¹¹ the HMM learns the characteristics of the different characters and the transitions between them. For training, HMMs use the **Baum-Welch algorithm** [Dempster et al., 1977]. It adjusts the parameters of an HMM and is an implementation of the iterative **expectation maximisation** algorithm. It maximises the probability of observing a training set of sequences according to the maximum likelihood estimation criterion.

Once the model has been trained, it can be used to recognise characters in new images or documents. This decoding step uses the **Viterbi algorithm** [Forney, 1973]. It calculates the most likely sequence of characters in a new image based on the observations (image segments, or rather, extracted features) in the input data.

Segmentation-based Methods

Segmentation-based methods for ATR involve dividing an image into primitive segments and attempting to match these segments with the characters of the word (see

⁹They are characterised by a two-dimensional matrix which describes how likely it is to reach a state, in our case a character, from another state. E. g., we can assume the transition probability of a t following a q in the English language to be 0 since qt is an impossible combination, at least if do not consider acronyms.

¹⁰see Suen [1979] for an overview of using n -grams in NLP

¹¹We only briefly mention training strategies here since HMMs will not play a role in this thesis. For a more concise overview of the inner workings, see Dietterich [2002].

Section 2.2.1). These methods are flexible concerning the size and nature of the lexicon because they rely on character models. Various constraints must be satisfied in the matching process, such as maintaining the spatial order of the primitive segments and ensuring that different segments are disjoint and cover all of the primitive segments of the word. Segmentation-based methods can be implemented as HMMs (other popular choices are dynamic programming or graph search methods, which we do not further elaborate on here). Some HMM-based methods do not explicitly determine the borders of a character before the evaluation phase and rely on estimating the probability of a feature given a possible character. Other methods match every single primitive segment with a complete character but may require additional processing to overcome imperfections in the segmentation algorithm.

Segmentation-free Methods

Segmentation-free ATR involves interpreting an observation sequence derived from a word or line image without the need for segmentation beforehand. This observation sequence can be based on various types of features, such as low-level features that are extracted directly from the image, aggregated low-level features that serve as primitives, or higher-level holistic features that are prominent and invariant with respect to writing styles. These higher-level features, known as symbols, can be further categorized based on their size, location, or orientation. In the upcoming section, we will gain an in-depth understanding of other segmentation-free methods commonly used in ATR.

2.2.3 Convolutional Neural Networks × Recurrent Neural Networks

Segmentation clearly poses a problem, which is why architectures that infer features from the line or word images themselves are advantageous. Certain neural networks are designed for exactly this purpose. How can we now reformulate the general problem of ATR from Equation 2.2 as a (deep) neural network? The general term persists while we add non-linearities on the one hand and additional layers on the other. Together, these measures help to overcome the weakness of linear models (namely that they only learn linear functions) and HMMs that are based on an *a priori* segmentation. As such, a deep neural network is supposed to be able to approximate every function. To introduce non-linearities, we pass Equation 2.2 through

an **activation function** (see below):¹²

$$\mathbf{h} = g(f(\mathbf{x}; \boldsymbol{\theta})) = g(\mathbf{W}^T \mathbf{x} + \mathbf{b}) \quad (2.4)$$

The function g introduces the non-linearity, while $\mathbf{W}$ represents the weights of the particular layer $\mathbf{h}$ and $\mathbf{b}$ acts as bias. By adding several hidden layers, we can create a deep neural network:

$$f(\mathbf{x}) = \mathbf{h}^{(3)}(\mathbf{h}^{(2)}(\mathbf{h}^{(1)}(\mathbf{x}))) \quad (2.5)$$

The superscripts indicate the layers. Equation 2.5 implies that input $\mathbf{x}$ goes through layer (1) and is altered by its weights and the non-linearity, serving as input for layer (2), which alters the modified input again by its weights and the non-linearity and so on. The chain-like structure guarantees that information flows through the whole network, from the input, over one or several hidden layers, to the **output layer**. This output layer maps the outputs from the last layer to a probability distribution over all possible labels in $\mathbf{y}$, i. e., $Pr(\mathbf{y}|\mathbf{x})$. With the resulting prediction $\hat{\mathbf{y}}$ and the real labels $\mathbf{y}$, it is possible to compute the **loss**, which is then used to adjust the parameters in the network. But the chain-like structure also means that these networks can grow quickly in the number of parameters that need to be trained.

The ideas of neural networks like **Convolutional Neural Networks** (CNNs) and **Recurrent Neural Networks** (RNNs), the combination of which we use for ATR, are much older and have their origins in the 1980s [Fukushima, 1980, LeCun et al., 1989] and the 1990s [Hochreiter and Schmidhuber, 1997], respectively. From 2010 on, however, the improved hardware – together with further developments in the research of neural networks (like different activation functions and normalisation procedures) – allowed for training models 1) more efficiently and 2) on larger amounts of data. It became possible to train (deep) neural networks, which up to this point were notoriously difficult to train, with the **backpropagation**¹³ [Linnainmaa, 1976] algorithm.¹⁴

This development led to a huge improvement in model performance in classification tasks in CV. For CNNs, this is best illustrated by the near-perfect performance on the MNIST dataset [LeCun et al., 2010]. The MNIST dataset contains 70k 28×28 binary images of handwritten digits. Although approaches using CNNs (like Ciresan et al. [2011]) could raise the benchmark only slightly or perform on par (with test set errors between 0.35% and 0.4%), they proved much more efficient in

¹²Note that for simplicity, we rewrite $\mathbf{x}$ as one feature vector of a segment contained in $\mathbf{X}$.

¹³Backpropagation stands for propagating the error back during the backward pass using automatic differentiation after the forward pass. This procedure guides how to adapt the network weights and is where the “learning” occurs.

¹⁴see Goodfellow et al. [2016] for more details

training. CNNs have remained competitive until today, reaching similar results as more recent model architectures on, e. g., the *ImageNet* [Deng et al., 2009] task, which is about image classification.¹⁵

For ATR, CNNs and RNNs can be combined as shown in Figure 2.6. In the following, we will highlight the most important features of these two components, illustrating parts of the architectures with the help of Figure 2.6. When we keep the original problem formulation from 2.3, we now find two different networks responsible for the two sub-tasks: the CNN extracts the features from the image, and the RNN generates the text sequence. The intriguing appeal of CNNs is that we no longer need to handcraft features, but so-called *kernels* assume this task for us.

Convolutional Neural Networks

CNNs have been used for recognising handwritten digits from their very beginning [Fukushima, 1980] and have then been adapted to recognise longer strings of texts from images. Like other neural networks, it contains an input layer, one or several hidden layers, and an output layer. In the following, we mention the most important building blocks for the CNN architecture.¹⁶

Convolution The main idea of the CNN is the transformation of an image of dimensions $\text{height} \times \text{width} \times \# \text{ of channels}$ to so-called *feature maps* via several layers. A set of learnable kernels of possibly varying dimensions slides over the input image to achieve this. The *stride* parameter determines how much the kernels overlap. The stride value impacts the output dimensions of the feature map. Assuming a stride value of 2, we multiply the values of a certain kernel of dimension, e. g., $\text{height}=3 \times \text{width}=3 \times \text{channels}=3$ (for an RGB coloured image) at each stride with the pixels of the image at this position, which results in a new pixel value. Depending on the values of the kernel, the resulting image may be sharpened, blurred, skewed, or undergo any other linear transformation.¹⁷ Convolutional layers can be stacked on top of each other, where the kernels of a subsequent layer would work on the previously produced feature maps. The more convolutional layers we stack,

¹⁵see, among others, Cai et al. [2023]

¹⁶For a more comprehensive introduction to how CNNs work, please see Zeiler and Fergus [2014] and Goodfellow et al. [2016].

¹⁷For a comprehensive explanation of kernel functionality, please see Victor Powell's blog: <https://setosa.io/ev/image-kernels/>.

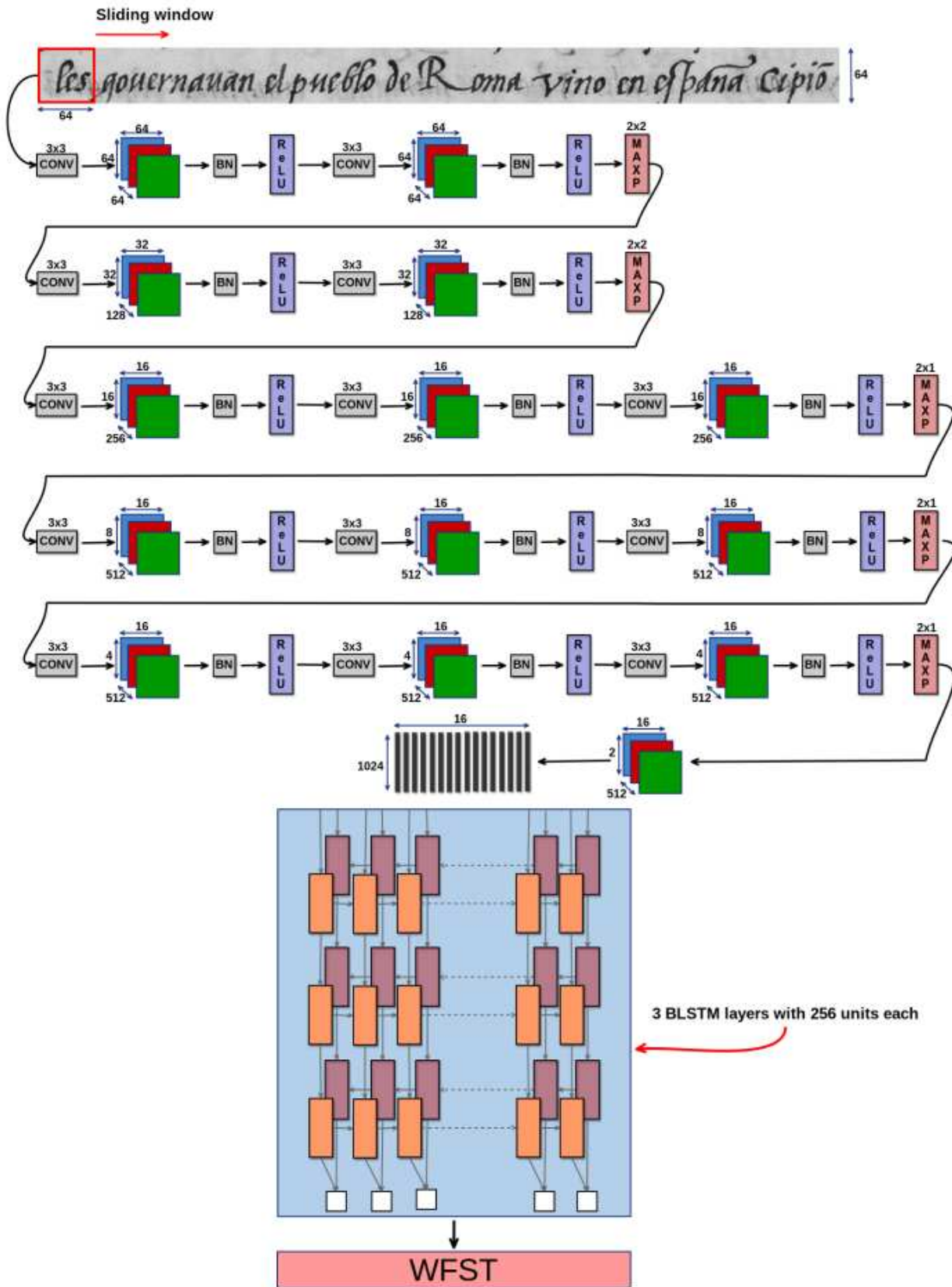


Figure 2.6: Example of a combination of a CNN and an RNN for ATR taken from Granell et al. [2018].

the more we increase the *receptive field*¹⁸ of the network. In Figure 2.6, we find 13 convolutional layers named CONV. Granell et al. [2018] used a consistent kernel size of 3×3 on a black-and-white image. They let the CNN slide over the image in a square of 64×64 pixels, having normalised the line height to 64.

Batch normalisation Batch normalisation is a means to enable the training of deep neural networks *per se*. As a side effect, it also prevents *overfitting*. As is true for standard machine learning models (also, e. g., the HMM), models may learn the training data by heart, i. e., they overfit, which harms their generalisation properties. During training, the data is fed into the model in batches (e. g., 8, 16, or 128). By normalising the batch, it is possible to *recentre* the values in the feature maps, which could otherwise lie far apart and drift apart even further during the following convolutions. In Figure 2.6 we find a batch normalisation BN after each convolution.

Activation function The *activation function* introduces non-linearities into the (deep) neural network and determines the output of each node (hence, it controls the flow of information that is put forward through the network). For CNNs, **Rectified Linear Unit** (ReLUs) are most often used [Fukushima, 1969]. The ReLU function eliminates values smaller than 0 and thus also prevents overfitting and vanishing gradients. In Figure 2.6, we find a ReLU layer after each convolution and batch normalisation.

Pooling After a convolutional layer we can add a *pooling layer*. Pooling shrinks the dimensions of the feature map and is thus an efficient way for downsampling. It selects the most dominant values for further computation. In Figure 2.6, we find five max-pooling layers (MAXP). They either have dimensions 2×2 or 2×1 . We see that whenever the 2×2 pooling is applied, the dimension of the whole feature map shrinks. It is divided by two and decreases first from 64×64 to 32×32 and then to 16×16 . The 2×1 poolings only affect the height of the feature maps and shrink them to 8, 4, and finally 2.

¹⁸To be more concrete: The kernels in the first layer are applied to the image directly. They produce a certain number of feature maps, which we could describe as a “summary” of the most important features of the original image. The subsequent kernels act on these feature maps and layer for layer get access to other pixels of the image (or “featurised” versions thereof) they could not take into account during the first pass.

Fully connected layers After several iterations of convolution and pooling layers, a *fully connected layer* enables the network to make predictions based on the extracted feature map sequence. For ATR, however, we need to infer character sequences from the feature map sequence, which is why an RNN is attached to the CNN (see below).

Recurrent Neural Networks

RNNs [Rumelhart et al., 1986] are suitable for sequence modelling. Their task in ATR is to generate the most likely character sequence from the feature maps provided by the CNN. The information flows from one cell to the next in a simple feed-forward fashion. In this manner, the RNN learns a language model capable of catching long-range dependencies. It is advantageous if a language model can rely on the first character of a word for determining the last character. An n -gram language model loses this information and only conditions on the preceding n characters. To counteract this weakness, an RNN establishes connections between the layers. More concretely, and as is evident from Figure 2.6, ATR makes use of so-called **Bi-directional Long-Short-Term Memory (BLSTM)** layers, which are a special type of RNNs, combining bi-directional RNNs [Schuster and Paliwal, 1997] and Long Short-Term Memory [Hochreiter and Schmidhuber, 1997] cells. Figure 2.7 is an example of a sequential RNN architecture. The input $\mathbf{x}$ gets encoded from left to right in the hidden layer $\mathbf{h}$ and from right to left in the hidden layer $\mathbf{g}$. At the same time, information flows from step t to the next step $t + 1$. Lastly, we see that $\mathbf{x}^{(t-1)}$ is connected to $\mathbf{g}^{(t-1)}$, which contains all the information that has been put through the network from the end of the sequence.

What information exactly reaches each state in the network is controlled by the Long Short-Term Memory (LSTM) cell. Figure 2.8 illustrates the inner workings of the LSTM cell. It contains several gates which either pass the information or do not. E. g., the *forget gate* controls whether the information of the actual step t should be considered to compute the output of the actual step $t + 1$, i. e., it determines if it is advantageous to give more weight to the current input or the memory.

When a line image traverses the CNN, it does not get segmented. The BLSTM, however, expects a sequence. The problem of labelling unsegmented data not only exists for ATR but also for speech or gesture recognition. This is why Graves et al. [2006] introduced RNN networks with **connectionist temporal classification (CTC)**. The CTC is a special output layer which computes a loss function that allows the labelling of unsegmented data. It permits to output an

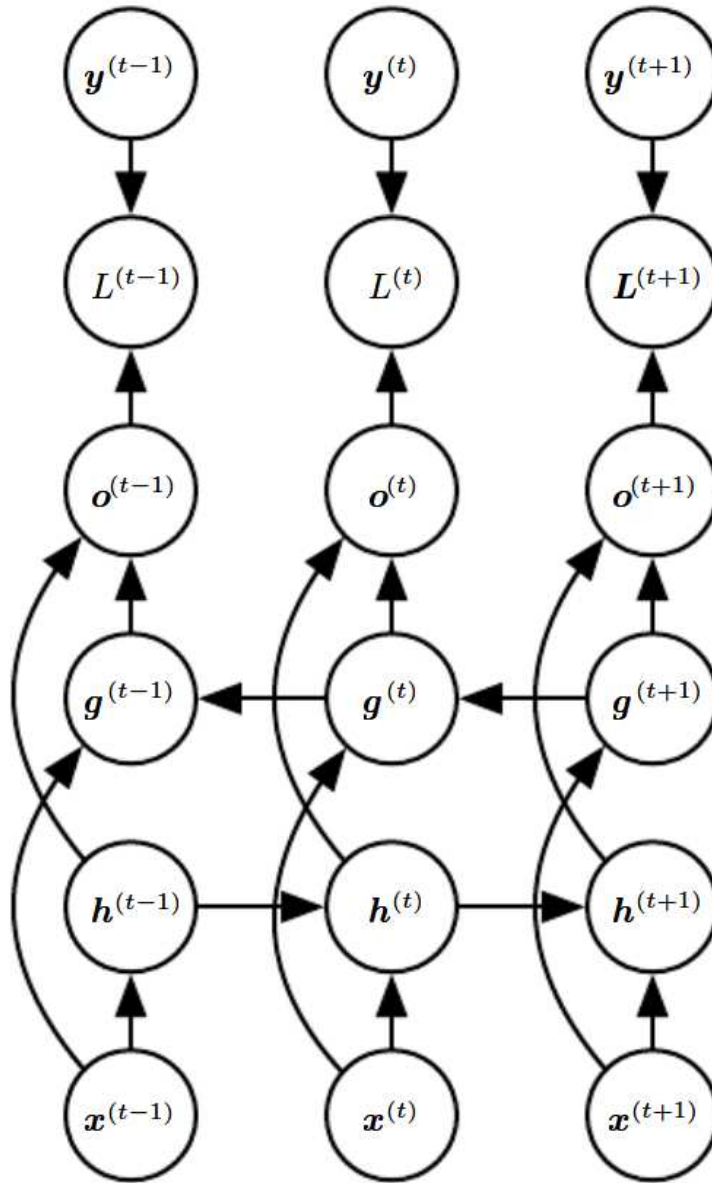


Figure 2.7: Illustration of a bi-directional RNN taken from Goodfellow et al. [2016].

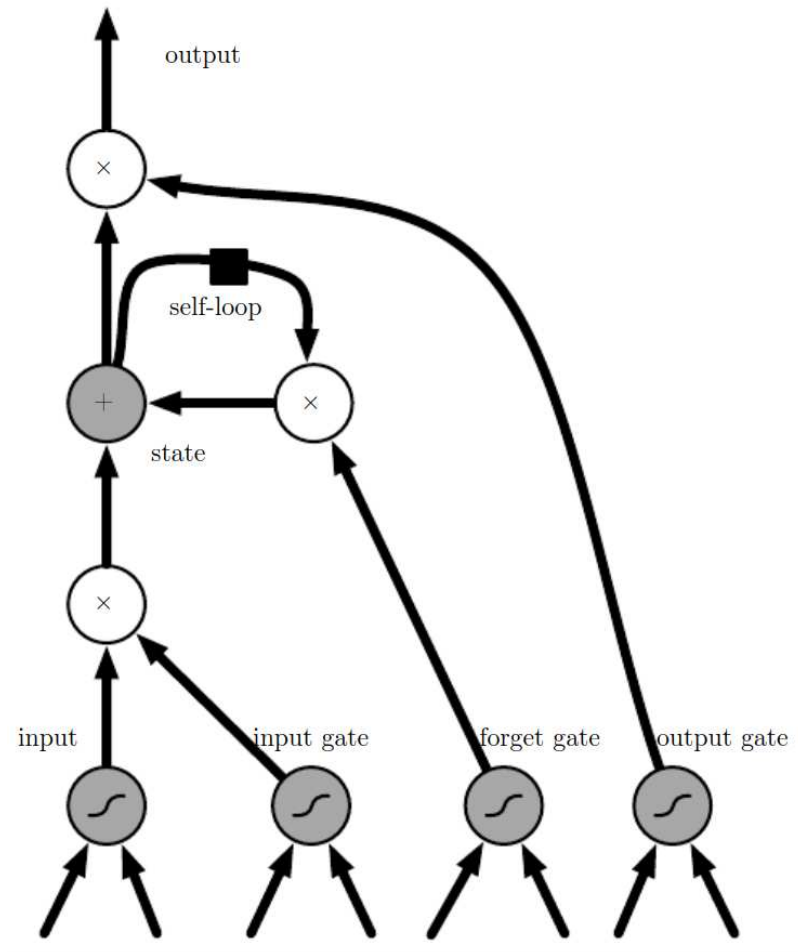


Figure 2.8: Scheme of the LSTM cell, taken from Goodfellow et al. [2016].

“empty” label albeit being bound to a certain number of time steps, in which case the model outputs an empty string (in the context of ATR). Using this approach, Graves and Schmidhuber [2008] applied multi-dimensional BLSTMs with a CTC output layer to recognise Tunisian city names written in Arabic handwriting. In a further study, Graves et al. [2009] combined BLSTMs and a set of handcrafted features to recognise text from lines written in English and performed ten percentage points better in terms of word accuracy than HMMs.

Figure 2.6 shows the application of BLSTMs after the CNN. Since the 512 feature maps of dimensions 2×16 after the last MAXP layer is unsuitable for standard BLSTMs, Granell et al. [2018] flatten the feature maps by concatenating the

columns, resulting in 16 feature vectors of dimension 1×1024 . They then predict the character(s) for each 64×64 square in the input line image. Since they use a sliding window approach with an overlap of two pixels, the model could predict a certain character twice (or even more often). This is why they added a Weighted Finite State Transducer (instead of applying the CTC loss) to map the output of the BLSTM to the final character occurrence.

In Chapter 3, we take a closer look at how to employ combinations of CNNs and RNNs for the use of ATR for printed historical documents. In addition, we also evaluate their suitability on handwritten documents in Chapter 4 and use them as baselines.

2.2.4 From RNNs to Transformers

With the emergence of **sequence-to-sequence** (seq2seq) models [Cho et al., 2014, Sutskever et al., 2014], RNNs became capable of mapping variable-sized input to variable-sized output. This development alleviated the restrictions imposed by CNN $\times$ RNN models using CTC. We thus briefly explain the improvements initiated by seq2seq models. We also touch upon the most valuable advances of seq2seq models, which resulted in the Transformer architecture and are central to this thesis. We thus recommend reading Section 2.2.4.

Sequence-to-Sequence Models

While originally developed for automatic machine translation, seq2seq models are also suitable for ATR. Seq2seq models work with an encoder-decoder architecture. The **encoder** produces a hidden representation from the input and the hidden states of the encoder as

$$h_e = \text{encoder}_{\text{RNN}}(\mathbf{x}, \mathbf{h}^{(n_e)}) \quad (2.6)$$

which is then fed to the **decoder** to generate the output sequence:

$$h_d = \text{decoder}_{\text{RNN}}(h_e, \mathbf{h}^{(m_d)}, \mathbf{y}) \quad (2.7)$$

During the decoding process, the decoder uses information from the preceding output (i. e., y_{t-1}) at each step t . Figure 2.9 is an example of an encoder-decoder structure, where C corresponds to h_e from Equation 2.6. In encoder-decoder networks, there is a similar task allocation as seen in the general problem definition in Equation

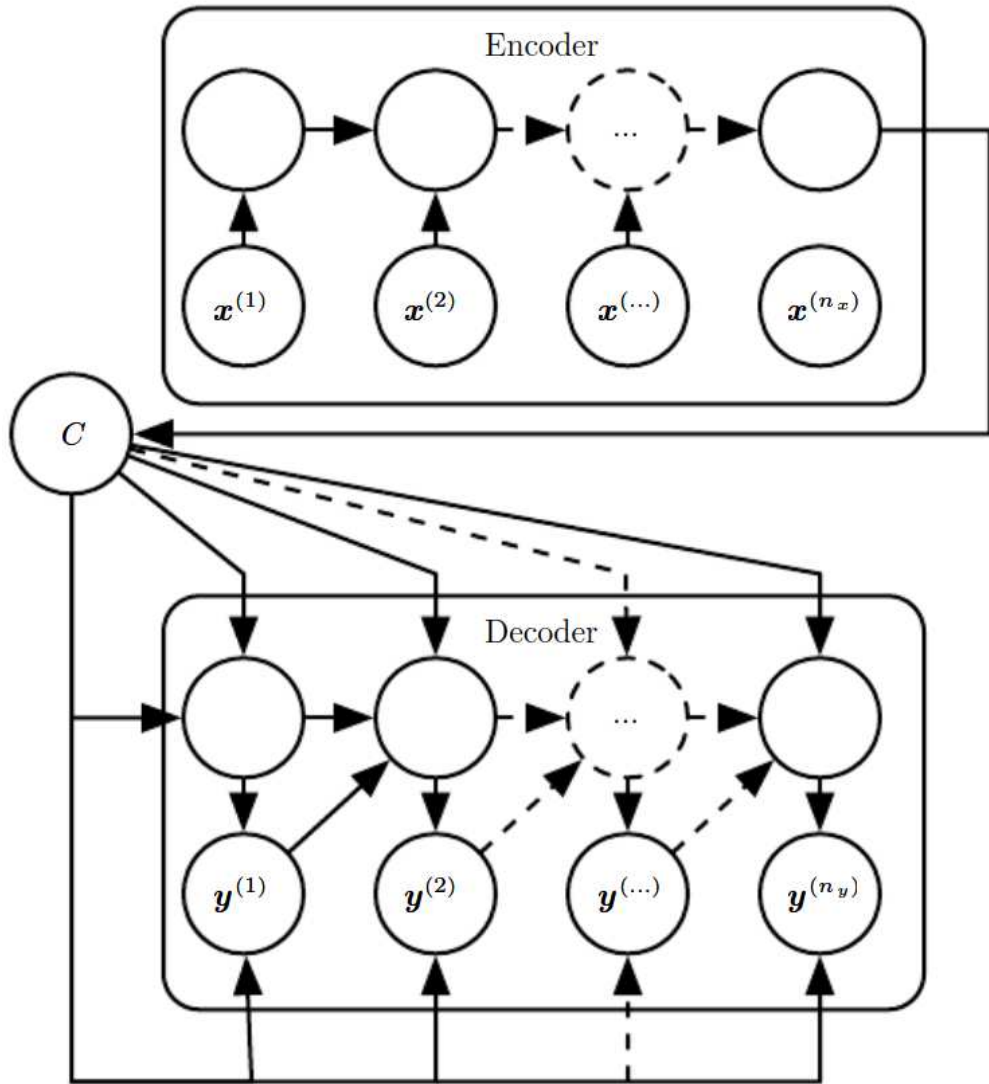


Figure 2.9: An encoder-decoder architecture taken from Goodfellow et al. [2016].

2.3: the encoder processes the visual input, while the decoder maps the visual signal to a language string. This contrasts the CTC approach introduced above, where only one (but possibly multi-layered) RNN processes the visual signal and outputs a character sequence simultaneously.

One weakness of early seq2seq models was that during the computation of h_e (or C in Figure 2.9), a lot of information gets *squashed* into a fixed length vec-

tor. This is problematic for long inputs, which are usually present in ATR. Bahdanau et al. [2015] alleviated this problem for machine translation by allowing for a hidden representation variable in length and adding an **attention mechanism**. The attention mechanism decides for each hidden representation h_t of an input x_t how important it is to generate any y in the output sequence. It adds weights to the hidden representation of h_t at each step t before the different h_t s of the encoder get combined to h_e . These weights are learnt and updated during the training process.

The Transformer

Vaswani et al. [2017] saw potential in the attention mechanism, but to them, the RNN proved to be the bottleneck, especially for longer sequences, which meant a heavier computational load. Thus, they designed an architecture called the **Transformer** which removes the RNN and adds **self-attention**. Figure 2.10 depicts the Transformer architecture.

While still organised as an encoder-decoder system, the main feature of the Transformer is the **Multi-Head Attention**. It occurs at three places in the Transformer architecture (see the orange boxes in Figure 2.10):

1. There is Multi-Head (self-)attention in the encoder block. While traversing the N blocks of the encoder, the encoder learns to put self-attention on inputs. It projects the input into queries, keys, and values. Given a query, e. g., a token¹⁹ in an input sentence, it is compared to all the other words, which act as keys, in the input. The values then are measures of the relation between query and key, and hence determine on which word each word in a sequence puts its attention.
2. The Masked Multi-Head (self-)attention in the decoder block works similarly, with the exception that it is shifted to the right to be able to predict the next output token. Moreover, the inputs, keys and values to the right of the output to predict are masked so the model cannot “peak” into the future.
3. Between the last encoder block and the decoder blocks is an additional Multi-Head Attention layer. The keys and the values from the last encoder layer are matched with the queries from the Masked Multi-Head Attention layer of the decoder. In this way, the decoder of the Transformer pays attention

¹⁹The input is mostly tokenised into sub-words. We elaborate on this in the following section.

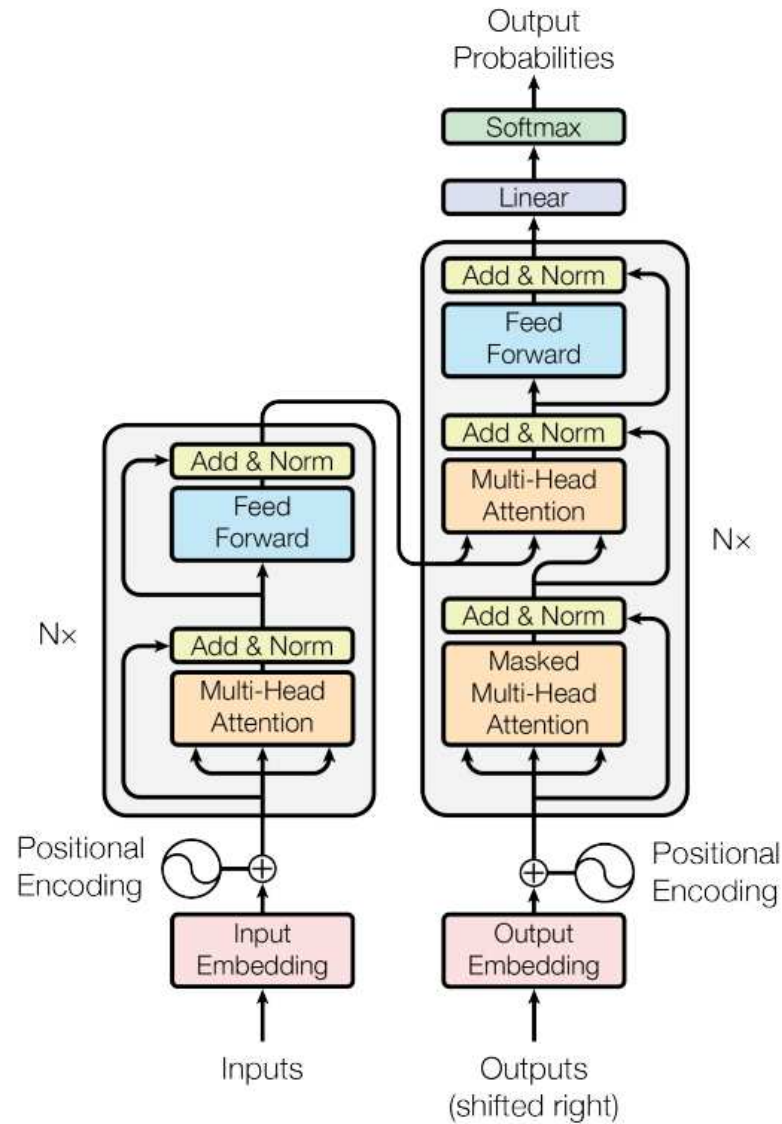


Figure 2.10: The Transformer architecture taken from Vaswani et al. [2017]. On the left, the encoder blocks, and on the right, the decoder blocks.

to the whole encoder input sequence and learns which encoder inputs are important to make a prediction at a certain state.

Another speciality of the Transformer is the processing of inputs. It does not input the sequence one by one, as an RNN would, but instead processes the whole

input in parallel. This results in a loss of positional information, which is crucial for many sequence-to-sequence tasks. The encoding of the position of each item in the input is thus crucial, which is why a **positional encoding** is added to the input embeddings.

TrOCR

Ever since the emergence of the Transformer architecture in 2017 [Vaswani et al., 2017] and the introduction of so-called **Bi-directional Encoder Representations from Transformers** (BERT) [Devlin et al., 2019] one year later, the approach in NLP from training fully-fledged systems from scratch has changed to using large **pre-trained** models and **fine-tune** them for specific tasks. Examples of this procedure comprise question-answering systems, summarisation, and machine translation. As such, BERT-style models *per se* are strong **transfer learners**²⁰ [Ruder et al., 2019] and capable of adapting to many different tasks. For this reason, Transformers have also found their way into image processing [Dosovitskiy et al., 2021, Touvron et al., 2021], leading to the development of large pre-trained Transformers for images [Bao et al., 2021].

The classical approach for ATR consists of a CNN to extract features from the word or line images, while an RNN translates the image features into a sequence of characters (as laid out in Section 2.2.3). The recent developments in language and image Transformers suggest that combining these two architectures would eliminate the CNN backbone on the one hand and replace the RNN with a stronger language model on the other. Li et al. [2021] realised exactly this approach by introducing *TrOCR*, a Transformer-based OCR model. The TrOCR architecture combines an image Transformer serving as an encoder and a language Transformer acting as a decoder. It has been trained on synthetically generated line images, of which 687.3M are printed and 17.9M handwritten data, together with the corresponding English transcription to train the language transformer. In summary, depending on the number of layers in the image transformer, the pre-trained TrOCR model has between 334M (TrOCR_{BASE}) and 558M (TrOCR_{LARGE}) parameters. Especially TrOCR_{LARGE} is considerably bigger than the common BERT-style models, of which most top at about 340M parameters, but still smaller than, e. g., GPT-2 models [Radford et al., 2019] with 1.5B parameters or GPT-3 [Brown et al., 2020] with even 175B parameters.

²⁰Transfer learning occurs when a model trained on task A is applied to task B , e. g., by adding a classification layer for sentiment analysis on top of BERT, which is trained as a masked language model.

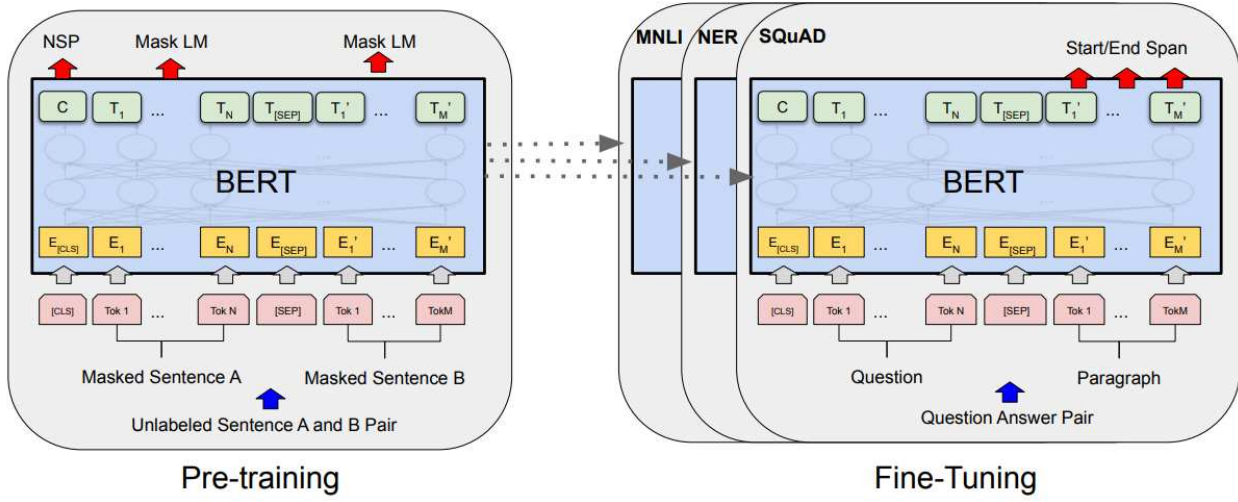


Figure 2.11: The BERT architecture taken from Devlin et al. [2019]. The inputs are masked, and there is a next-sentence prediction (NSP) for each input sentence. The pre-training step is illustrated on the left. The right-hand side of the image shows the fine-tuning step for specialised tasks like, e. g., named entity recognition (NER).

Since we use TrOCR throughout this thesis (especially in Chapters 3 and 4), we present the building blocks in more detail. As mentioned above, TrOCR consists of an image encoder and a language decoder.

TrOCR decoder Because the language model represented by the decoder is a slightly modified version of the original Transformer decoder by Vaswani et al. [2017], we start with a short explanation of the decoder. The decoder in TrOCR consists of a large, pre-trained **masked** language model called BERT [Devlin et al., 2019]. The BERT model (see Figure 2.11) is optimised for two tasks:

1. Masked language modeling: during training, 15% of the input **tokens**²¹ are masked, and the task of the model is to predict the masked tokens.
2. Next-sentence prediction: BERT tries to predict whether the following sentence is a continuation of the input sentence.

²¹It is important to note that BERT operates on sub-words (or WordPieces, see Wu et al. [2016], inspired by Sennrich et al. [2016]), i. e., a token corresponds to a **sub-word**, where the optimal sub-word tokenisation is learnt from a reference text.

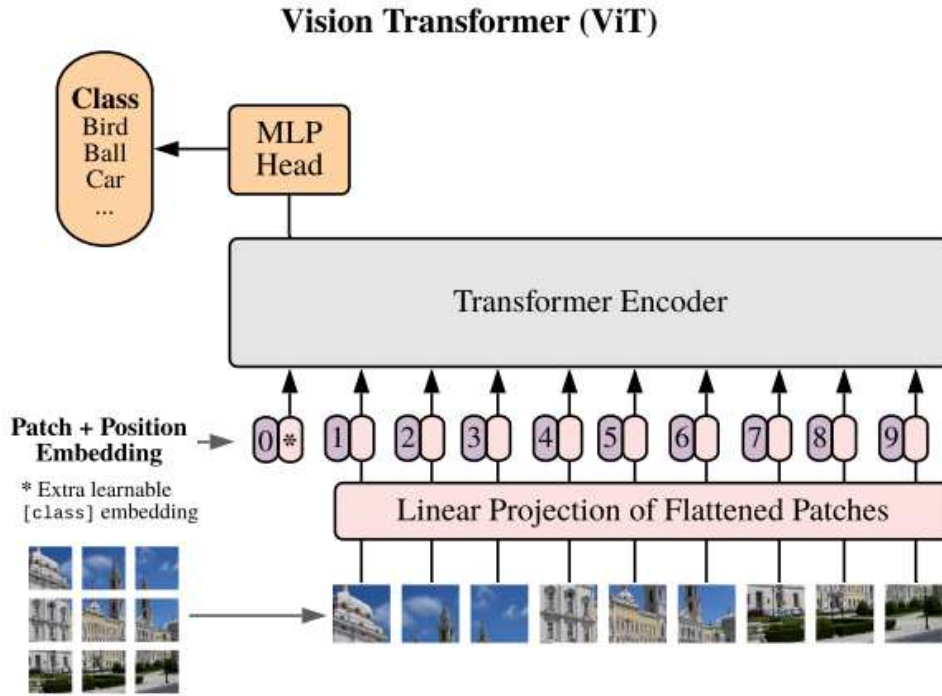


Figure 2.12: Architecture of the ViT taken from Dosovitskiy et al. [2021].

BERT is pre-trained on the BookCorpus [Zhu et al., 2015] (almost 1B words) and a part of the English Wikipedia (2.5B words), corresponding to 16GB of text data. Li et al. [2021], however, used the *RoBERTa* model [Liu et al., 2019] as decoder in TrOCR. In contrast to BERT, RoBERTa was trained on ten times more data. Moreover, RoBERTa dismisses the next-sentence prediction. Overall, Liu et al. [2019] showed that RoBERTa is more robust and BERT was “significantly undertrained.”

TrOCR encoder In TrOCR, the image encoder is a Vision Transformer (ViT) first introduced by Dosovitskiy et al. [2021]. A classical language Transformer learns the attention between all the (sub-)words represented as vectors in a sentence. An image, however, consists of pixels. A 128×128 pixel image already contains 16,384 pixels, and learning the attention between all the pixels would be computationally impractical. This is why Dosovitskiy et al. [2021] propose to patch the image into squares of the same size, each representing an *image token*. They argue that an image is worth 16×16 words, i. e., 256 such image tokens, or “words”. This means that a 128×128 gets split into tiles of 8×8 pixels. After going through a linear projection

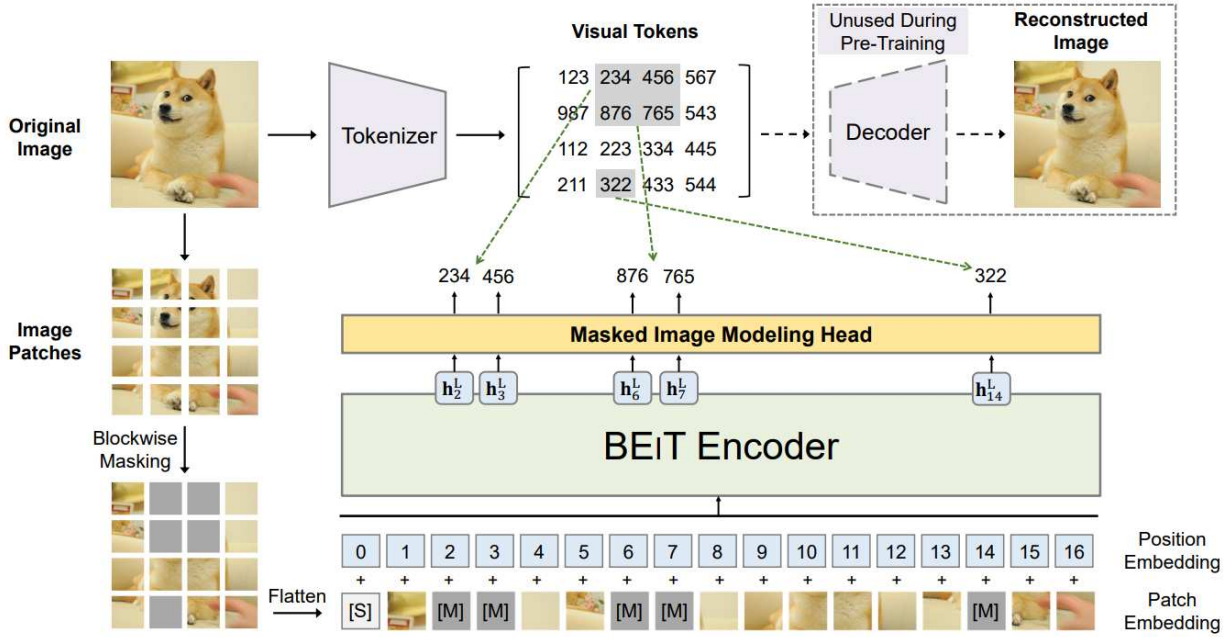


Figure 2.13: The full BEiT architecture taken from Bao et al. [2021]

to flatten the patches, each patch gets a positional embedding like in the language Transformer. Now the image is ready to go through a Transformer encoder, which learns the attention between the image patches. Dosovitskiy et al. [2021] first experimented with image classification, so they appended a dedicated [CLS] token to the image representation. For our toy example, this means that instead of 256 vectors representing the image, we have 257 vectors. To get the class of the image, [Dosovitskiy et al., 2021] used a multi-layer perceptron attention head on the [CLS] token. Figure 2.12 shows the ViT architecture.

The representation of images as a sequence of patches now allows for the large-scale pre-training of image Transformers in a BERT-like fashion. Bao et al. [2021] introduced *BEiT* – BERT Pre-Training for Image Transformers –, which is the closest to BERT-style training (see Figure 2.13). They first trained a *Variational Auto-Encoder* [Kingma and Welling, 2013] to learn so-called *visual tokens* from the image. This image tokeniser then represents each patch as a discrete *visual tokens*. The task of the Transformer is to predict the masked visual tokens (i. e., patches) during the pre-training phase. Bao et al. [2021] pre-trained BEiT on nearly 1.2M images of the ImageNet-1k dataset [Russakovsky et al., 2015].

Having introduced the conceptual idea behind TrOCR, we can now combine the

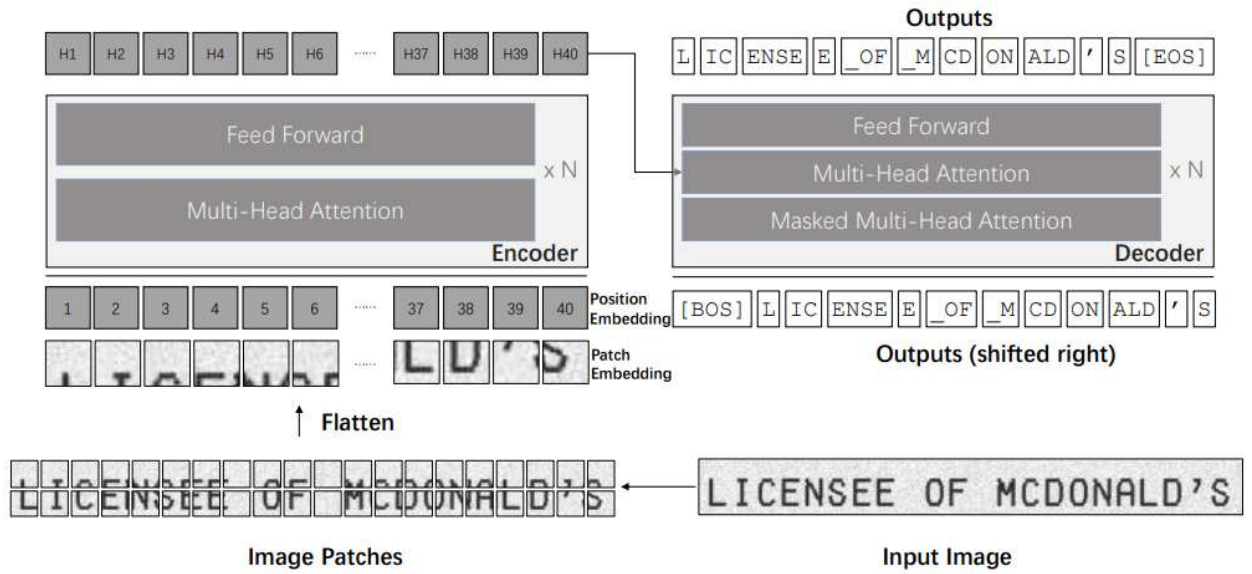


Figure 2.14: The TrOCR model architecture taken from Li et al. [2021]

encoder and decoder and arrive at the final architecture (see Figure 2.14). Li et al. [2021] used 687M of printed and about 18M of synthetically generated handwritten text lines in English to pre-train the TrOCR model.²² During this phase, the model learns to extract relevant features from the images and decode them into English text, therefore training the language model from scratch. The authors initialised the RoBERTa decoder with 6 and 12 layers, referring to them as BASE when paired with the pre-trained 12-layer BEiT instance and LARGE when paired with the 24-layer BEiT model, respectively. Finally, they fine-tuned their pre-trained TrOCR instances on real-world data, like the *IAM dataset* [Marti and Bunke, 2002]. The IAM dataset consists of handwritten English lines from different authors. TrOCR_{BASE} reaches a CER of 3.42% and TrOCR_{LARGE} a CER of 2.89% on this dataset. The score of TrOCR_{LARGE} is only 0.14 percentage points behind the best score of Diaz et al. [2021] who reported that a combination of a self-attention encoder and an RNN with CTC layer as decoder worked best (with a CER of 2.75%).

Note that Li et al. [2021] were not the only researchers to exploit Transformers for ATR. Kang et al. [2022] used a CNN in combination with the Transformer (instead of a ViT as Li et al. [2021] do). They reach a CER of 7.62% on the IAM dataset and manage to decrease it to 4.67% by adding synthetic training data. We will see how

²²This results in a pre-trained model for printed text and one that is additionally pre-trained on handwriting.

to apply Transformers to the ATR problem in Chapters 3 and 4.

2.3 ATR in the Digital Humanities

It is beyond the scope of this thesis to provide a detailed summary of digitisation endeavours all over the world, but we will nevertheless highlight a few projects in which ATR played an important role. Let us recall from Chapter 1 what the aims of the digitisation of historical text documents are:

1. **Conservation** of historical documents that is subject to degradation.
2. Improve **accessibility** of documents in archives.
3. Obtain a **faithful** representation of the digitised object. *Faithful* means in terms of image quality and text quality.

A digitisation workflow should thus include ATR after the documents have been scanned. However, ATR is often applied for printed documents, since good results can be expected (see Chapter 3), but for handwritten collections, we cannot expect results of the same quality with existing HTR models since the handwriting styles of manuscripts differ more than fonts in printed documents.

If ATR is applied, however, it should be in the best interest of institutions and projects which provide digitised documents to place particular emphasis on quality estimation and assurance. For a few years now, there have been mass digitisation initiatives during which libraries and archives alike digitise their data collections (or material from contractors) containing printed or handwritten documents. Some provide online access to the digitised content, like the *HathiTrust Digital Library*,²³ with more than 17 million digitised items, the project *Google Books*,²⁴ with approximately 40 million available titles, or the *Project Gutenberg*,²⁵ with about 60,000 titles.

Besides digitising material independently, some projects collect already digitised data to make it accessible to various user groups. One of the biggest initiatives has been the *europaena*²⁶ project, which has aggregated over 50 million digitised objects,

²³<https://www.hathitrust.org/>

²⁴<https://books.google.ch/>

²⁵<https://www.gutenberg.org/>

²⁶<https://www.europeana.eu/de>

among which we find many historical newspapers. Similar efforts are made by the projects *impresso*,²⁷ *OceanicExchanges*,²⁸ and *NewsEye*,²⁹ albeit at smaller scales.

On an even smaller scale, we find cultural heritage projects like the *Text+Berg*³⁰ corpus [Bubenhofer et al., 2015], curated by the Department of Computational Linguistics at the University of Zurich. It mainly consists of the digitised and partially parallel yearbooks from the Swiss Alpine Club. It contains roughly 53M tokens, of which German and French each contribute about 22M tokens. The rest goes to Italian, Romansh, Swiss German, and English.³¹ The Text+Berg project is special regarding the crowd-sourcing initiative during which yearbooks with especially bad OCR were manually corrected [Clematide et al., 2016]. Moreover, Volk et al. [2011] examined approaches for automatically correcting OCR errors. Thus, the project can provide almost error-free text.

The more recent *bulletin4corpus*³² [Volk et al., 2018] by the same institution, could profit from the Text+Berg project, especially in terms of OCR and error correction. It comprises 700 issues from Credit Suisse’s *Bulletin* magazine³³ and amounts to about 21M tokens. Like the Text+Berg corpus, the Credit Suisse corpus is multilingual and attributes 4.6M tokens to English, 16.1M to German, 14.2M to French, 4.6M to Italian, and 0.9M to Spanish, respectively. The author of this thesis was part of the project team in both the Text+Berg and the *bulletin4corpus* project.

Although the goal of these initiatives is the same, namely to provide digital representations of physical objects and to make them searchable, the approaches to achieve this aim are unlikely to be the same. Digitisation is far from a standardised procedure, and the preconditions vary from project to project. More concretely, the digitisation process depends on the source documents: 200-year-old books need different treatment than 500-year-old letters. Any restorations may influence the manner and speed of digitisation. Moreover, there is a variety of scanners available which produce different qualities of images. For successful ATR, research suggests a minimum resolution of 300 dpi, and scans as regular as possible (i. e., no angles, scans from originals instead of microfilms, measures against text shining through, etc.)

²⁷<https://impresso-project.ch/>

²⁸<https://oceanicexchanges.org/>

²⁹<https://www.newseye.eu/>

³⁰<https://textberg.ch/>

³¹The corpus also contains yearbooks of the British publication *The Alpine Journal*, which is why we also find about 6.3 million English tokens. Moreover, the French *Echo des Alpes* forms a subcorpus in the Text+Berg collection.

³²<https://pub.cl.uzh.ch/projects/b4c>

³³It is the world’s oldest banking magazine.

[Terras, 2022a]. Last but not least, it is likely that every institution uses different ATR software which is tailored to its needs.

All these factors influence the outcome of the ATR. Institutions should make their procedures transparent and conduct a systematic quality evaluation of the text obtained by ATR systems. A frequent method to arrive at an estimate is the evaluation of small ground truths (GT), i. e., manually transcribed texts with corresponding line images, generated from the archived documents [Verheusen, 2008, Tanner et al., 2009, Anderson et al., 2010, Christy et al., 2017, Dannélls et al., 2019]. The caveat of such evaluation procedures is that they do not provide a quality estimation for the entirety of the text recognised in a collection but only for a very small portion of the documents for which a GT has been produced. In the case of Verheusen [2008], who described the mass digitisation of around 40 million pages from the *Koninklijke Bibliotheek*³⁴ in the Netherlands, there were manual checks of 1% of each batch consisting of 50,000 pages. Due to the high efforts involved, they abandoned this procedure. In addition, the costs that the evaluation with a GT generates are especially noteworthy. Verheusen [2008] estimated that the development of the quality estimation tool, which checks everything from the quality of the images, the text extracted by the OCR, to the final XML representation cost €200,000 and the maintenance thereof €80,000 per year.

Yet another approach to provide estimates is to use the ATR systems' confidence score of characters or words. Since ATR models are probabilistic in nature (as we have seen above), we can use the probabilities that the models produce for each character in a sequence to estimate a model's confidence in a word or character [Mor and Wolf, 2018]. Such confidence scores then inform users about the estimated quality of the text, and can even help in industrial applications to verify or reject documents automatically in a way to reduce human intervention [Sarkar et al., 2001].

In the end, we can use the scores of ATR models on small GTs or their confidence scores to answer one of the most important questions: which ATR system to choose? Digitisation processes in heterogeneous settings are especially demanding and libraries and archives typically contain sources from a variety of backgrounds in a multitude of different styles. At first glance, it is therefore not obvious which ATR system produces the best output. Holley [2009] mentioned that the use of multiple OCR systems can help to improve the quality. Going one step further, combining the output from different ATR systems would guarantee the best possible result (as is done in the Calamari system Wick et al. [2020]).

³⁴<https://www.kb.nl/>

The difficulty of choosing the best possible ATR systems comes from the growing availability of such systems. With ATR techniques growing more and more mature, users in need of ATR software have several options to choose from. They can, for example for OCR, choose a fully ready and fee-based system like the *ABBYY FineReader*.³⁵ If they wish to train models for either OCR or HTR themselves, they can choose the also paid-for *Transkribus platform*,³⁶ which emerged from the *READ*³⁷ project. It allows the creation of GT data and the training of HTR models [Mühlberger et al., 2019]. Next to chargeable solutions, we find open-source projects for both OCR and/or HTR with *Tesseract*,³⁸ *ocropus*³⁹ and its fork *kraken*,⁴⁰ and *ocr4all*.^{41,42} Additionally, there is *OCR-D*⁴³, which offers customisable OCR and HTR workflows including wrappers of some of the aforementioned software and a few additional tools.

In a digitisation project, it is thus likely that experiments with several different models are performed. However, if the number of such systems continually grows, and especially when we obtain further systems by training models ourselves we quickly generate a model zoo. The challenge is then to pick the model which produces the best results. The basis on which we make this decision is the recognised text alone since no GT is available. In Chapter 5, we take a closer look at this problem.

Hence, not only the ATR quality but also transparency is critical when it comes to displaying the results to users. This is especially true for digital editions, where there is scepticism about the incorporation of HTR results mainly due to a lack of trust [Nockels et al., 2022] towards *the machine* and the result it outputs.

³⁵<https://www.abbyy.com/finereader-server/>

³⁶<https://readcoop.eu/de/transkribus/>

³⁷<https://readcoop.eu/>

³⁸<https://github.com/tesseract-ocr/tesseract>

³⁹<https://github.com/ocropus/ocropy>

⁴⁰<http://kraken.re/>

⁴¹<http://www.ocr4all.org/de/home.php>

⁴²We use some of these frameworks in the following chapters, where we introduce them in more detail.

⁴³<https://ocr-d.de/en/>

Chapter 3

OCR for Blackletter in Historical Newspapers

As mentioned in Section 1.2.1, OCR models still experience difficulties when applied to historical documents, although they have improved in recent years. In one of the earlier approaches, Breuel et al. [2013] showed that BLSTMs achieved excellent results for Antiqua and blackletter fonts with character error rates (CERs)¹ of 0.6% on the Antiqua test set and 0.15% on one of the blackletter test sets, respectively. Considering these results, we could think of OCR not as a solved but as a very well-handled problem. However, Breuel et al. [2013] raised concerns about whether their techniques will generalise to new data. In this chapter, we show that these concerns are valid and that not all OCR models perform equally well. To illustrate this, we focus on a ground truth dataset built from a **historical newspaper** and introduced in Section 3.3. This dataset serves as an ideal case in point since there already exists an OCRed version of the whole dataset. By comparing the performances of other OCR models to these already existing versions, we will be able to answer the following research questions:

RQ 3.1 How successful are ATR methods for recognising blackletter text in historical newspapers?

¹See Section 1.2 again for a conceptual definition of the CER, or Section 3.5.2 for a formal definition.

RQ 3.2 Can we use ATR methods developed for HTR to recognise the text from printed documents?

RQ 3.3 How much ground truth (GT) data, i. e., manually annotated material, do you need to train reliable models?

RQ 3.4 How well do ATR models perform on documents from other newspapers (termed out-of-domain data in machine learning)?

3.1 Challenges in Historical Newspapers

Although briefly mentioned in Chapter 1, we hint at the challenges historical newspapers pose. The quality of the OCR results depends on several factors (with examples in Figure 3.1):

- (a) Low distinctiveness of characters, e. g., the “long s” *ſ* and “f” in older German texts.
- (b) Change over time regarding vocabulary and spelling, e. g., *Commando* (EN *command*) versus *Kommando*, both of which have the same meaning, but come in different forms, thereby enlarging the diversity of possible character sequences.
- (c) The use of small font sizes (and changes in the fonts themselves).
- (d) Mixing of Antiqua and blackletter on the same page.
- (e) Poor paper quality (e. g., smears, smudges, sometimes due to conservation issues).
- (f) Text from the backside shining through.
- (g) Distortions of the image due to digitisation from bound newspaper collections or imprecise digitisation workflows.

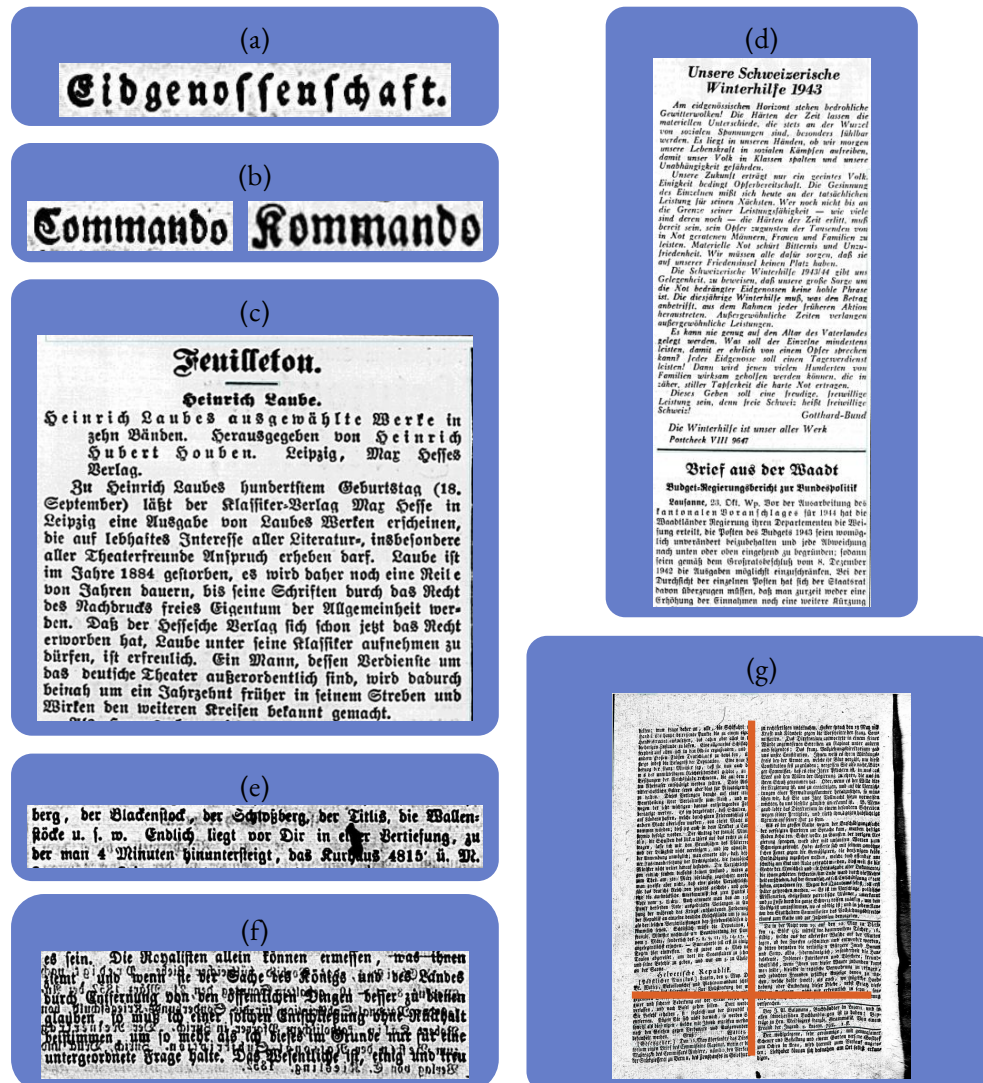


Figure 3.1: Different factors influencing OCR quality for newspapers: (a) distinctiveness of characters, e. g., an OCR model outputs *Eidgenossenschaft*, *Eidgenossen-fchast*, etc., for the German word *Eidgenossenschaft* (EN *confederation*) (b) spelling variations of a word (EN *command*), (c) small (and different) font sizes, different fonts, and small margins between lines, (d) black letter and Antiqua on the same page, (e) smears and smudges, (f) text shining through from the backside, (g) distorted image (orange line as references of the x- and y-axes).

Historical documents digitised a few years ago likely suffer from low-quality OCR. The possibility of OCR models to output a confidence score might give an idea about the quality of the extracted text. Still, such scores are not always *available*, and even if they are, they are often *unreliable*. Additionally, libraries and archives often downscale scanned images when distributing them over the web. This means that researchers in the Digital Humanities must deal with low- to medium-resolution images. If a researcher wants to re-run OCR to produce a better textual basis, she must work with this inferior material.

Having an idea about the quality of the extracted text is important for indexing and applying text mining techniques to digitised material. Working with OCRed resources means that researchers might base their discoveries on results which contain a certain bias introduced by the OCR [Traub et al., 2015]. Often, such errors only become noticeable during the actual work with the digitised corpora. The study by Chiron et al. [2017] took up this point and predicted the risk of missing relevant documents due to OCR errors given a user query in the OCRed [Gallica collection](#).² By using the search logs of four months of user queries, they found that 7% of the most common search terms potentially miss relevant documents due to OCR errors. As such, insufficient OCR impedes advances in Digital Scholarship [Smith and Cordell, 2018] due to erroneous text.

3.2 Related Work

The problems accompanying text recognition from images of historical documents are manifold and complex, but recent progress in neural OCR techniques has led to significant improvements. Springmann and Lüdeling [2017] analysed the performance of the [OCRopy](#)³ system on a diachronic book corpus ranging from 1487 to 1914 (all in blackletter fonts). They stated that OCRopy was the only OCR system that achieved character accuracies (CAs) consistently over 94% and that their models generalised well enough so that they performed equally well on a variety of books. They further claimed that based on a recommendation by Springmann et al. [2016], a training set of 100 to 200 lines suffices to produce comparable results to models which have seen a considerably larger training set.

The [READ](#)⁴ project focuses on the recognition of handwriting. Its core is the

²<https://gallica.bnf.fr>

³<https://github.com/tmbdev/ocropy>

⁴<https://readcoop.eu/>

*Transkribus*⁵ framework. Transkribus is a tool that helps researchers to transcribe manuscripts in the first place. It performs layout analysis which dissects a page into text regions, lines, baselines, and words. Moreover, it offered⁶ the possibility to run the *ABBYY FineReader Server 11*⁷ OCR software on images uploaded to Transkribus. With the transcriptions made within Transkribus, it is possible to train *Handwritten Text Recognition* (HTR) models [Weidemann et al., 2018].⁸ These models are elaborate neural network architectures that recognise handwriting with only a limited amount of training data. However, we found that HTR models can also be applied successfully to recognise printed documents, with CERs of 1.1% or better [Ströbel and Clematide, 2019].

Regarding the amount of training data, Martínek et al. [2019] showed that for German-language blackletter newspapers, a ground truth of 10 pages is sufficient to achieve a CER of 1.4%. They used a CNN×RNN architecture (in a similar, but much simpler fashion than Weidemann et al. [2018]) and different training strategies like, e. g., binarisation of the images, padding, and the use of synthetically generated training data for blackletter recognition.

Wick et al. [2018] investigated both the performance of different systems (OCRopy and *Calamari*⁹) on medieval printed books in German and Latin, as well as the amount of training data needed. They found that with an increasing number of lines in the training set (starting from only 60) the CER for both systems drastically improved. E. g., for data from 1488, their Calamari model managed to lower the CER from 4.9% when using 60 lines to train to 0.43% when using 3,000 lines. However, they do not indicate an optimal number of lines for a training set.

These recent studies show that researchers in Digital Humanities use several different OCR models to extract text from historical documents. Many models are freely available. However, it demands a lot of effort from the researchers to understand the different processing steps [Klein et al., 2016], since there is no standardised workflow. *OCR-D*¹⁰ [Neudecker et al., 2019] attempts to facilitate the training of OCR models for collections via creating standards for training and testing a variety of OCR software instead of limiting users to only one system. Comparisons between different systems are thus simplified and more transparent.

⁵<https://readcoop.eu/transkribus>

⁶This feature is not available anymore.

⁷<https://www.abbyy.com/de-de/finereader-server/>

⁸For more details, see Section 3.4 below.

⁹<https://github.com/Calamari-OCR/calamari>

¹⁰<https://ocr-d.de/>

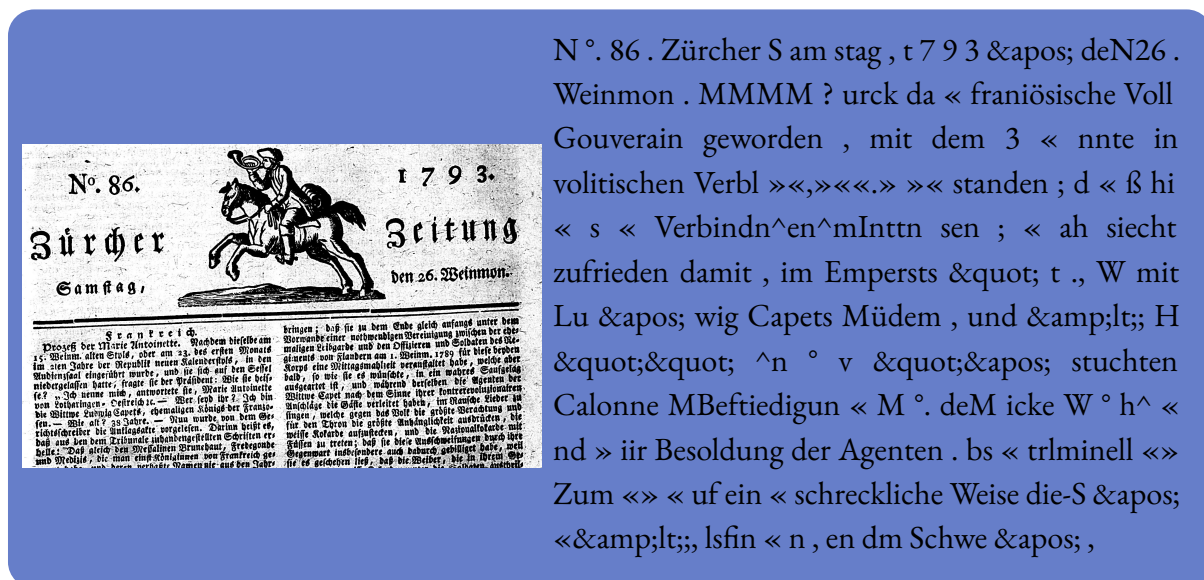


Figure 3.2: Example OCR from 2005 as produced by the Fraunhofer Institute of an NZZ page from 1793. The resulting text is nearly illegible.

3.3 Data for OCR

Section 1.2.2 briefly mentions the *Neue Zürcher Zeitung* (NZZ) and its availability on the e-newspaperarchives platform. In 2017, however, high-quality OCR output was not available. The only possibility to search through the whole NZZ was the NZZ's own [archive](#)¹¹. It offered – and, as a matter of fact, still does – limited search possibilities (keyword search only) and mostly highly illegible texts (as an example in Figure 3.2 shows).

The origin of this low-quality OCR is a digitisation effort by the Fraunhofer Institute, which dates back to 2005. The NZZ, first published in 1780, wanted to digitise its entire archive for its 225th anniversary. Jacob [2005] described that the source for the digitisation process was the NZZ's microfilm collection. Not digitising from paper sources, as usually is recommended [Holley, 2009], is cheaper [Verheusen, 2008] but results in low-quality images and impacts subsequent OCR. The Fraunhofer Institute tried, first and foremost, to correct the distortion of images and to enhance unfocused images. These issues were not resolved completely and thus severely impacted the OCR quality (see Figure 3.2 again). Due to monetary restrictions, the

¹¹Note that since 2017, the archive's website has received a facelift.

large amount of data, and the limited time frame, no manual corrections were possible. The Fraunhofer Institute also noticed that the quality of the microfilms differed. This is because the NZZ started to microfilm their archives in the 1950s, and there have been technological advances since then. Moreover, there was no designated quality control process to evaluate the output of the OCR system.

One of the goals of the *impresso*¹² project [Ehrmann et al., 2020] was to collect available newspapers from Switzerland and Luxembourg in German, French, and Luxembourgish and to make their contents accessible via a search interface. A major role in supporting queries played the automatic inference of topics present in an article on a page using topic modelling [Blei et al., 2003] and word embeddings to provide alternative queries (also cross-lingual) [Mikolov et al., 2013, Bojanowski et al., 2017]. Correct text representations are thus essential because search indexing and text mining methods provide incomplete or erroneous results on low-quality OCRed text (as mentioned in Section 2.3).

We thus wanted to profit from recent advances in machine learning that resulted in increased performance, even on difficult¹³ training material. The elaborate deep learning models (as introduced in Section 2.2.3 and onward) created new standards in OCR. However, like any supervised machine learning method, deep learning models also need training material. Although creating a GT for OCR can be costly and time-consuming, we felt the need for a dedicated GT to train models tailored to the NZZ first to improve the OCR results and second to be able to evaluate existing and future OCR versions. Moreover, OCR models for German blackletter or datasets were not readily available then.

3.3.1 Creating a Ground Truth for Blackletter-typed Newspapers

The NZZ was published in blackletter from its beginnings in 1780 until 1947.¹⁴ We randomly selected one title page per year from this period for manual transcription. By restricting ourselves to title pages, we ensured the page’s main content was textual. This heuristic guarantees excluding pages loaded with advertisements, stock

¹²*impresso – Media Monitoring of the Past* is a project funded by the Swiss National Science Foundation under grant number CRSII5 173719.

¹³With *difficult* we mean applying OCR on *low-resolution* images of newspaper pages printed in *blackletter*.

¹⁴Still, we find newspaper pages – and especially pages containing advertisements – in Antiqua before 1947, and some blackletter text has been published even after 1947.

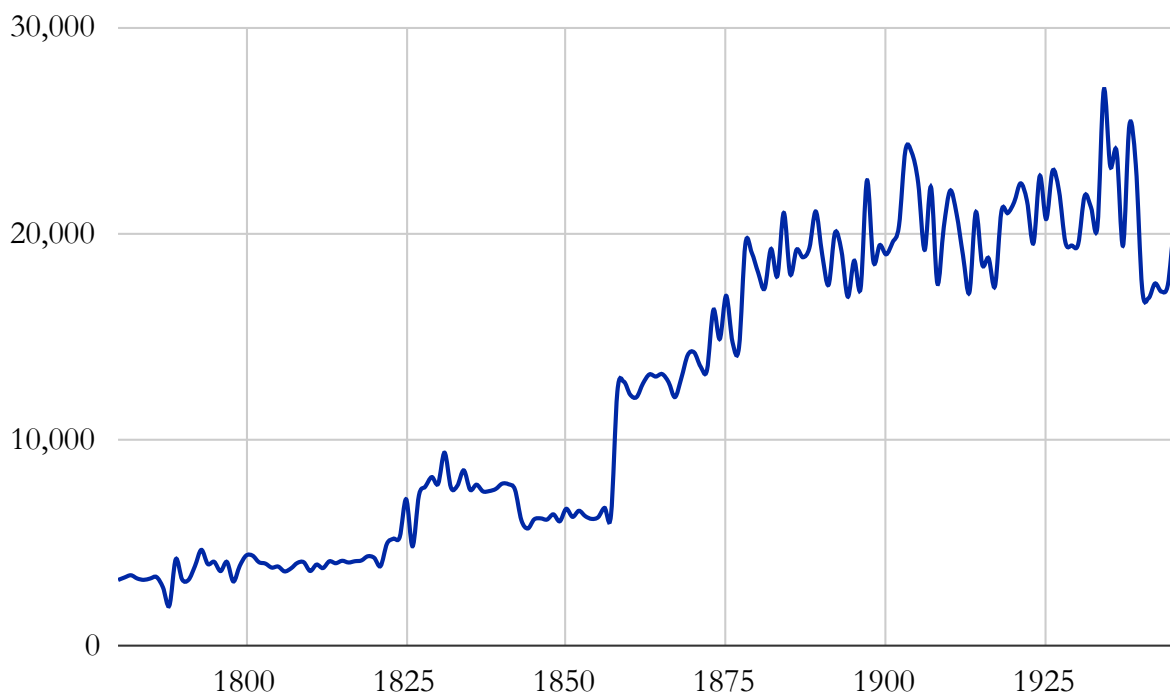


Figure 3.3: Character frequency on the front pages in the GT per year from 1780 until 1947. The steep increase at the end of the 1850s is due to the switch from two to three columns.

reports, etc. However, it should be noted that the amount of text on the title page from 1780 (3,180 characters) is much smaller than on a title page from 1939 (23,316 characters), as Figure 3.3 shows. The more recent data in the ground truth is thus overrepresented, a fact we will take up again in Section 3.6.1. The main reason for the increase of characters per page is the evolution of the NZZ’s layout, as shown in Figure 3.4. Not only did the number of columns change, but also the size of the paper sheet the newspaper was printed on.

We extracted the images from the PDFs produced by the Fraunhofer Insitute in 2005 with *PDFlib TET*¹⁵ and loaded them into the *Transkribus*¹⁶ application. We applied the Transkribus-internal OCR software *ABBYY FineReader Server 11* to the uploaded front pages. We used the output text as a basis from which we started the manual correction. We also corrected regions and added baselines required to train an HTR model (see Figure 3.5). The resulting GT of 167 pages contains 304,286

¹⁵<https://www.pdflib.com/de/produkte/tet/>

¹⁶In Section 3.2, we provide further details about Transkribus and ABBYY.



Figure 3.4: Examples of three different front pages from the NZZ (1780, 1880, 1898). Notice the layout change over time from two to four columns.



Figure 3.5: Example of the Transkribus tool. Green boxes correspond to regions, and the blue lines to baselines. The lines need to be adjusted if they are too far from the imaginary line on which the characters sit.

words in German and 43,151 lines. Depending on the amount of text on a page, the manual correction of a page took between 1.5 and 3 hours. We published the GT on [Zenodo](https://zenodo.org/record/3333627#.ZEWGEdJBxkh)¹⁷ and [GitHub](https://github.com/impresso/NZZ-black-letter-ground-truth)¹⁸.

With the layout recognition information and the controlled text, we could crop the lines and extract the corresponding text to create image-text pairs (see Figure 3.6), which are needed as input for model training and testing.

¹⁷<https://zenodo.org/record/3333627#.ZEWGEdJBxkh>

¹⁸<https://github.com/impresso/NZZ-black-letter-ground-truth>



Figure 3.6: Line image-text pair used as training and testing data.

3.4 OCR Models

We evaluate seven different OCR systems in our experiments:

1. *ABBYY FineReader XIX*¹⁹ (FRXIX) results from the Fraunhofer Institute provided in 2005,
2. *ABBYY FineReader Server 11* (FRS11) results, available from within Transkribus,
3. a Transkribus HTR+ model²⁰ [Weidemann et al., 2018] (see below),
4. a Transkribus PyLaia model,
5. *kraken*,²¹
6. *Tesseract*,²² and
7. TrOCR (see Section 2.2.4).

Figure 3.7 shows an example output from five OCR systems. ABBYY’s OCR software is commercial, and thus, it is difficult to provide more details at this point. We extend the category of non-commercial OCR software with HTR+, PyLaia, kraken, Tesseract, and TrOCR, each of which we present here briefly.

HTR+ HTR+ models were part of the Transkribus framework until 2022. Users could train such models given a GT. The model proposed by Weidemann et al.

¹⁹<https://www.frakturschrift.com/de/products/finereaderxix>

²⁰This model architecture is not available anymore.

²¹<http://kraken.re/>

²²<https://github.com/tesseract-ocr/>

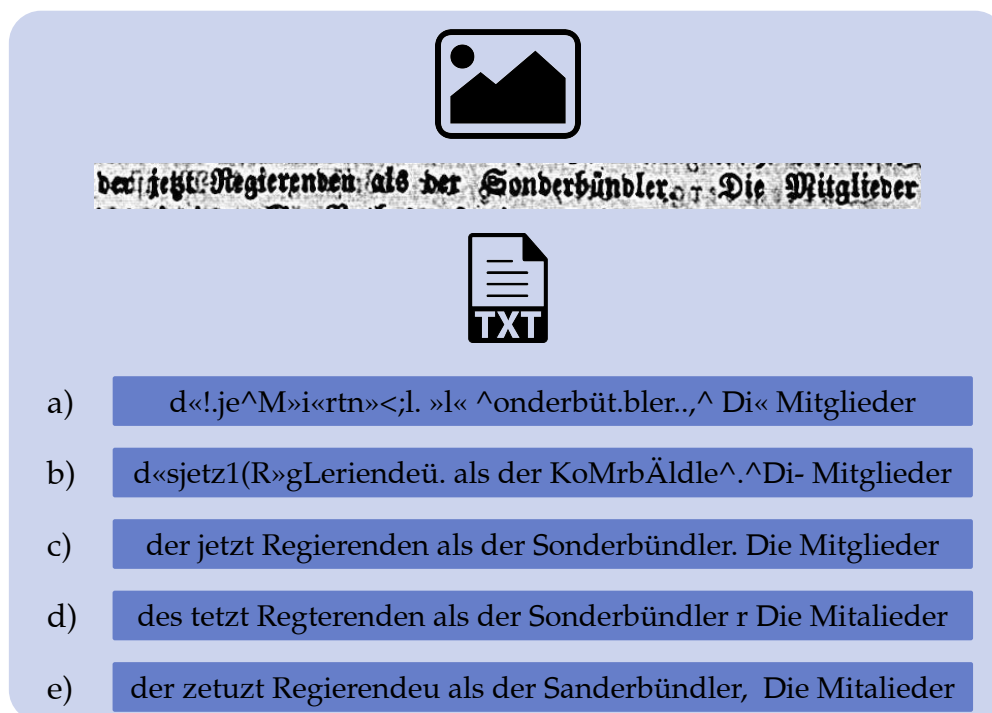


Figure 3.7: Example outputs from five OCR models for a line from the NZZ from 1850. a) FRXIX, b) FRS11, c) HTR+, d) kraken, e) Tesseract.

[2018] is a deep neural network which combines CNNs and BLSTM layers (see Section 2.2.3). Figure 3.8 summarises the HTR+ model. We notice that it is much simpler than the model proposed by Granell et al. [2018]. We used Transkribus version 1.8.0.

PyLaia The [PyLaia](https://github.com/jpuigcerver/PyLaia)²³ software for HTR superseded the HTR+ architecture in Transkribus. It is more flexible because it lets users define the exact architectures themselves. In contrast to HTR+, PyLaia is open-source software.

kraken kraken is a forked version from OCRopy and supports the training of OCR models with neural networks. In contrast to an HTR+ model, kraken allows the user to determine the architecture of the neural network with the help of the

²³<https://github.com/jpuigcerver/PyLaia>

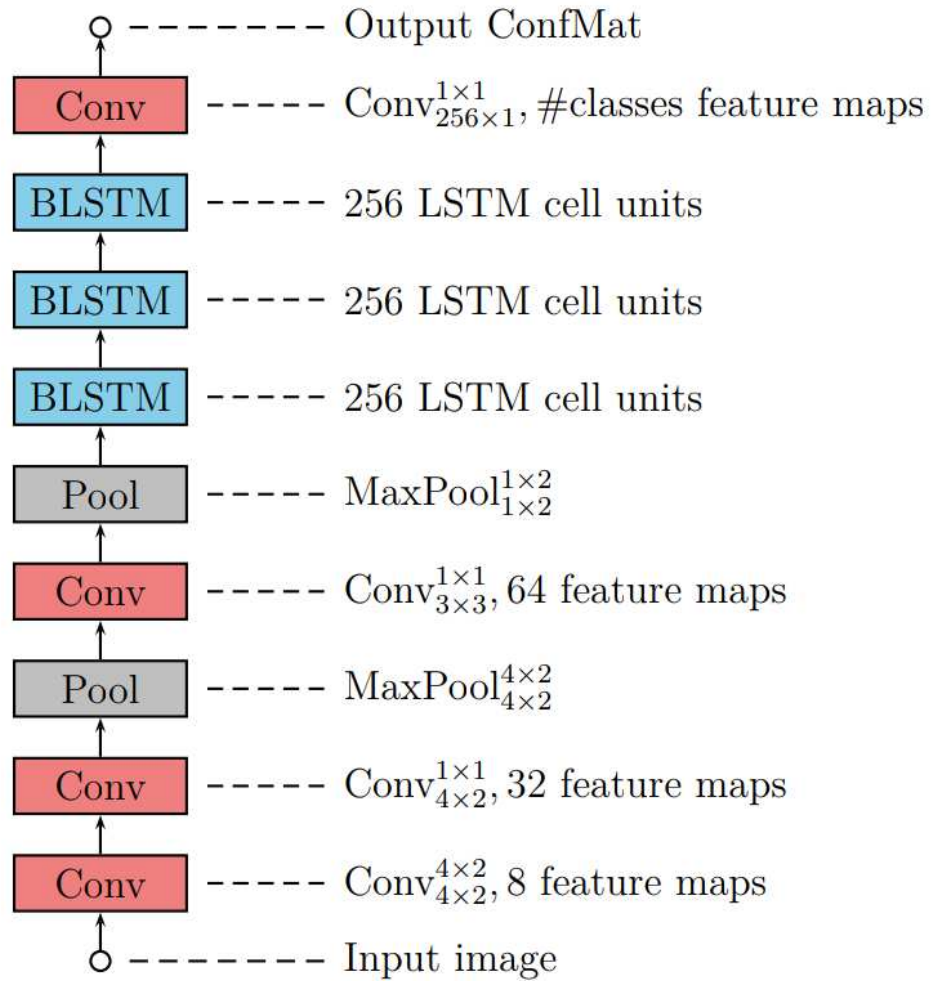


Figure 3.8: The HTR+ model architecture as presented in Weidemann et al. [2018].

Variable-size Graph Specification Language²⁴ (VGSL).²⁵ This means a user has more control over the structure, allowing for tuning a model until it delivers good enough results. kraken runs on GPUs, which speeds up training time. We used version 2.0.5.

Tesseract Tesseract is Google’s solution of a neural OCR system. Like kraken, it allows the user to specify the structure of the neural network with VGSL. In contrast

²⁴<https://github.com/mlmbai/tensorflow-models/blob/master/street/g3doc/vgslspecs.md>

²⁵We provide an example of VGSL in Section 3.6.1.

to kraken, Tesseract is not GPU-enabled. We used version 4.00.

TrOCR We describe the TrOCR architecture in detail in Section 2.2.4.

3.5 Evaluation

3.5.1 Bag-of-Words Evaluation

We use the bag-of-words F1-measure metric of the PRImA TextEval 1.5 software package [Clausner et al., 2016] to evaluate the OCR models. By applying a bag-of-words approach, possible differences in layout recognition or word order of the different systems cannot distort the results.

Considering the example in Figure 3.9, we first determine the bag-of-words for each page in the GT and for the output of the OCR model, which is kraken. We build the bag-of-unique-words from the respective bags-of-words. This bag determines the number of words the OCR system identified correctly, i. e., true positives (TP). We can now compute *Recall* R , which is the percentage of words the model should have identified as true positives divided by the sum of true positives and false negatives (FN):

$$R = \frac{TP}{TP + FN} \rightarrow \frac{4}{8} = 0.5. \quad (3.1)$$

Precision P , on the other hand, gives us the percentage of correctly identified words, which is the difference between true positives and the sum of true positives and false positives (FP):

$$P = \frac{TP}{TP + FP} \rightarrow \frac{4}{9} = 0.44. \quad (3.2)$$

The *F1-measure* is the harmonic mean between Precision and Recall, as given in Equation 3.3:

$$F1 = \frac{2 \times R \times P}{R + P} \rightarrow \frac{2 \times 0.5 \times 0.44}{0.5 + 0.44} = 0.46. \quad (3.3)$$

The numbers reported in this section correspond to the average F1-score of all pages in the test set.

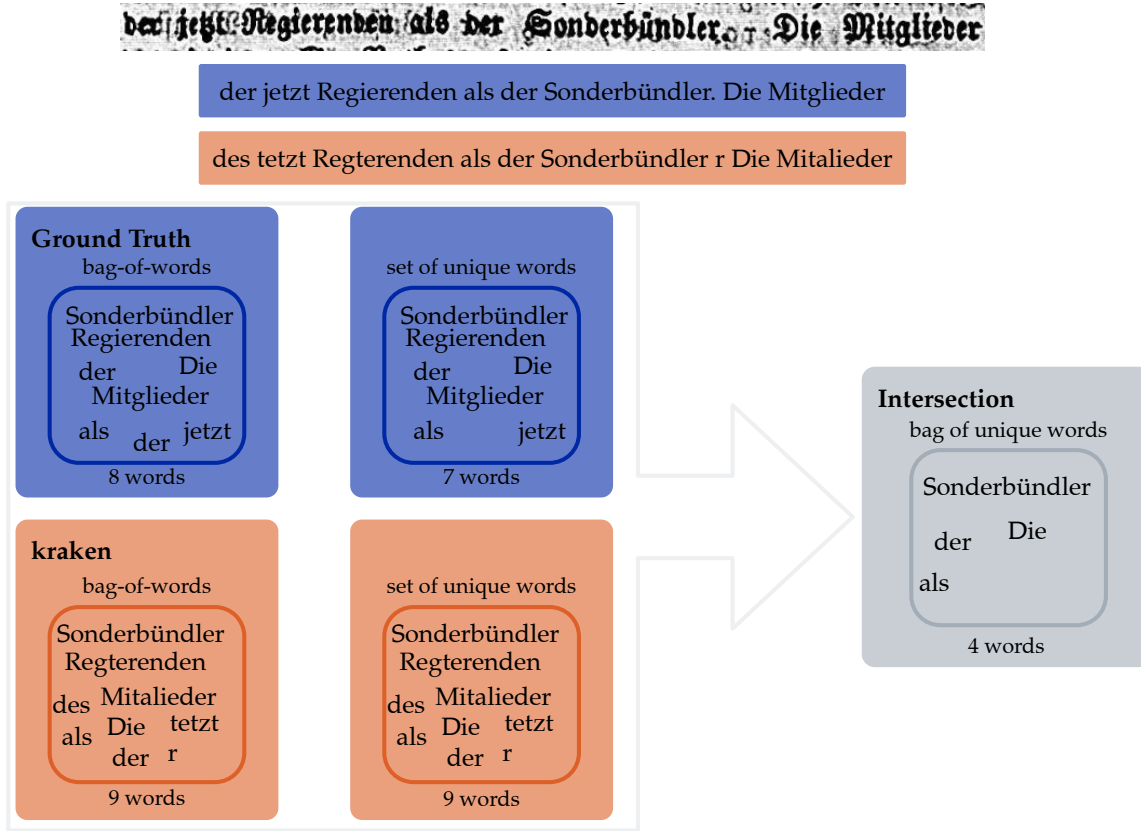


Figure 3.9: Example for bag-of-words evaluation with one line from which the kraken model extracted the text.

3.5.2 Character Error Rate and Character Accuracy

Character accuracy (CA) and word accuracy (WA) are traditional measures for evaluating ATR results with GT material. Rice [1996] defined CA and WA as follows:

$$CA/WA = \frac{n - E}{n} \quad (3.4)$$

In Equation 3.4, n refers to the total number of characters, or words, in the GT, while E describes the number of errors when we compare the GT against the ATR result. Following Handley and Hickey [1991], Rice [1996] determined E to be the error value described in Equation 3.5, which compares two strings $A = a_1, a_2, \dots, a_i$ and $B = b_1, b_2, \dots, b_j$ and depends on a cost function γ .

$$E = d_{A,B,\gamma} \quad (3.5)$$

The cost function γ is expressed as $\gamma = (\gamma_I, \gamma_D, \gamma_S)$, where each position in the tuple specifies the cost for insertions (I), deletions (D), or substitutions (S), respectively. If $\gamma = (1, 1, 1)$, i. e., the unit cost function, we generally speak of the *Levenshtein distance* [Levenshtein, 1966]. As an example, we take the correct string $A = \text{ingenium}$ (EN *temper, disposition*) and the erroneous string $B = \text{ingeniiim}$, where the u has been wrongly recognised as double i . We construct a directed edit graph to determine the cost we suffer by applying the cost function γ . Figure 3.10 shows the edit path as dark grey boxes.

With the described error function and the Levenshtein cost of 2 for the example word B consisting of 9 characters, we arrive at a CA of $\frac{9-2}{9} = 0.78$. The CER is just $1 - CA$, or 0.22 in this case.

To date, these measures, together with their counterparts CER and word error rate (WER), are still the most applied in evaluating ATR models. CA and WA are straightforward to compute given an ATR result and the corresponding GT text. Or so you might think. Neudecker et al. [2021] compared different implementations of OCR evaluation metrics (more concretely, CERs and WERs) and tools against each other and found that the tools provide different results. This is mainly due to two factors: firstly, ATR systems with integrated layout analysis sometimes provide different reading orders of paragraphs or lines, which distorts the quality measures, and secondly, the different tools apply different normalisation strategies, which again influences the final scores. We examine this problem in Chapter 5.

3.6 Experiments

We conduct the following experiments to find answers to our research questions:

1. We compare different existing ATR architectures by training them on the NZZ dataset and comparing the performances according to the CER and bag-of-words F1-measure.
2. We conduct a size ablation experiment to determine how many pages of GT we need to train satisfactory models.
3. We apply the best-performing model to check for transferability to other data.

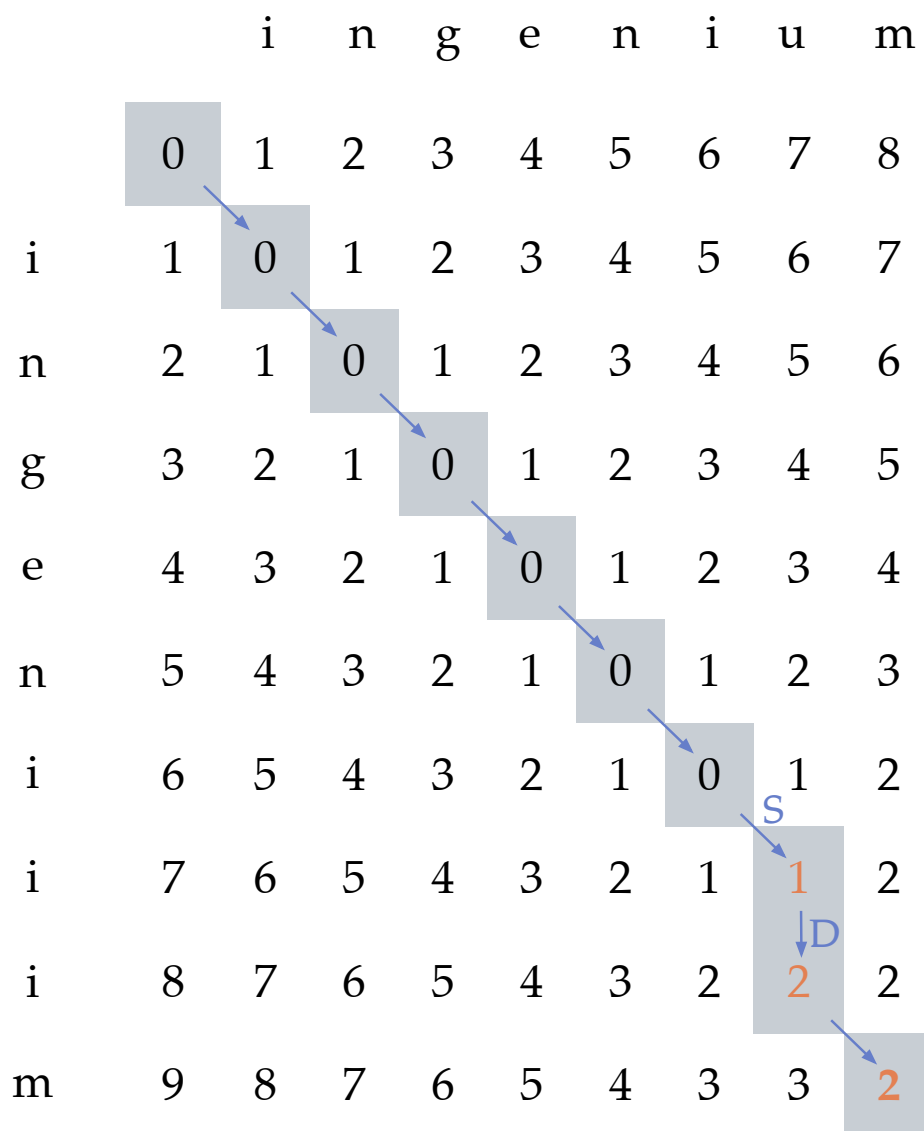


Figure 3.10: The Levenshtein distance for turning *ingeniim* to *ingenium*. We follow the grey path and incur 0 cost if the characters match. Substituting the third *i* in *ingeniim* with an *u* and deleting the fourth *i* each cost 1. The resulting Levenshtein distance, in this case 2, is the sum of all the costs suffered along the path.

3.6.1 Experiment 1 – Comparison of ATR Architectures

This experiment aims to get an overview of the performances of different ATR models. To this end, we train the following models on the NZZ dataset (see Section 3.3): HTR+, PyLaia, Tesseract, kraken, and TrOCR.

To compare the results, we train each model on the same data and validation split and set aside a separate test set. We use 135 pages for training, 16 for validation, and 16 for testing, corresponding to an 80/10/10 dataset split. The test set contains a page from roughly every decade to get an apt representation of the diachronicity of the GT, and we use this test set throughout our experiments to evaluate the performances of the different systems. Here we present the setup for this experiment:

ABBYY FRXIX and FRS11 To evaluate ABBYY’s FRXIX solution, we extracted the texts from the PDFs produced by the Fraunhofer Institute for the test pages. Similarly, we exported the texts from test pages generated by ABBYY’s FRS11 version in Transkribus.

HTR+ We use the standard settings provided by Transkribus and train 50 epochs without early stopping. Unfortunately, the interface provides no insights regarding learning rate, batch size, etc. We learn from Weidemann et al. [2018] that they used the Adam optimiser [Kingma and Ba, 2015].

PyLaia The PyLaia module in Transkribus allows users to adjust several settings, among which we also find the structure of the model. We decided to apply the standard configuration, similar to the network architecture in Figure 3.8. On the side of the BLSTM layers, PyLaia uses the same number of layers (i. e., 3) with the same number of units (i. e., 256) and a **Dropout**²⁶ rate [Srivastava et al., 2014] of 0.5. On the side of the CNN, PyLaia works with four instead of three layers and generates 12, 24, 48, and 48 feature maps, respectively. They use the CTC loss (see Section 2.2.3). Moreover, the settings allow for defining the pre-processing steps of the images, including desloping, deslanting, stretching, and a general enhancement which includes the binarisation of the image. These steps are always carried out in

²⁶Dropout is a means to prevent the network from overfitting by randomly “deactivating” nodes in the network.

the HTR+ model, as is **data augmentation**²⁷ by dilation and distortion [Weidemann et al., 2018], for which PyLaia allows control. We also trained the model for 50 epochs without early stopping.

Tesseract and kraken For Tesseract and kraken we used the standard settings. The network specifications in VGSL were:²⁸

```
Tesseract = [1,36,0,1 Ct3,3,16 Mp3,3 Lfys48 Lfx96 Lrx96 Lfx256 01c]
```

```
kraken = [1,48,0,1 Cr3,3,32 Do0.1,2 Mp2,2 Cr3,3,64 Do0.1,2 Mp2,2 S1(1x12)1,3 Lbx100 Do  
01c]
```

In their essence, both systems work similarly. Differences lie in the normalisation of the input (48 pixels for kraken and 36 pixels for Tesseract), the addition of dropout Do in kraken, and the use of BLSTM layers Lb in kraken. Tesseract, on the other hand, uses two forward LSTMs Lf and two reverse LSTMs Lr as default settings. Mp stands for max pooling and S reshapes the output of the previous layer. Regarding hyper-parameter settings Tesseract uses a default learning rate of 0.002, while kraken uses 0.001. The default optimiser for both systems is Adam.

Additionally, we rebuilt the HTR+ model from Figure 3.8 in VGSL in kraken (we call the rebuild kraken+) in the following way:

```
kraken+ = [256,64,0,1 Cr4,2,8,4,2 Cr4,2,32,1,1 Mp4,2,4,2 Cr3,3,64,1,1 Mp1,2,1,2  
S1(1x0)1,3 Lbx256 Do0.5 Lbx256 Do0.5 Lbx256 Do0.5 Cr255,1,85,1,1]
```

We used kraken because of the much easier handling as compared to Tesseract. We applied dropout after every BLSTM layer to prevent overfitting (similar to the default PyLaia architecture). The number of filters in the last convolutional layer of the HTR+ depends on the number of different characters. Since we cannot control for this number in VGSL, and since we could not find this information in Weidemann et al. [2018], we examined the frequency distribution of the characters in the training set. We determined that limiting the number of filters to 85 (out of 144 different characters, among which we also find symbols) is a reasonable threshold.

²⁷Data augmentation is a common means in machine learning to increase the number of training samples and to make the model more robust. It can also prevent overfitting. We introduce a specific method for data augmentation in more detail in Section 4.3.2.

²⁸C=convolutional layer, Mp=max pooling, L=LSTM cell, Do=dropout, S=re-scaling, the first set of four comma-separated numbers defines the batch size, height normalisation, length normalisation and number of channels

We used a batch size of 256 and the same normalisation of the input height. We reduced the learning rate to 0.0001. Moreover, we used a cyclical learning rate [Smith, 2017], which should result in faster convergence of the model and lead to better results than a fixed learning rate or a learning rate schedule without optimising for the best learning rate. With the cyclical learning rate, the kraken model did not show any significant improvements on the validation set after 18 epochs.

TrOCR We initialised the pre-trained LARGE version of the printed version and trained for 0, 1, 5, 10 and 15 epochs. Since we are in a fine-tuning setting here, it is not necessary to train as many epochs as for HTR+, PyLaia, Tesseract and kraken.

Results and Discussion of Experiment 1

model	F1	SD
ABBYY FineReader XIX	0.678	0.111
ABBYY Fine Reader Server 11	0.811	0.073
HTR+	0.970	0.013
kraken	0.865	0.064
Tesseract	0.861	0.055
kraken+	0.954	0.024
PyLaia	0.969	0.018
TrOCR	0.990	0.015

Table 3.1: Averaged F1-scores and standard deviation (SD) of the performance of eight OCR systems on the NZZ test set based on the bag-of-words evaluation.

Table 3.1 summarises the performance of the eight OCR models as the average F1-scores (as introduced in Section 3.5.1) over the 16 pages in the test set. Unsurprisingly, the result of the 2005 ABBYY FineReader XIX (FRXIX) model is far worse than those of more recent systems. E. g., for the test set page from 1820, the F1-measure of the FRXIX output is at only 0.38 and thus totally illegible. A closer



Figure 3.11: Example page from the NZZ test set posing problems for FRXIX.

look at the page (see Figure 3.11) shows that it poses a problem due to the bad image quality and large portions of the text from the backside shining through.

With the ABBYY FineReader Server 11, i.e., a pure re-OCR'ing using ABBYY's most recent version when we performed this experiment in 2017, achieves an improvement of about 13 percentage points (= F1-score of 0.51). The result of HTR+,

however, leads to an additional improvement of 16 percentage points (= F1-score of 0.67).

That the overall F1-scores for the standard kraken and Tesseract are almost the same was to be expected since they use comparable architectures. Tesseract’s results are slightly more consistent, though, but kraken’s use of a BLSTM layer is beneficial in terms of character recognition. If we look at the HTR+ rebuild kraken+, we find that it performs only 1.6 percentage points short of the HTR+ model. We assume that the difference in performance is due to the image pre-processing and data augmentation. As such, kraken is an important candidate for building OCR models, not least because it is completely open-source. PyLaia performs on par with HTR+ and TrOCR delivers the best results with an F1-score of 0.99. The results are interesting because if we look at the CERs of the best-performing models in Table 3.2, we find that PyLaia performs best, while TrOCR is on par with kraken+, but still behind the HTR+ model. Hence, we conclude that it is important to consider more than one quality measure, which we elaborate on in Chapter 5. Moreover, the difference in the CERs likely stems from different computations, as stated by Neudecker et al. [2021].

model	CER
ABBYY FineReader Server 11	3.85
HTR+	0.67
kraken+	0.89
PyLaia	0.60
TrOCR	0.89

Table 3.2: Averaged CER on the NZZ test set for some of the models, all except ABBYY showing near-perfect performance.

This requires a qualitative analysis. According to the F1-measure, PyLaia performs worst on the test set page from 1904 with an F1 score of 0.95. On the same page, TrOCR achieves an F1-score of 0.99. Table 3.3 shows a random sample of ten lines containing errors from the PyLaia result and the GT and the TrOCR result of the same line. Errors are highlighted in orange, underscores represent missing characters and characters marked in blue errors in the GT. The errors of the PyLaia model are

nothing out of the norm. It mistakes an *e* for a *c*, an *u* for an *n*, or a capital *F* for a capital *R*. We also notice TrOCR and PyLaia never commit the same errors.

GT	TrOCR	PyLaia
Nebentälern, die Zeutral- und Westschweiz ohne Jura	Nebentälern, die Zeutral- und Westschweiz ohne Jura	Neventälern, die Zentral- und Westschweiz ohne Jura
und Genfersee. Dem in seiner Kunst so weit und voll-	und Genfersee. Dem in seiner Kunst so weit und voll-	und Genfersce. Dem in seiner Kunst so weit und voll-
Künstlern längst das größte Interesse gefunden, wird	Künstlern längst das größte Interesse gefun ^{ee} n, wird	Künsti ^{er} n längst das größte Interesse gesunden, wird
Ein Genfer Humanist.	Ein Genfer Humanist.	Ein Genfer H ⁿ manist.
gestern abend im Hotel Metropole von der Faculté des	gestern abend im Hotel Metropole von der Faculté des	gestern abend im Hotel Metropole von der Raculté des
Nicoles eine neue Richtung und einen neuen Imput ^s .	Nicoles eine neue Richtung und einen neuen Imput ^s .	Nicoles eine neue Richtung und einen neuen Impu's.
Teil der Schweiz und in Basel entstanden sind, werden	Teil der Schweiz und in Basel entstanden sind, werden	Teil der Schweiz und in Ba ^e l entnanden sind, werden
und Bremgarten sind in diesem Kapitel berücksichtigt.	und Bremgarten sind in diesem Kapitel berücksichtigt.	und Bremgarten sind in diesem Kavitel berücksüchtigt.
praktischen Rücksichten ihre Existenz verdanken; künst-	praktischen Rücksichten ihre Existenz verdanken; künst-	praktischen Rücksichten ihre Exin ^{en} z verdanken; künst-
des Hochgebirges, jene einfachen Häuser, die lediglich	des Hochgebirges, jene einfachen Häuser, die lediglich	des Hochgebirges, jene einfachen Hänser, die lediglich

Table 3.3: Small qualitative analysis of the OCR errors made by TrOCR and PyLaia models (errors in orange).

That TrOCR reaches almost perfect F1-scores in the bag-of-word evaluation deserves further scrutiny. For this purpose, we look at the training regime of TrOCR. Table 3.4 lists the CERs for a different number of training epochs. We note that for a zero-shot performance (i. e., using the pre-trained TrOCR model without any fine-tuning) CERs differ greatly depending on whether the TrOCR model pre-trained on printed texts or the one pre-trained on handwritten texts is used. From the experiments in Chapter 4 we learnt that fine-tuning for ten epochs is a reasonable choice. Adding five epochs, no matter whether we use the printed or the handwritten pre-trained model, leads to slightly worse results. And since the fine-tuning on the printed base model for ten epochs performed better, we did not continue fine-tuning the handwritten pre-trained model.

The results of fine-tuning the TrOCR model pre-trained on printed data are impressive. After fine-tuning on only one epoch, the CER decreases from a zero-shot performance of 88.92% CER to 0.99% CER. This is an improvement of 8,982%. These numbers show that TrOCR is a highly adaptable model and can compete with traditional OCR methods.

epochs	model type	CER
zero-shot	printed	88.92
1		0.99
5		0.89
10		0.97
15		1.22
zero-shot	handwritten	33.41
10		1.21
15		1.3

Table 3.4: Averaged CERs over the pages of the NZZ test set for different numbers of training epochs for TrOCR.

Last but not least, and referring back to the imbalanced word distribution over the years mentioned in Section 3.3.1, the over-represented data from later periods of the period for which we have constructed a GT does not pose a problem for the recognition of text from pages belonging to the part of the underrepresented early data.

Regarding **research questions 3.1** and **3.2** this experiment shows that – given a large enough GT – the most recent HTR architectures successfully recognise texts from historical newspapers. The small qualitative evaluation showed results that are easy to be corrected automatically, therefore leading to near-perfect OCR (or, according to Holley [2009], “good” OCR).

3.6.2 Experiment 2 – Training Size Ablation

The ablation experiment determines the number of pages needed while training an OCR system to achieve the best results possible. The results from Section 3.6.1 suggest focusing on TrOCR alone. However, we introduce a size ablation experiment for the TrOCR architecture in Chapter 4, so we use one of the best-performing open-source CNN×RNN models for the ablation experiment. Table 3.1 suggests that kraken+ is a good candidate.

For this experiment, we split the corpus into different training set sizes while the test set remains the same. Table 3.5 provides an overview of the different training set sizes. We decided to test training set sizes of 12, 25, 50, 100, and 150 pages, where bigger training sets always contain the pages from smaller training sets. One exception is kraken+ 200L, where we sampled 200 lines at random from the training set pages. We chose to train a model on 200 lines because Springmann et al. [2016] reported from their experiments that the character accuracy ranges from 95% to sometimes even 99%, even for small training sets. We then trained a kraken+ model²⁹ for each training set size for 50 epochs with early stopping and again compared the performance of each model on the test set using the bag-of-words F1 measure. kraken+ took more than two days to run on the largest training set on our infrastructure.

Training set size	# words	# lines
150 pages	273,440	38,756
100 pages	183,246	25,904
50 pages	89,705	12,740
25 pages	45,134	6,404
12 pages	22,081	3,157
200 lines	1,407	200

Table 3.5: Summary of all training sets used for the training of different models divided into number of words and number of lines.

Results and Discussion of Experiment 2

We first notice from Table 3.6 that a training set of only 200 lines is **insufficient**, although they are in the same range as reported by Springmann et al. [2016]. We also point to the fact that as soon as the CER reaches a certain threshold, word accuracy suffers significantly. Judging from the values in Table 3.6, this threshold lies between models trained on 12 and 25 pages. As such, we deem the bag-of-words F1-measure

²⁹We also trained HTR+ models on the same training set sizes. We will use these models in Section 3.6.3.

to be more telling than CER, but we nevertheless include the CER of the kraken+ models for informative purposes. The kraken+ 200L model, with a CER of 5.04%, achieves an average bag-of-words F1-score of 0.81, which is only slightly worse than FRS11 (see Table 3.1). In general, we note an upwards trend, i. e., the more training lines we include, the better the models perform, both in terms of CER and bag-of-words F1-measure. Also, the SDs are so low that we conclude that the models output consistent results over all periods.

	Models					
	kraken+ 150	kraken+ 100	kraken+ 50	kraken+ 25	kraken+ 12	kraken+ 200L
CER in %	0.89	0.99	1.40	1.62	3.61	5.04
F1	0.95	0.95	0.92	0.91	0.82	0.81
SD	0.02	0.03	0.04	0.05	0.07	0.07

Table 3.6: Averaged performances of kraken+ on the NZZ test set in terms of CER and F1-measure using different training set sizes (200 lines, as well as 12, 25, 50, 100, and 150 pages). The standard deviation (SD) refers to the SD within the F1-measure.

With CERs between 0.89 and 1.65 for kraken+ models trained on 25 to 150 pages, we are well above the threshold after which we can call OCR results *good* [Holley, 2009]. The gain in F1 score of four percentage points when comparing models trained on 25, 100, or 150 pages suggests that a GT consisting of any number between 25 and 50 pages is sufficient to achieve good OCR results.

Hence, these results show that concerning **research question 3.3**, we recommend a GT of roughly 6,000 lines for good OCR. However, this experiment also showed that average OCR is achievable with only 200 lines of GT. Chapter 4 examines size-ablation capabilities for TrOCR with equal amounts of lines.

3.6.3 Experiment 3 – Transfer

This experiment tests whether the application of the models we have trained on medium-quality NZZ (i. e., images scanned with less than 300dpi) images remains stable when we apply them to high-quality images on the one hand and to pages printed in blackletter from other newspapers on the other hand. The source ma-

terial we trained on is of medium quality because the pages of the NZZ have been scanned from microfilm and not from paper. Moreover, the resolution of the images is less than 300 dpi, which is generally recommended to obtain good OCR results.

To test the transferability, we took high-quality scans digitised directly from paper for four pages from the NZZ test set and loaded them into Transkribus. We then extracted the text from these pages using the HTR+ models trained on 12 to 150 pages and also compared these results to OCR results from FRS11. We opted for the HTR+ models for this scenario because we also evaluated against the FRS11, which was also available within Transkribus. Using only the Transkribus environment was thus more convenient. We tested the transfer to other German-language and black letter publications by taking five high-quality scans each from the *Bundesblatt* and the *Neue Zuger Zeitung*³⁰ covering the last 50 years of the 19th century and extracted the text using the same models. Again we used the bag-of-words F1-measure to evaluate the performance on the slightly different data.

Results and Discussion of Experiment 3

Models	Paper scans		Microfilm	
	F1	SD	F1	SD
HTR+ 150	0.982	0.002	0.985	0.005
HTR+ 100	0.983	0.004	0.983	0.005
HTR+ 50	0.977	0.004	0.978	0.006
HTR+ 25	0.972	0.009	0.969	0.008
HTR+ 12	0.969	0.005	0.956	0.011
FRS11	0.891	0.031	0.847	0.024

Table 3.7: Mean F1-scores and standard deviation (SD) of the HTR+ models and FRS11 on four high-quality images (1780, 1830, 1880, 1929) of the NZZ, compared to performance on scans from microfilm.

³⁰The *Bundesblatt* is a collection of announcements by the Swiss government. The *Neue Zuger Zeitung* is another German-language Swiss newspaper, not to be confused with the NZZ.

First, we looked at the average of each model on a reduced test set (all of which are from the original test set). The overall picture (see Table 3.7) is that the performance stays the same. For the recognition of text in high-quality images there are slight deteriorations of 0.03 and 0.01 for the HTR+ 150 model and the HTR+ 50 model, respectively. FRS11 profited the most from the better image quality and improved by almost five percentage points. This shows that HTR+ models, and by extension also the kraken+ models, are more robust to the quality of the input image. The results in Table 3.7 further suggest that the models are transferable and that the performance will not suffer, although we have trained the models on medium-quality images.

This discovery is important for GLAM institutions because it advocates for partial retro-digitisation of already existing digitised documents. With “partial,” we mean that it is not necessary to re-scan the documents because applying a more recent and potent OCR model produces better text results.

Regarding the transferability to other newspapers, we see average F1-scores above 0.95 for all HTR+ models (see Table 3.8). Moreover, FRS11 scores over 0.9 for the first time on the Bundesblatt. It is, however, still five percentage points away from the best-performing HTR+ model for the same data. The results show that a transfer between publications is possible and that the OCR is sometimes even better than on the data the systems saw during training. We should mention, though, that the standard deviation (SD) is a little higher in these evaluations, which stems from the smaller test set size.

	Bundesblatt		Neue Zuger Zeitung	
Models	F1	SD	F1	SD
HTR+ 150	0.986	0.015	0.990	0.007
HTR+ 100	0.986	0.013	0.991	0.006
HTR+ 50	0.982	0.010	0.989	0.007
HTR+ 25	0.982	0.015	0.985	0.009
HTR+ 12	0.976	0.018	0.982	0.010
FRS11	0.924	0.027	0.884	0.037

Table 3.8: Average performance on five pages each from other publications

3.7 Conclusion

We were able to show that state-of-the-art neural OCR systems like HTR+, kraken, and Tesseract trained on in-domain data performed better than standard commercial systems at the time when we conducted the experiments. It is most likely that the commercial systems have closed the gap, although this would need verification. In terms of **research questions 3.1** and **3.2**, we can state that both CNN×RNN architectures and Transformer-based techniques deliver high-quality results. The kraken+ results are on par with Transkribus but kraken+ provides more freedom when training an OCR model. The same goes for TrOCR, while we need to stress TrOCR’s ability to adapt to new data. We further discuss this issue in the next chapter.

For Transkribus, we should mention a few restrictions: e. g., it is impossible to download the model to apply it independently from the Transkribus infrastructure. Moreover, although creating a GT within Transkribus is simple, annotators must add baselines for an HTR+ model to work. This is obsolete for kraken,³¹ Tesseract and TrOCR which only require line images without baseline information. An additional hurdle of Transkribus is its pricing system. While transcribing, annotating, and training of PyLaia models are still free components within Transkribus, applying models for text recognition has become a paid service.

We furthermore provided an estimate of the number of training pages necessary to provide solid OCR results kraken+. Responding to **research question 3.3**, our experiments show that a GT of between 25 and 50 pages is sufficient to provide good OCR results. This insight significantly reduces the time needed to create a GT to train models on new data. The TrOCR experiment furthermore suggests that this number could be lowered due to the fact that the training from scratch becomes unnecessary.

Lastly, we have shown that the models are transferable to other periodicals or newspapers. We can thus reply to **research question 3.4** that at least HTR+ and kraken+ models adapt well to changes in quality and domain. Even better, because the results suggest that the most recent ATR architectures can even cope with medium-quality images, GLAMs can evade expensive re-scanning processes.

³¹However, more recent versions of kraken offer the possibility to train with baseline information.

Chapter 4

Transformer-based ATR for Handwritten Documents

As we have seen in Chapter 3, the TrOCR model outperformed the traditional CNN×RNN approaches. Moreover, fine-tuning for only one epoch resulted in huge performance gains. Since the TrOCR model is still very recent, this new approach to ATR deserves further scrutiny. Especially in today’s regime of pre-training and fine-tuning, we must ask ourselves whether we can fine-tune TrOCR to adapt not only to historical newspapers but also to other historical data, e. g., handwriting. For one, TrOCR has never seen handwriting styles from the 16th century, and secondly, it has only seen English data during pre-training. We fine-tuned it already on the German-language NZZ, but we did not address the language issue in Chapter 3, which we do in this chapter. For these reasons, we investigate the following research questions (RQ):

RQ 4.1 Can TrOCR recognise historical handwritings in various languages like Latin, Italian, and different variants of German?

RQ 4.2 How much new data does TrOCR need to adapt to historical handwriting? And what is the optimal hyperparameter setting?

RQ 4.3 Does data augmentation for TrOCR boost performance for handwriting recognition?

RQ 4.4 Does TrOCR outperform traditional architectures?

4.1 Transformer-based ATR with TrOCR

We have already introduced TrOCR in Section 2.2.4. Nonetheless, we present a simplified view of its architecture again because we want to focus on what happens between the encoder and the decoder. Perhaps the most important feature of TrOCR is its cross-attention between the encoder and the decoder. It decides which image tokens correspond to which sub-words. In Figure 4.1, we see how the image line is re-sized before further processing. Next, the image traverses the visual encoder and gets passed on to the decoder, which decodes the image tokens to generate the sub-words. Figure 4.1 depicts the cross-attention between the visual encoder and the language decoder. We see 16 attention heads stemming from the BEiT [Bao et al., 2021] architecture for each decoding step. The darker a square in an attention head at a specific step, the more attention is put to that region in the image. The overall impression is that the heads in the centre pool the attention. By looking at the re-sized image, this is logical since the handwriting to decipher runs over the middle of the image.

Zooming in on the single layers of the visual encoder and summing up the attention over all heads along the x-axis results in Figure 4.2 (inspired by Kang et al. [2022]). The figure shows an interesting behaviour, namely that the attention in the first layer is spread across the whole image. The further the image traverses through the layers, the clearer the attention focuses on specific places in the image. As such, in layer 9, we observe the attention shifting from left to right in a way that mimics how a human would read a line.

4.2 Data for HTR

The main datasets used in this chapter stem from three sources, which we describe here. First, we introduce a benchmark dataset which serves to highlight differ-

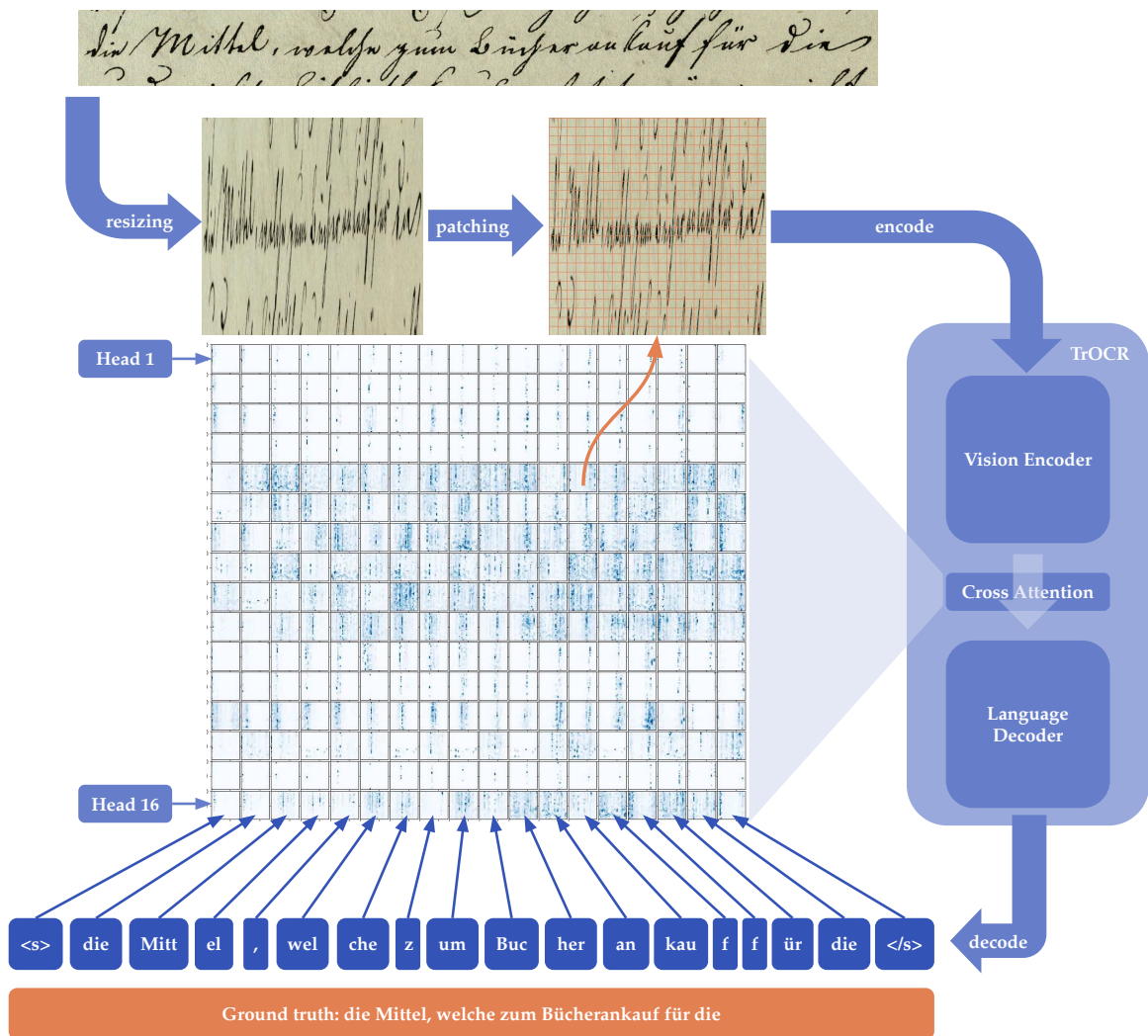


Figure 4.1: Flow of an input image through the TrOCR architecture to produce a text string with a focus on the cross-attention layer.

ences between traditional and Transformer-based HTR approaches. The second is a dataset compiled from correspondence data to test HTR approaches in settings with many different hands. The third is a smaller dataset containing handwriting from a manuscript by Rudolf Gwalther.



Figure 4.2: View of two layers (1 and 9) with attention summed over all heads along the x-axis.

4.2.1 Benchmark Dataset

ICFHR 2018 Competition Dataset

We take this dataset, compiled within the [READ](https://readcoop.eu/)¹ project, from the 2018 competition on automated text recognition. The competition was held at the 16th International Conference on Frontiers in Handwriting Recognition [Strauß et al., 2018]. We will refer to this dataset as *ICFHR 2018 Competition Dataset* or just *ICFHR2018* data. The task was to minimise the character error rate (CER) on a test set with the help of a pre-trained model trained on the “general data”. In the following, the competition participants needed to adapt to different smaller training sets containing handwriting styles the pre-trained model has not seen. We chose this dataset because it represents the state of the art at that time with traditional ATR architectures and because it has been used in later stages [Aradillas et al., 2021].

To be more specific, the dataset is composed of two parts. The first part² comprises 17 documents, with each document containing 25 pages. The second part consists of five different documents with five different scripts.³ Figure 4.3 shows an example from the five datasets on which fine-tuning is performed. A separate test set contains one document for each of the documents in the fine-tuning part. Table 4.1 taken from Strauß et al. [2018] summarises the dataset statistics.

¹<https://readcoop.eu/>

²We call this part *pre-training*.

³We call this part *fine-tuning*.

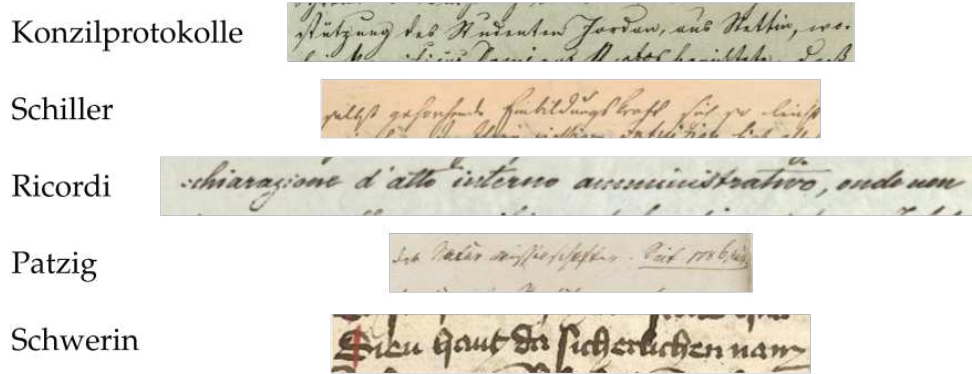


Figure 4.3: Example lines from the fine-tuning dataset.

	data part 1		data part 2	total
	pre-training	fine-tuning	testing	
documents	17	5	5	27
lines	11,925	2,878	2,181	16,984
words	75,145	16,399	6,695	98,239
characters	444,072	99,924	57,881	601,877

Table 4.1: ICFHR2018 dataset statistics.

Since the competition design included size ablation experiments to answer how many lines of a new document a pre-trained model needs to see to deliver satisfactory results, we can further break down the fine-tuning data into the different sizes we show in Table 4.2 (taken from [Aradillas et al., 2021]).

The data is, above all, very diverse as it concerns the languages. The manual inspection of the data led to the conclusion that there are documents in English (EN), (Early New High) German (DE), Latin (LA), Italian (IT), Swedish (SV), and Norwegian (NO), i. e., the dataset is **multilingual**. In the fine-tuning part, however, we only find German and Italian. This has implications for the pre-training and the specific fine-tuning, some of which we address in the experiments in this chapter.

To assess the language distribution of the two data parts, we used the Python lan-

fine-tuning set	# of lines for different fine-tuning set sizes in pages			# of lines in test set
	1	4	16	
Konzilprotokolle	29	116	469	358
Schiller	21	84	328	244
Ricordi	19	88	383	280
Patzig	38	156	641	576
Schwerin	68	264	1,057	718

Table 4.2: ICFHR2018 fine-tuning dataset sizes. The columns correspond to the number of lines in one, four, or 16 pages.

guage classifier `langid`⁴ [Lui et al., 2014] and restricted the range of languages to identify the languages mentioned above. This results in the language distributions depicted in Figure 4.4.

We see a clear dominance of (Early New High) German and English. We should also note that we did not perform a large-scale evaluation of the accuracy of `langid`, so there are wrongly identified languages. Especially for short lines (below 15 characters), the information we obtain from `langid` is unreliable. For example, we do not expect Danish, Norwegian, or Swedish lines in the fine-tuning part. A manual inspection of ten lines identified as any other language than German or Italian showed that indeed `langid` misclassified the language.

4.2.2 Creating a Ground Truth from Edited 16th Century Correspondence

The Bullinger dataset consists of scan images and corresponding text lines from the Bullinger correspondence. Heinrich Bullinger (1504–1575) was an important Swiss reformer and successor of Huldrych Zwingli (1484–1531) as pastor of the Gross-

⁴<https://github.com/saffsd/langid.py>

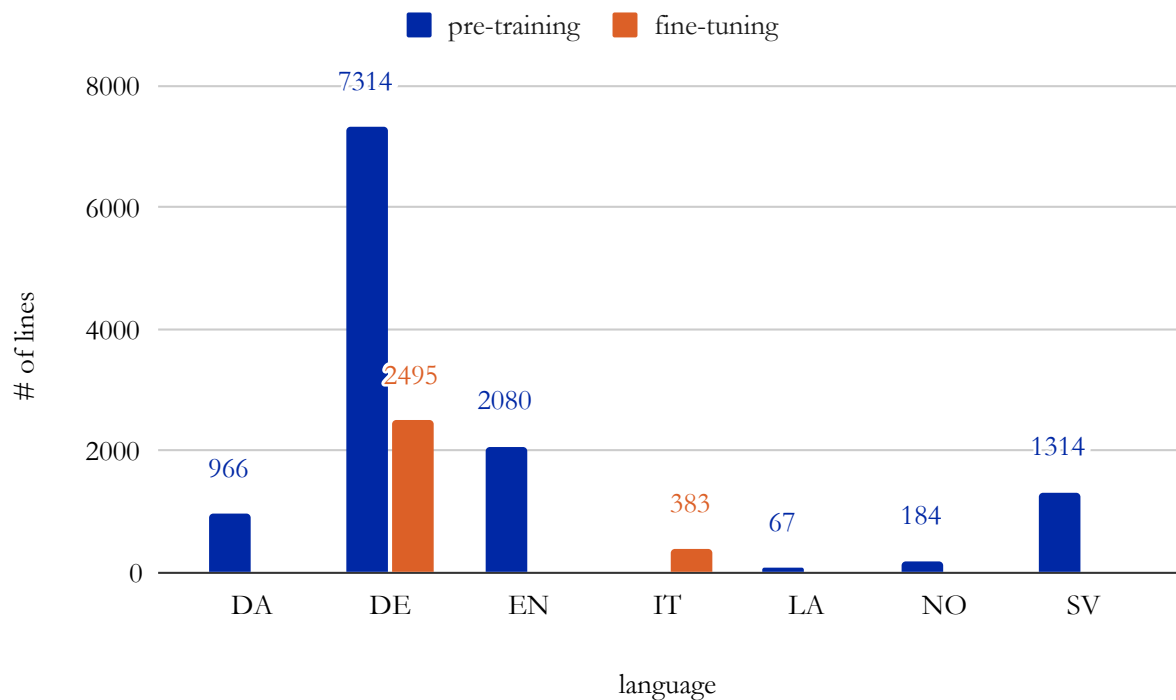


Figure 4.4: Automatically determined language distribution with langid for the two data parts measured in lines.

münster Zürich. He wrote to and received letters from a European audience during his lifetime. He not only corresponded with scholars and theologians from all over Europe, but also with royalty, like the Holy Roman Emperor [Maximilian II.](#),⁵⁶ as depicted in Figure 4.5

From summer 2021 until summer 2023, the Zurich State Archive and the Central Library of Zurich have digitised between 11,000 and 12,000 letters that have been preserved, making the Bullinger correspondence one of the largest letter networks of the 16th century. At the same time, there exist 3,141 edited transcriptions which have been published in 20 volumes, the first of which was published in 1974 [Gäbler et al., 1974–2020]. These transcriptions are available as TUSTEP or PDF files. There are an estimated 5,400 additional “working transcriptions”⁷ available as TXT files.

⁵<https://www.bullinger-digital.ch/letter/3472>

⁶Heinrich *Bullinger*, Dem durchlüchtigsten, großmächtigsten fürsten und herren h(ern) Maximilian, 1558; Staatsarchiv Zürich, Zürich StA, E II 338,1566.

⁷These “working transcriptions” have either been produced by students or scholars but have not

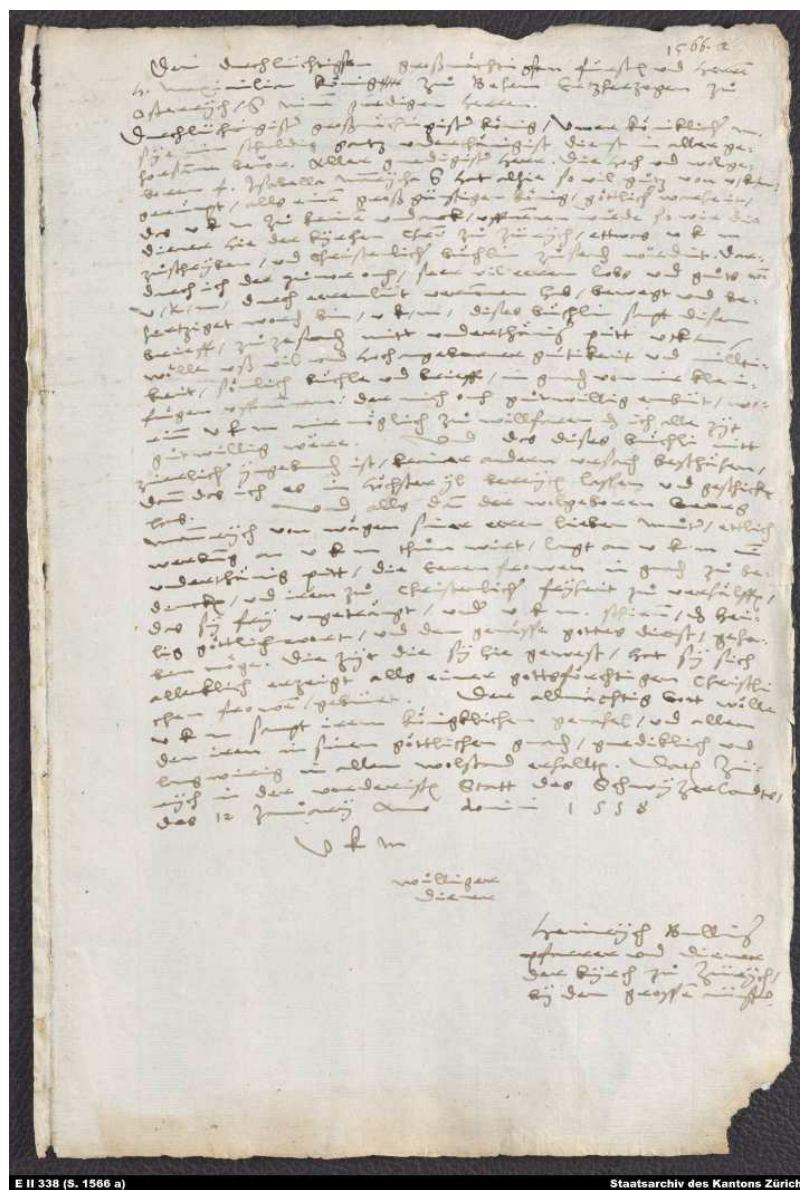


Figure 4.5: A letter from Heinrich Bullinger to the Holy Roman Emperor Maximilian II. dating back to January 12, 1538, written in Early New High German.

These working transcriptions are generally of high quality, but they are not ready to be published and sometimes only contain mere summaries of the letters. Hence, if we want to use this material to create training material for ATR models, we must

been published in the edition yet.

accept that we might introduce certain errors.⁸

Before we could use these data to train HTR models, however, several pre-processing steps were necessary. The first step to generating training data was to align the text we extracted from the different text file formats to the corresponding line in the scan of a letter. Our partners at the University of Bern used Transkribus and its *Text2Image* module,⁹ a tool developed by the [CITlab Rostock](#).¹⁰ The Text2Image process consists of an initial layout analysis to identify the text lines in the scan. A first text recognition step with a base HTR model produces confidence matrices of each image line. The CTC algorithm Graves et al. [2006] then determines the probability of these confidence matrices against the transcriptions. Text2Image allows the definition of a confidence parameter. The higher this parameter, the less text Text2Image will assign to the image lines. It is important to note that Text2Image is an iterative, semi-supervised approach, during which the base HTR is continually trained with the aligned material to provide better alignments in the subsequent steps. Still, this process is error-prone. Misalignments tend to increasingly occur at the end of lines, where a word either is missing or a word (or a part of it) of the next line is wrongly added. This is why we call this kind of aligned data as *silver truth*.¹¹

	# lines	%	# words	# words / line	# characters	# char. / line
LA	134,236	81.02	1,073,106	7.99	7,314,648	54.49
ENHG	31,437	18.98	253,372	8.06	1,547,725	49.23
Total	165,673		1,326,478	8.01	8,862,373	53.49

Table 4.3: Summary of the Bullinger correspondence data aligned via Text2Image.

The data we present here stems from May 2022. At that time, our partners at the University of Bern just finished aligning the text and images of all the available scans. Table 4.3 provides an overview of the training data produced via Text2Image. The whole dataset, including the training, validation, and test set splits, is available on [GitHub](#)¹².

⁸We leave it for future research to examine the quality of automatically aligned GTs.

⁹Unfortunately, this feature is not available anymore.

¹⁰<https://github.com/CITlabRostock/CITlabModule>

¹¹GTs are sometimes also referred to as *gold standards*. Since our alignment contains certain errors, we cannot refer to it as gold but as silver standard.

¹²<https://github.com/pstroer/bullinger-htr>

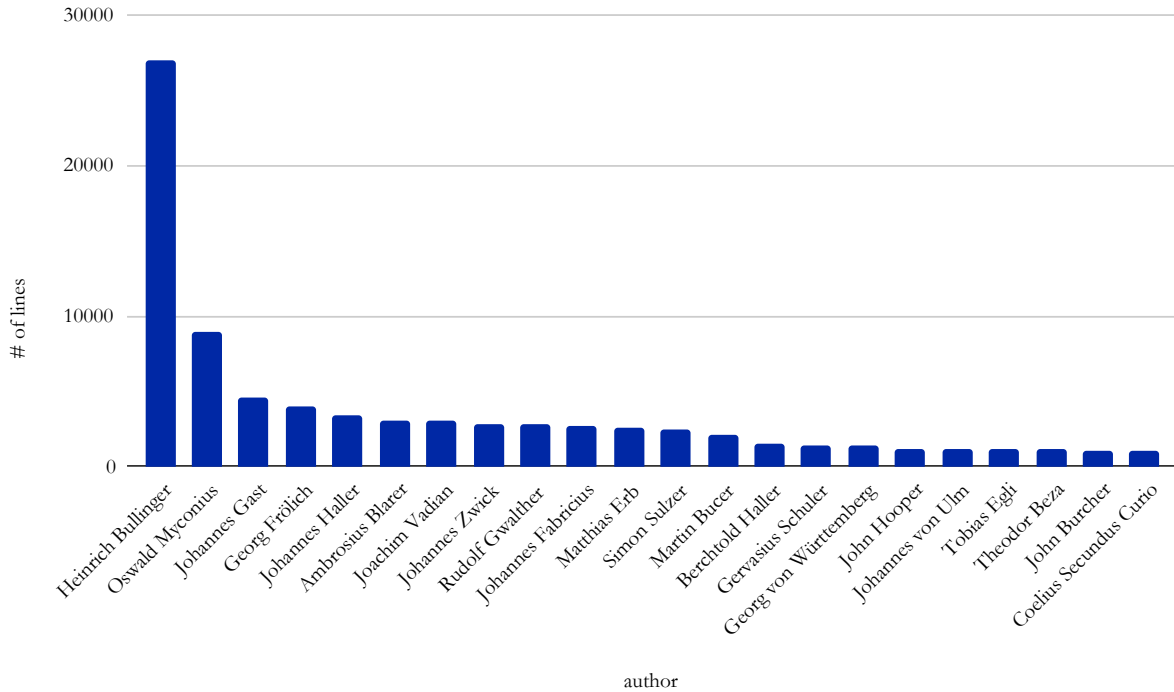


Figure 4.6: Author distribution in the Bullinger correspondence according to the number of lines in the training set. Here we see the authors with at least 1,000 lines.

Since most of the correspondence has been written in either Latin or Early New High German (ENHG), we applied a language identifier specifically trained by Volk et al. [2022] to distinguish between those languages to the text lines. All in all, we have more than 160k lines, 81% of which are in Latin and 19% in ENHG. We have over 1.3M tokens and 8.8M characters in our dataset. As such, we are not in a low-resource scenario anymore (like we are with the ICFHR2018 data, see Section 4.2.1). Nevertheless, if we look at the author distribution of the total set, we need to reassess the situation.

As is common for correspondence data comprising several hundred different authors, there is a heavy skew as concerns the author distribution. In the entirety of the Bullinger correspondence, 2,812 letters can be attributed to Heinrich Bullinger alone, which makes him the most represented author. With 672 letters, Johannes Haller is the runner-up. This skew was also projected onto the dataset with which we experimented. With Bullinger as the main character of the correspondence, he accounts for 33,675 lines, 9,208 in Early New High German and 24,397 in Latin.

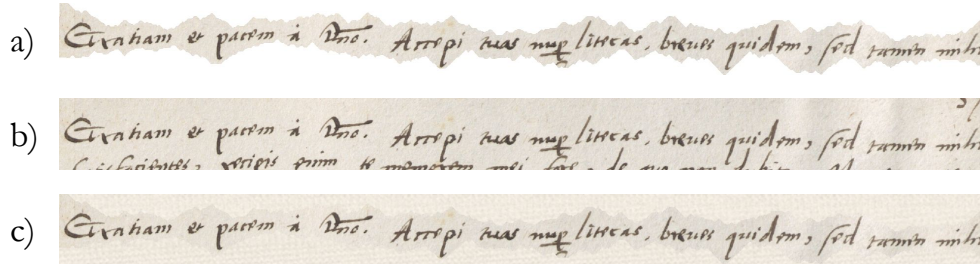


Figure 4.7: Different pre-processing strategies: a) polygon crop according to Transkribus’ layout analysis, b) rectangular crop, c) polygon crop with a randomly generated background.

Skewed data requires exploring different training strategies, some of which we introduce in this chapter. Figure 4.6 gives an impression of the distribution of 22 authors for which we have at least 1,000 lines in our training set.

Next in the processing pipeline is the image pre-processing. With the layout analysis from Transkribus, we had several different options, all of which we depict in Figure 4.7. The layout analysis detects the lines in an image and defines a polygon around the handwriting in the line. The cropped line in Figure 4.7 a) follows the contours defined by the polygon and puts the cropped line image on a white background. Version b) is a rectangular crop according to the highest and lowest points of the polygon in both the x - and y -direction. The third variant c) is the same as a), but instead of using a white background, we took the average colour of the whole line and created a random background with it.

The different pre-processing styles presumably have different consequences for the training procedure. Lines in the styles of version a) could tempt the model to focus on the outer edges instead of the actual writing. We also noted that the layout analysis is not always perfect; sometimes, it cuts especially long strokes of “g”s, “f”s, or bigger initials at the beginning of a line. Lines like in b), on the other hand, can contain handwriting of the preceding and succeeding lines, making it difficult for the model to focus on what handwriting to decode. In example b), we see two almost complete words from the line below, which we do not want to occur in the result. The problem accentuates itself for authors whose lines are severely slanted. Pre-processing the lines as introduced in c) could mitigate both of the problems of variants a) and b). We examine the impact of image pre-processing in Section 4.3.3.

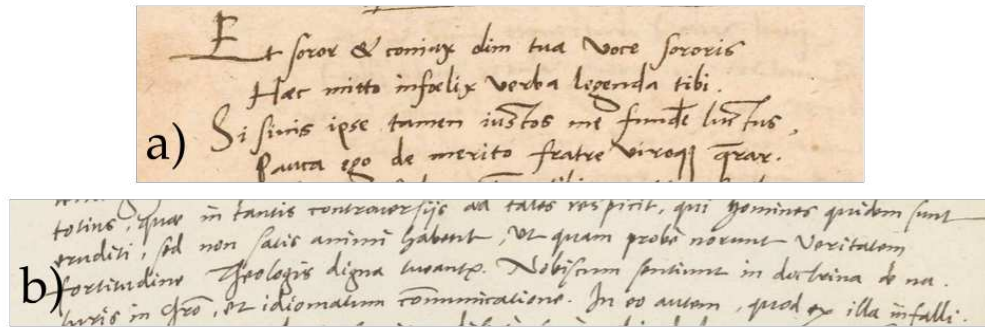


Figure 4.8: Examples from Gwalther's handwriting. (a) depicts his handwriting in the manuscript *Lateinische Gedichte*, (b) is from a letter to Heinrich Bullinger.

4.2.3 The Gwalther Dataset

Before we received scans from the Zurich State Archive and the Central Library of Zurich to compile a silver truth for correspondence data, we compiled a dataset from one of Bullinger's correspondents: Rudolf Gwalther (1519–1586). He wrote a volume called *Lateinische Gedichte*¹³ (*EN Latin Poems*), whose scan images and partial transcriptions by Prof. em. Dr. Peter Stotz (1942–2020) are available on [e-manuscripta](#).¹⁴ In May 2022, 142 pages of the 600 pages contained transcriptions. We loaded the images into the Transkribus interface, applied layout recognition to identify lines, and aligned the transcriptions with the lines. We made minor corrections during this process, mainly concerning inconsistent capitalisation and punctuation. The resulting **Gwalther** dataset consists of 4,037 lines amounting to 26,088 words. It is publicly available on [Zenodo](#)¹⁵.

This dataset will be useful because Gwalther wrote around 75 letters to Bullinger. As Figure 4.8 shows, Gwalther's handwriting differs from a) his manuscript *Lateinische Gedichte* and b) a letter he wrote to Bullinger. Hence, we design an experiment to check whether a model trained on Gwalther's handwriting from the book can be applied to his letters.

¹³Gwalther, Rudolf: [Lateinische Gedichte]. [Zürich], 1540-1580. Zentralbibliothek Zürich, Ms D 152.

¹⁴<https://doi.org/10.7891/e-manuscripta-26750>

¹⁵<https://doi.org/10.5281/zenodo.4780947>

4.3 Experiments

The first experiment considers whether TrOCR allows further fine-tuning on data it does not know and how well it can cope with such data. We, therefore, take the Gwalther dataset, the ICFHR2018 dataset, and the Bullinger dataset described in the previous section to fine-tune handwritten TrOCR models. We use the TrOCR model for printed data since initial runs showed slight performance gains over the TrOCR model pre-finetuned on handwritten data no matter how many epochs we fine-tuned.

4.3.1 Fine-tuning on the Gwalther Dataset

The first part of the experiment examines how TrOCR copes with the handwriting from Gwalther (see Section 4.2.3). We split the data into 3,603 lines for training and 433 lines for testing. We trained an HTR+ model within Transkribus as a baseline for 50 epochs and with a **base model**.¹⁶ Furthermore, we fine-tuned both the BASE and the LARGE instances of the TrOCR model pre-trained on handwritten data on the Gwalther training set. Table 4.4 shows the results.

Model	specification	fine-tuning epochs					epochs
		3	5	10	15	20	50
HTR+	-	-	-	-	-	-	2.74
TrOCR	BASE	3.84	3.72	3.18	3.31	3.62	-
TrOCR	LARGE	2.94	2.72	2.58	2.55	2.62	-

Table 4.4: CER of HTR+ versus TrOCR while TrOCR has been trained on a different number of epochs and with two configurations.

¹⁶Transkribus offers the possibility to provide a base model for training so that the model does not need to be trained from scratch. We used the *Acta_17 HTR+* as a base model. The data (chancery writings) and model were compiled by Alvermann et al. (see <https://readcoop.eu/de/erfolgsgeschichten/greifswald/>) from the University of Greifswald. Period: 1580-1705, size: ~600k words in Latin and Low German from 1k different authors, epochs: 1k.

Results and Discussion for the Gwalther Dataset

First, the baseline set by the HTR+ with a CER of 2.74% is low, i. e., we already achieve good text quality. As concerns TrOCR, the BASE model is incapable of beating the baseline. However, the LARGE model beats the baseline by 0.19 percentage points after only 15 epochs of fine-tuning. We conclude from this that the more powerful vision encoder benefits the results. Moreover, the CER decreases with an increasing number of fine-tuning epochs, although after a certain number of epochs, the CER increases again, which is a sign of overfitting. Overfitting occurs faster for TrOCR than it does for HTR+ models. The differences in CER between models fine-tuned for different numbers of epochs are small, but in terms of **relative error reduction**, TrOCR pushes the CER by 18.87%. Judging from these results, there is a clear preference for the LARGE model.

Our results are surprising because the pre-trained TrOCR model never saw any Latin data before these experiments. For example, our model only sees 23k Latin words during fine-tuning on the Gwalther dataset. The vocabulary overlap of the training and test set is 68.9%,¹⁷ and still, TrOCR beats the baseline. Moreover, TrOCR has never been confronted with early modern manuscripts. We, therefore, postulate TrOCR as a highly flexible model that adapts quickly to new data (at least if it contains one author only). With the following ablation experiment containing multilingual data, we want to test TrOCR’s abilities further.

However, the flexibility has its limits. We get an entirely different picture when we apply the Gwalther model trained on the book volume to a letter containing 42 lines (see Figure 4.8 again). The best model is the LARGE model trained for three epochs with a CER of 39.3%. Although the writer is the same and the handwriting is similar, we do not achieve CERs near the ones presented above. This means that although TrOCR is a versatile model, it would still need some fine-tuning if there is much intra-writer variation.

4.3.2 Fine-tuning on the ICFHR2018 Dataset

The challenges for further fine-tuning the pre-trained TrOCR model are the uncertainty about the number of epochs to fine-tune further and the “correct” setting of

¹⁷It would also be informative to compute the sub-word overlap since TrOCR is trained on sub-words.

fine-tuning set	fine-tuning set size	epochs				
		1	3	5	10	15
Konzilprotokolle	0	15.81	12.09	12.23	12.31	12.65
	1	13.86	13.40	10.19	12.28	11.61
	4	11.99	9.43	7.99	8.62	8.62
	16	7.84	4.89	4.62	5.84	6.44
Schiller	0	23.50	20.24	19.84	20.16	19.15
	1	30.85	20.43	20.55	20.58	20.46
	4	23.19	19.79	18.18	18.71	18.21
	16	18.08	13.70	12.54	12.07	13.46
Ricordi	0	35.24	37.78	34.38	37.66	38.52
	1	33.46	32.01	30.30	30.84	31.33
	4	28.68	22.86	23.55	23.45	25.79
	16	17.69	14.11	13.53	13.75	15.72
Patzig	0	29.85	27.94	28.11	27.74	26.99
	1	27.80	22.89	22.15	22.55	23.52
	4	21.80	16.88	15.72	16.50	16.71
	16	15.62	10.65	10.96	11.00	11.21
Schwerin	0	28.49	28.89	28.04	27.58	27.31
	1	19.61	14.89	13.71	14.02	14.21
	4	12.10	8.30	7.57	7.68	8.53
	16	5.34	4.41	4.20	3.99	4.32
Average		21.04	17.78	16.92	17.37	17.74

Table 4.5: CERs on the test sets of the different fine-tuning sets. Fine-tuning size 0 means zero-shot evaluation of the pre-fine-tuned model. Best performance per dataset in bold.

the hyper-parameters like dropout rates and **warmup steps**.¹⁸ Our general assumption is that given the size of the pre-trained TrOCR models, the model would overfit to new data quickly. Moreover, the performance of a further fine-tuned model also depends on the amount of data used during fine-tuning. We argue that the diverse ICFHR2018 dataset is the most suitable one to examine the effect of further pre-fine-tuning and specialised fine-tuning to specific handwritings on varying dataset sizes. We also find ourselves in a low-resource setting, which is realistic for GLAM institutions because they, too, do often not have the resources to create GTs for all the collections they host.

¹⁸Warmup describes a procedure by which a very small learning rate is used at the beginning for a certain number of steps to adapt the model to new data slowly.

The procedure for finding the optimal number of epochs to train foresees the pre-fine-tuning of the TrOCR model for handwritten data on the ICFHR2018 pre-training data (see Section 4.2.1) for 1, 3, 5, 10, and 15 epochs. Each of these pre-fine-tuned models serves as a base model in the next step, where we further fine-tune these models on 1, 4, or 16 documents for each fine-tuning set (see Table 4.2) for the same number of epochs. For example, we pre-fine-tune a model on the pre-training data for three epochs and then proceed with the specific fine-tuning (also for three epochs) on the differently sized training set portions of the fine-tuning part. Moreover, we apply the pre-fine-tuned model without fine-tuning on the five different test sets, i. e., we evaluate its zero-shot performance.

fine-tuning set	fine-tuning set size	Reference by Aradillas et al. (2021)
Konzilprotokolle	0	14.50
	1	10.80
	4	6.51
	16	4.32
Schiller	0	24.60
	1	17.31
	4	12.22
	16	9.38
Ricordi	0	34.23
	1	22.71
	4	21.02
	16	11.14
Patzig	0	38.21
	1	26.70
	4	16.11
	16	10.00
Schwerin	0	31.32
	1	10.80
	4	5.52
	16	3.41
Average		16.54

Table 4.6: Reference CERs from Aradillas et al. [2021]

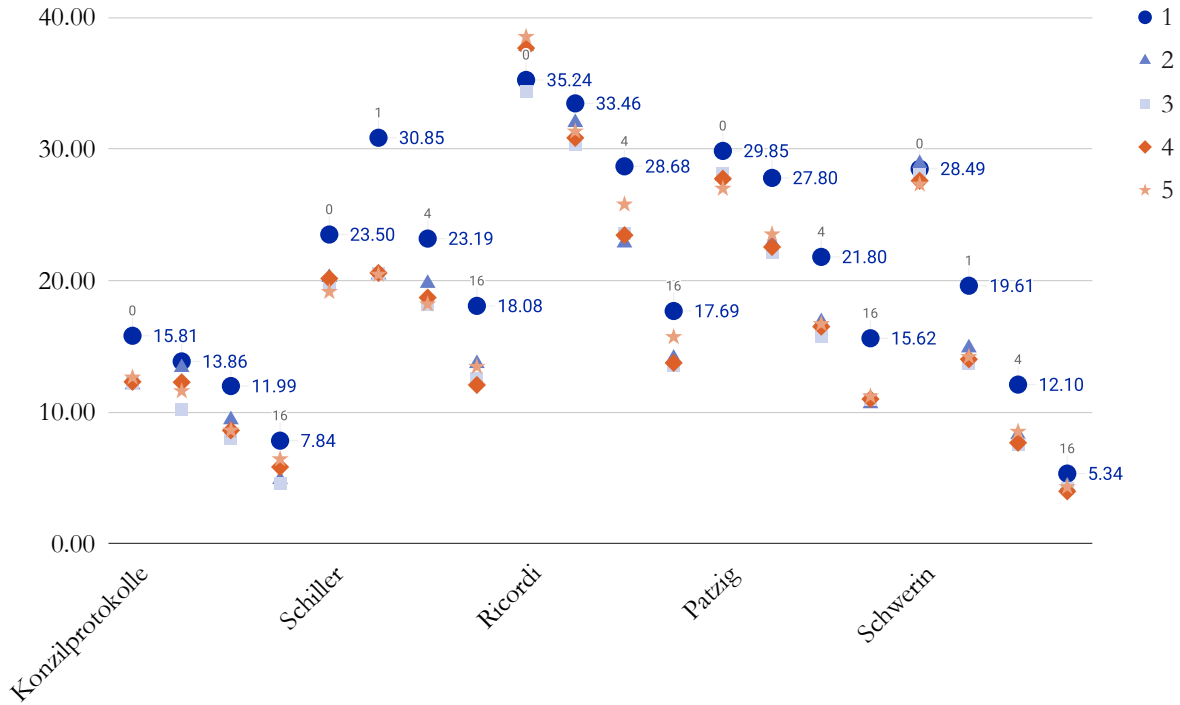


Figure 4.9: Plot showing CERs from Table 4.5. The symbols refer to the number of epochs.

Results and Discussion for the ICFHR2018 Dataset

We evaluate each model separately and conduct the same procedure for the different number of epochs. When we average the CERs over document sizes and the zero-shot performance of a pre-fine-tuned model on a specific dataset, we get the CERs presented in Table 4.5.

While the performance steadily improves the more epochs we fine-tune, there is an increase from 5 to 15 epochs, suggesting that the best-performing model is fine-tuned for something between 5 and 10 epochs (for a dataset of around 11k images). For a graphic representation of the results, see Figure 4.9. Compared to the average CER of Aradillas et al. [2021] of 16.54%, TrOCR performs slightly behind the state-of-the-art (see also Table 4.6 for the reference values).

The best-performing approaches in Strauß et al. [2018] and Aradillas et al. [2021] report significant performance gains thanks to data augmentation strategies, especially grid distortion. We thus followed the procedure by Aradillas et al. [2021], who used



Figure 4.10: Example of RWGD applied to a line image (a) from the ICFHR2018 dataset to produce different versions (b). We stacked ten different versions on top of each other to make the distortions visible. For real training, we do not generate the green lines.

random warp grid distortion (RWGD) [Wigington et al., 2017] to produce copies of lines with slight variations in them. RWGD works by putting a grid of squares over a line image. The intersections between the horizontal and vertical lines act as control points that we perturb in a random direction by a random distance, where the exact amount of movement in the x and y direction is determined by a normal distribution. We used the same grid size and standard deviation as detailed in Wigington et al. [2017] but linearly scaled them to the size of our line images as suggested by the authors. We produced ten different versions per line image resulting in ten times the number of lines to pre-fine-tune on (i. e., 114,681). Figure 4.10 shows an example of RWGD applied to a line image. The variations introduced by RWGD aim to mimic real variations in handwriting.

We repeated the pre-fine-tuning on the augmented data from the pre-training part and the specific fine-tuning on the different training set sizes from the fine-tuning part from above, resulting in CERs as presented in Table 4.7.

We noticed a steady CER increase from 19.94% to 27.1% for 1 to 15 epochs. The best result we obtained with one epoch of pre-fine-tuning falls 3.02 percentage points short of the best pre-fine-tuned models that did not see augmented data. The CERs hint at suboptimal learning rates rather than overfitting because we should see a decrease in CER.

One remedy for this problem is the integration of a warmup epoch with the Adafactor learning rate schedule [Shazeer and Stern, 2018]. We added a warmup period for the models pre-fine-tuned on one and three epochs and re-trained the models with and without data augmentation. We used 10% of all the optimisation steps as

fine-tuning set	fine-tuning set size	epochs				
		1	3	5	10	15
Konzilprotokolle	0	11.28	12.18	12.94	14.10	14.52
	1	9.76	10.14	10.54	11.60	13.66
	4	10.05	8.36	8.91	9.31	11.90
	16	7.69	7.75	7.10	6.56	9.48
Schiller	0	18.50	19.67	20.02	21.10	23.96
	1	18.45	18.85	18.35	20.51	22.31
	4	17.24	18.32	16.94	17.42	22.43
	16	14.74	14.32	14.24	13.22	18.23
Ricordi	0	37.33	43.67	44.84	50.95	53.37
	1	35.93	37.72	40.34	41.60	52.83
	4	30.93	34.51	35.12	33.53	49.53
	16	22.78	26.00	29.66	22.34	48.56
Patzig	0	25.79	27.36	28.03	29.62	28.18
	1	20.55	22.00	22.97	22.10	25.81
	4	19.06	18.76	20.80	17.08	24.10
	16	24.53	16.94	14.30	12.36	18.21
Schwerin	0	31.99	34.65	33.48	35.38	37.47
	1	19.25	25.01	22.87	18.30	32.15
	4	14.08	14.10	14.54	10.00	23.11
	16	8.89	8.38	6.97	5.06	12.27
Average		19.94	20.93	21.15	20.61	27.10

Table 4.7: CERs on the test sets of the different fine-tuning sets. Fine-tuning size 0 means zero-shot evaluation of the pre-fine-tuned model. In the pre-fine-tuning step, data augmentation has been used.

warmup period. In our case, we were left with 114,681 lines after data augmentation (we removed a few very short lines that posed problems during training). When using a batch size of 8, the number of optimization steps is 14,336, which entails a warmup period of 1,433 steps. Warming up resulted in significant performance boosts for both models. The 3-epoch model without data augmentation achieved a CER of 17.08%, while the 1-epoch model with data augmentation resulted in a CER of 16.31%, therefore beating the best model from above. We summarise all the results in Table 4.8 and provide them in a chart in Figure 4.11.

We note that warmup for one and three epochs almost has no effect when pre-fine-tuning on the original data. However, as soon as the model is fed the augmented

fine-tuning set	fine-tuning set size	epochs		
		without DA		with DA
		1	3	1
Konzilprotokolle	0	19.66	12.25	10.42
	1	11.17	10.19	8.03
	4	8.61	7.11	6.67
	16	5.70	5.86	5.56
Schiller	0	29.29	20.61	17.40
	1	31.92	18.59	15.25
	4	20.29	16.39	14.39
	16	15.03	13.91	11.33
Ricordi	0	41.29	34.01	36.65
	1	31.35	27.52	30.66
	4	22.93	23.07	23.04
	16	15.61	15.50	15.47
Patzig	0	33.52	26.35	24.83
	1	25.73	20.62	19.46
	4	17.89	16.23	15.36
	16	12.55	12.53	10.75
Schwerin	0	32.22	29.68	32.65
	1	15.89	16.30	15.08
	4	8.67	9.42	8.73
	16	4.44	5.37	4.54
Average		20.19	17.08	16.31

Table 4.8: CERs on the test sets of the different fine-tuning sets. For all models, we trained a combination of warmup and data augmentation (DA).

data, we see a performance gain of 3.63 percentage points in CER for the model pre-fine-tuned on one epoch. This performance gain is only due to the warmup, showing that slowly adapting an already powerful model like TrOCR to new data is beneficial. Especially for relatively small datasets, data augmentation in combination with warmup needs to be considered. We also checked whether the number of augmented versions of a line image for pre-fine-tuning makes a difference. We evaluated this for one, three, and 15 epochs and noted differences to the overall averages (see Table 4.7) of 3.06, 2.56 and 4.25 percentage points, respectively. With warmup, however, the difference of the model pre-fine-tuned on one epoch was 0.1 percentage point. This means that a limited number of augmented versions is sufficient to improve the CER on a diverse dataset significantly. Adding dropout of 0.2 to the

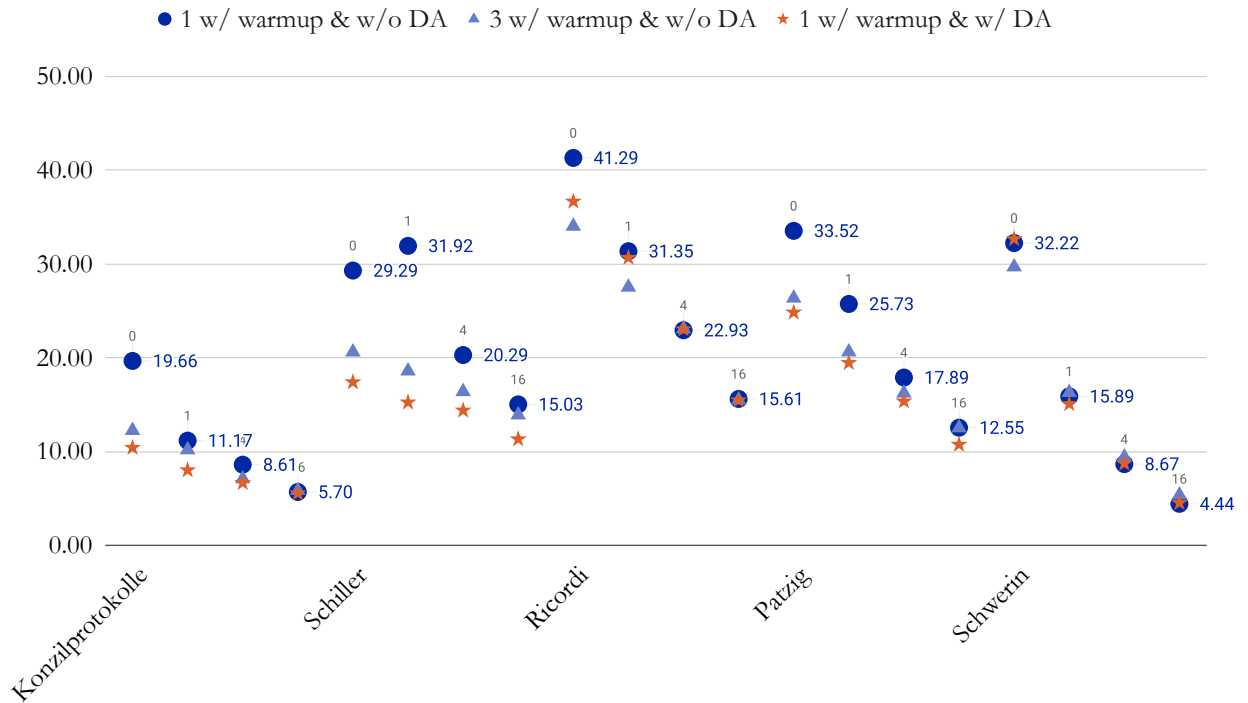


Figure 4.11: CERs for pre-fine-tuned models from Table 4.8.

best-performing model (training of one epoch, with warmup and data augmentation of ten different versions) to further regularise the model did not help. It led to a worse CER of 20.39%.

To summarise the results so far, we saw that for a relatively small dataset, it is advantageous to fine-tune on a rather small number of epochs. Furthermore, TrOCR benefits from warmup periods, especially with data augmentation. We also see that the composition of languages in the pre-fine-tuning data has an influence. Isolating the Ricordi dataset momentarily, we note that it is the only set containing lines in Italian. The data for the pre-fine-tuning does not contain lines in Italian (but in Latin, which is closely related). Comparing the zero-shot performances of the models pre-fine-tuned for one epoch from Table 4.8, the model pre-fine-tuned with data augmented lines performs 4.64 percentage points better than the one without data augmentation. Although this is generally true for most fine-tuning sets, we hypothesise this is due to the number of Latin lines in the pre-fine-tuning data. This hypothesis is justified when we look at Table 4.7: All these models also see much more Latin data. However, the zero-shot performance for the Ricordi dataset degrades to

a much greater extent than it does for the other datasets. Having stabilised TrOCR with warmup, the only difference for the 1-epoch model must be an adaption to the language, in this case, to a related language.

4.3.3 Fine-tuning on the Bullinger Dataset

We use the Bullinger dataset presented in Section 4.2.2 for the following experiments. More concretely, we use the pre-processed version c) as shown in Figure 4.7. We want to examine the behaviour of TrOCR concerning the following issues:

1. The findings of the previous section suggest that TrOCR should be fine-tuned for a restricted number of epochs. One variable to check is thus the number of epochs needed to achieve acceptable results.
2. We mention different pre-processing regimes in Section 4.2.2, which is why we investigate whether pre-processing influences TrOCR.
3. Because data augmentation plays an important role in the previous section, we study the effect of training on data-augmented lines.
4. The previous section also hinted at the fact that the languages present in the (pre-)fine-tuning sets influence the performance of TrOCR. Therefore, we scrutinise the factor *language* closer in these experiments.

For the experiments, we split the data in training, validation, and test split at a ratio of 80/10/10. In addition to this test set, we evaluate on two additional test sets, one representing the *frequent writers* and the other the *infrequent writers*. The frequent writer test set consists of 1,235 randomly chosen lines from writers who have written at least ten letters to Bullinger. The test set consisting of lines from infrequent writers comprises 1,013 lines. Due to the almost Zipfian distribution of writers¹⁹ we must expect that all the models we fine-tune are biased towards these frequent writers. By evaluating on these additional test sets, we gain further insights into the behaviour of TrOCR. Moreover, we split the training data of the frequent and infrequent writer test sets into Latin and Early New High German (ENHG) with the tool mentioned in Section 4.2.2. This will test whether TrOCR is influenced by the uneven language distribution of the training data (see Table 4.3). Warmup proved

¹⁹By *Zipfian* writer distribution we mean that very few authors have written to Bullinger very frequently, and many have written to him rarely or even only once.

a must from the experiments on the ICHFR2018 dataset, which is why it is always included.

Results and Discussion on the Bullinger Dataset

Number of epochs Figure 4.12 shows the results concerning issues 1 and 4. From the performance on the test set, we can infer that performance stabilises after five epochs. Fine-tuning TrOCR for ten epochs resulted in a CER of 9.7%, which is why we expect the CER to increase again after six epochs. We also fine-tuned a TrOCR model on the Latin and ENHG data for five epochs based on these experiments. We also see that the performance on frequent writer data is generally better across all models, thereby confirming our hypothesis that there is a bias towards frequent writers. However, we consider the gap to be small. In general, we attest TrOCR good performance on recognising 16th century handwritings.

Multilinguality The multilingual model outperforms models specifically fine-tuned on a certain language. E. g., the multilingual model achieves better CERs for the ENHG test splits both on the frequent and infrequent writers test sets. The difference of one percentage point or more is significant and a sign that the quantity and variety of the training data are more important than an even language distribution. Figure 4.13 shows example outputs of three different TrOCR systems (multilingual, Latin, ENHG). The line is an instance of code-switching, i. e., the author switches the language in the middle of the line. The orange lines show the language boundaries. We see that the multilingual model successfully makes a switch from recognising Latin to recognising ENHG, but with a buffer of one word (it recognises the ENHG *Denn*, EN *then*, as LA *deum*, EN *god*). The ENHG model “germanises” the Latin part, while the Latin model “latinises” the ENHG part. Code-switching occurs often in the Bullinger correspondence, which is why it is of interest to us to enable the model to switch languages at the right time. This could be a direction of future research.

Data augmentation For testing the effect of data augmentation on the fine-tuning of TrOCR, we generated additional line versions in the same fashion as in Section 4.3.2. However, since we have considerably more data in the Bullinger dataset, we did not generate more than six different versions. Table 4.9 shows the results of the data augmentation experiment. It becomes evident that the more data-augmented lines we use, the better the performance gets. With six augmented ver-

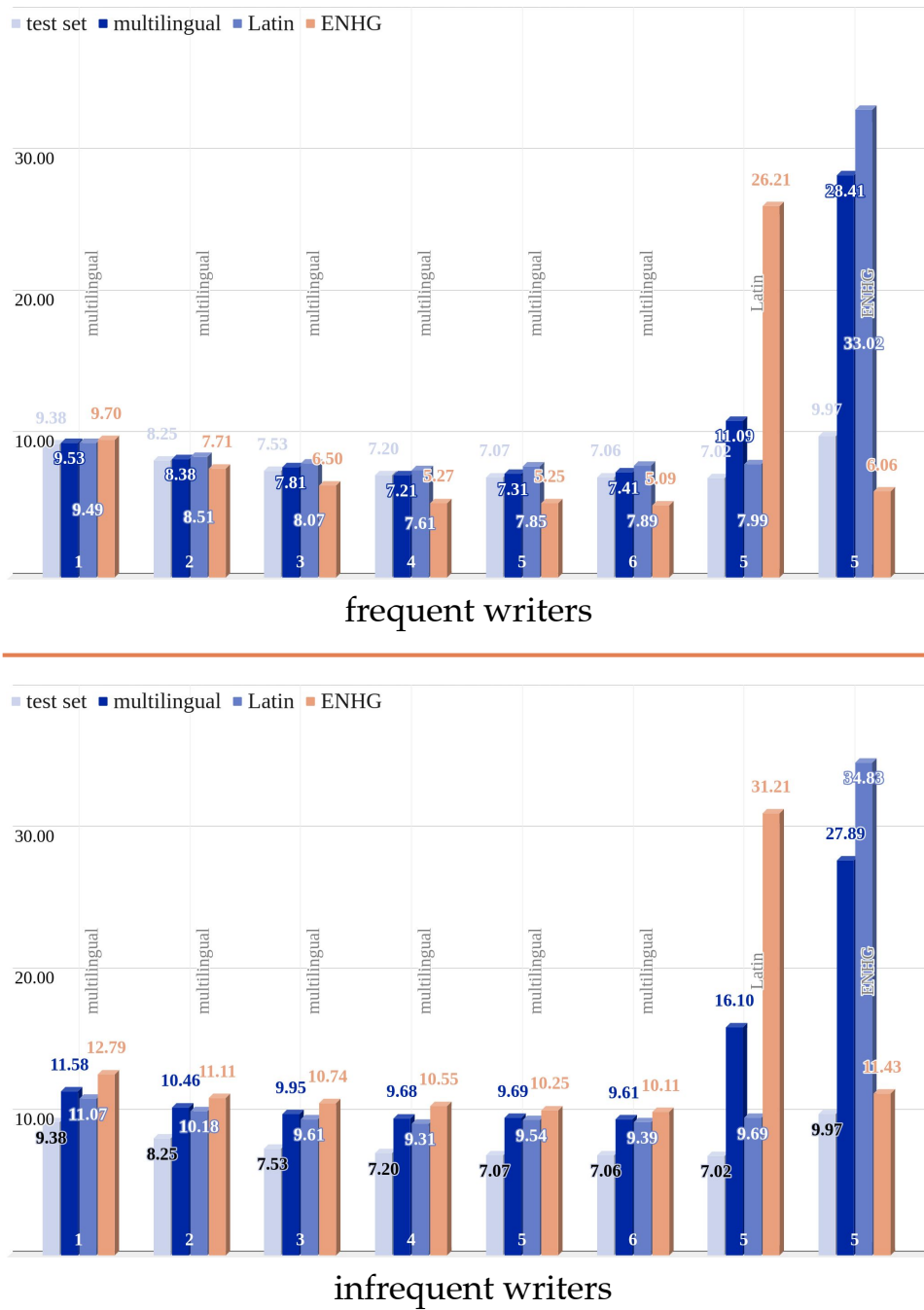


Figure 4.12: CERs of different TrOCR models trained on different language splits and evaluated on different datasets (general test set in light blue, frequent and infrequent writers test sets). On the x-axis, we noted the number of epochs we fine-tuned the model for. The y-axis corresponds to the type of model that has been trained.



Figure 4.13: Example output for three different TrOCR models (multilingual, Latin, ENHG). The orange line marks the language switch.

sions of a line and two fine-tuning epochs, TrOCR delivered the best results, beating the model that has been fine-tuned for six epochs by 0.4 percentage points, which corresponds to a relative error reduction of 5.67%. This shows that even for bigger datasets, data augmentation is a valuable means to boost performance.

Effect of pre-processing Lastly, we want to shift the focus to the effect that data pre-processing has on the fine-tuning of TrOCR. Since we have used the cut-out lines (following a detailed polygon generated by the layout recognition of Transkribus), we are interested in whether that influences the performance of TrOCR. As stated in Section 4.2.2, we hypothesise that using the exact polygon cut-out on a white background negatively affects TrOCR, since it focuses on the contours of the line. We thus fine-tuned TrOCR on images as shown in 4.7 a) for one to three epochs, resulting in CERs of 20.51%, 8.38%, and 7.61%, respectively. TrOCR is clearly influenced by being presented with clean cut-outs of line images, reflected by the high CER of 20.51%. The considerably lower CER after two and three epochs show that continuous fine-tuning helps TrOCR to focus the attention on the relevant places in the line images. This could raise hopes when it comes to the application of TrOCR models: Due to its ability to learn the location of actual handwriting in a line that it needs to decode, it is possible for ATR to successfully perform

# epochs	CER on test set		
	# of data-augmented lines		
	2	3	6
1	8.03	7.38	6.80
2	7.02	7.38	6.66

Table 4.9: CERs of different TrOCR models fine-tuned with different amounts of data-augmented lines. Overall best performance is in bold.

on lines that have not been pre-processed in the same manner. As such, TrOCR takes away the focus on pre-processing and rather places it on the development of writer-adapted models.

Bullinger model The author distribution shows a heavily skewed image. This requires exploring different training strategies. One could be to train specific models for the authors for which we have the most training material. Starting with Bullinger himself, who contributes roughly 20% to the overall Bullinger dataset, we fine-tuned a model just to Bullinger’s handwriting. We achieved the best CER performance of 9.2% while training for ten epochs. The model trained on one epoch achieved a CER of 11.43%. We note that there were no considerable differences in performance between fine-tuning one to ten epochs, but it is advisable to check for the number of epochs.

4.4 Conclusion

This chapter has shown how TrOCR can be applied to different handwriting datasets. TrOCR models demonstrate comparable performance to HTR+ models on the frequent writers’ test set of the Bullinger dataset while achieving a slight improvement of 0.45 percentage points in CER on the test set of infrequent writers. With respect to **research question 4.4**, we note that it is a tight race between TrOCR and traditional architectures. We also have only very briefly covered methods to customise TrOCR for new datasets, and there remain several additional tech-

niques to be investigated. We leave this for future work. However, the main insights when it comes to fine-tuning TrOCR models are the following:

- TrOCR has no problem adapting to historical data, therefore positively answering **research question 4.1**.
- TrOCR handles multilingual data well, except in a few-shot setting (e. g., the ICHF2018 dataset, see Section 4.3.2), where the model is only presented with a limited number of lines to decode. To answer **research question 4.2**, we note that with as little as 3,600 lines (in the case of the Gwalther dataset) we can achieve good ATR results.
- Always use warmup when fine-tuning on a new dataset (refers to **research question 4.2**).
- Start with as little as one epoch and slowly increment the number of epochs rather than fine-tuning on too many from the beginning (refers to **research question 4.2**).
- Use data augmentation to boost performance (refers to **research question 4.3**).
- TrOCR is robust to noise (e. g., line segmentation artefacts) in the input image (as shown by the experiment on the pre-processing method).

As such, TrOCR is a promising and extremely flexible model which is suitable to be applied to a wide variety of data. It proves as a valuable alternative to commercial or other open-source software in current digitisation endeavours, especially since it is easily adaptable to new material (not least because it is also available on the [Hugging Face](https://huggingface.co/docs/transformers/model_doc/trocr)²⁰ model hub). For additional ideas about how to further develop ATR models with TrOCR, see Chapter 6.

²⁰https://huggingface.co/docs/transformers/model_doc/trocr

Chapter 5

Evaluation Methods for ATR in the Absence of a Ground Truth

We see in Chapter 4 that TrOCR provides results for HTR in a CER range between 7% and 10% for difficult data. At the same time, the CER can fall below 3% for “easier” datasets (e. g., the Gwalther dataset). Becoming more and more reliable, HTR results will find their way into online representations of GLAM institutions [Terras, 2022a].

Hence, we consider ATR a “research facilitator” as long as it performs within reasonable accuracy ranges. But several studies have shown that inaccuracies in OCRed documents harm information retrieval and text mining techniques like named entity recognition and linking, topic modelling, and language modelling [Alex and Burns, 2014, Chiron et al., 2017, van Strien et al., 2020, Hamdi et al., 2020, Pontes et al., 2019, Hill and Hengchen, 2019].

So what do we mean by “reasonable accuracy ranges”? Holley [2009] shared her experiences of the *National Library of Australia Newspaper Digitisation Program*, which found that OCR contractors and libraries usually agree that “good” OCR means $> 98\%$, “average” OCR between 90% and 98% , and “poor” OCR any score $< 90\%$ character accuracy, respectively. Springmann et al. [2016] defined ranges of $> 95\%$ and between 90% and 95% character accuracy, but one time referring to the former as “good” and later as “excellent”, and to the latter as “reasonable” and in another place as “good”, respectively. Although opinions of what good OCR is differ, there is consensus that any score $< 90\%$ means poor quality.

While Holley [2009] and Springmann et al. [2016] were evaluating OCR for printed texts, these accuracy ranges naturally extend to text recognised from handwritten documents. However, HTR for personal handwriting styles is particularly challenging since a personal hand has much more variance than printed fonts or regular scripts like blackletter. Nonetheless, HTR has seen considerable improvements over the last decade and it is continuing to make improvements, again referring to the previous chapter.

Although HTR has made progress, it has not yet found its way to a broader application in libraries and archives. If we again venture a look at e-manuscripta, we see that in the category *correspondence* containing $\sim 52k$ titles, only 529, or 0.1%, contain transcriptions (see Table 1.1). With the output quality of modern HTR, libraries and archives can further process the scans they accumulated over the years of mass digitisation and thus provide researchers and the public access to the content of manuscripts of various sorts.

However, deploying HTR models to produce this content comes with uncertainties. While we can evaluate the performance of HTR models during the development phase via accuracy or error scores on validation and test sets, we cannot do so during the application phase. Hence, there are only a few options to estimate the quality of the resulting texts:

1. We compile a new ground truth (GT) from a representative sample of the new data.
2. We rely on confidence scores.
3. We find GT-free evaluation metrics.

As concerns the first option, compiling a GT is costly, time-consuming, and requires human experts. Moreover, we suspect that we only obtain a rough estimate of the overall quality [Hodel et al., 2021]. As for the second option, confidence scores are not always available, and if they are, it is unclear whether and to what extent they correlate with actual accuracy scores across different documents and distinct models [Springmann et al., 2016]. This leaves the third option as the only remaining one, which we will address in this chapter.

In this context, we pose the following research questions (RQs) which are essential to this chapter:

RQ 5.1 What surrogate metrics can help to choose the best ATR output?

RQ 5.2 How do the measures correlate with CER?

We propose different metrics in Section 5.2.1 and investigate whether there is a significant correlation between the introduced metrics and the actual CER.

The scrutiny of metrics for performance evaluation has practical implications for GLAMs. On the one hand, they can estimate the text quality, which is relevant for researchers working with this data. On the other hand, GT-free metrics allow for model selection when several models have produced hypotheses. Factors like the number of authors in a particular collection, the size of the GT, the skew of the author distribution, scan quality, and the like justify the training and testing of multiple models with different parameters and training data. To produce the best possible text during the application phase, it is necessary to apply all models and decide which model produced the best result for each document. In such a case, GT-free evaluation metrics can assist libraries and archives in making this choice. Hence, we also test whether GT-free evaluation metrics suit model selection.

Thus, this chapter investigates a collection of GT-free evaluation metrics to provide a quality estimation of HTR model results. Moreover, we examine whether we can use the metrics to select the best-performing models.

5.1 Related Work

Studies examining the evaluation of HTR output are few, which might be because baselines on popular datasets like the *IAM* database [Marti and Bunke, 2002] for handwritten English sentences are indicated in standard measures (CERs or word error rates) [Sánchez et al., 2019]. Neudecker et al. [2021] pointed out that although measures like the CER standardise the evaluation of OCR'd output, different implementations in tools that assist the evaluation make it difficult to compare model performances. More concretely, Neudecker et al. [2021] found that different tools deliver different CERs when evaluated on the same data. They mentioned that this is mainly to different assumptions the implementations make about the results, e. g., the layout recognition (reading order) or upper-cased and lower-cased characters (i. e., a tool might lower-case all results or ignore punctuation). Additionally, Schambach [2010] stated that such standard measures to rank models “may be of limited

significance for choosing a recognition system for a real-life application.” In his research, he demonstrated that various model architectures do not necessarily have a consistent correlation between the final performance metric and error limitation (e. g., rejection of uncertain results) for recognising postal addresses.

Whenever confidence scores are available, we can exploit them to estimate the text quality. For example, Sarkar et al. [2001] used confidence score outputs by an OCR model to build a triage system for OCRed documents. They classified documents to bypass manual inspection, which they successfully achieved for 41% of all documents in their validation set. Overall, they could show that their triage method significantly sped up the document verification process.

Clausner et al. [2016] suggested an OCR quality prediction system that assesses the accuracy of OCR results by any given system. Their classifier relies on 28 features, including metadata, image and layout information, textual features, and confidence scores. Their model was trained on newspaper data and provided reliable OCR quality estimates with an average error rate of 6.1% on a bag-of-words basis, indicating suitability for OCR quality estimation. In addition, a more detailed feature analysis of their classifier showed that the confidence scores are the third most important feature, only topped by lexical features.

Work by Springmann et al. [2016] on the automatic OCR quality estimation of historical prints also showed that the relationship between accuracy and confidence scores is significant (in contrast to Schambach [2010]).¹ Their approach investigated two measures: a mean token lexicality score based on the text-channel-model by Refle and Ringlstetter [2013] and confidence values that the OCR model assigns to its output characters. Moreover, they further improved their “standard” models that they had already trained by continuous training with different amounts of “pseudo ground truths” consisting of text output where standard models return high confidence values. Lastly, Springmann et al. [2016] used the confidence values to rank the models during training, which enabled them to choose the best one according to the confidence values after all training runs had been completed.

Alex and Burns [2014] illustrated the usefulness of lexicality in OCR outputs and advocated for a much simpler evaluation procedure. They examined the influence of OCR errors on named entity recognition in historical documents and found that recall is especially affected. They suggested measuring the quality of a text by simply computing the ratio of all words in a document and words occurring in an English

¹We believe that how much we can trust a model’s confidence scores strongly depends on the model itself.

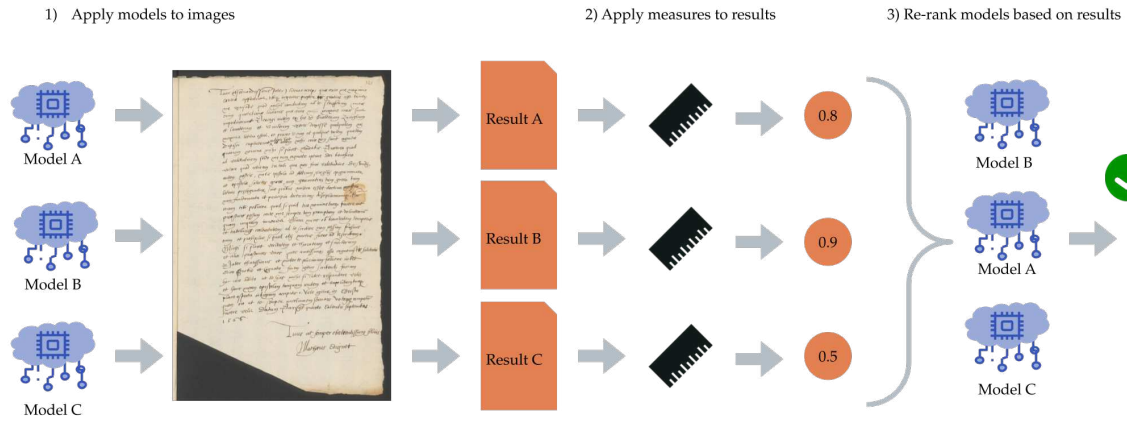


Figure 5.1: Pipeline to evaluate and rank the performance of HTR models. Here, we evaluate with the same measure. This chapter will explore various options to identify the most appropriate one.

dictionary. The test of their metric against human evaluation showed a correlation between their score and human judgements.

Salah et al. [2015] developed an approach that operates without external language resources. Their method relied on two OCR outputs: an OCR result and a second OCR output by another system called the reference (i. e., an uncertain GT). They used word-based character agreement rates and mutual information scores as features for a Support Vector Regressor that predicts the word recognition rate. They found their approach reliable, with an R^2 score of 0.95.

We can also use error classifiers in OCR'd texts as quality estimators. Nguyen et al. [2019] proposed an error detector that uses several word- and character- n -gram measures as features to train an OCR error detector. They also introduced a so-called *peculiarity index* which they defined as the root-mean-square of the indices of its trigrams. The higher the peculiarity index of a word, the more unusual its trigram composition is and, consequently, the more unlikely it is to be correct. Their error detector works best with all features included, reaching an F1-score of 0.79 on monographs and 0.7 on periodicals.

5.2 Method

Our goal is to evaluate different GT-free metrics and to check for each of them if it is suitable to rank ATR models.² In the first step, we train many models on datasets we specify in more detail in Section 5.3. Next, we apply all models to different test sets and compute all metrics introduced in this section. This procedure results in a ranking for each metric. At the same time, we rank all the models we have trained according to the CER they achieve on the test set, which we use as our reference ranking. Ideally, a metric’s ranking is the same as the ranking according to the CER. We use Spearman’s rank correlation measure to learn whether the reference ranking correlates with the metrics’ rankings. The coefficients provide information on the significance level of correlation between each pair of rankings. Figure 5.1 summarises the pipeline, which we describe in more detail in this and the forthcoming subsections.

5.2.1 Metrics

Lexicon-based Metrics

Clausner et al. [2016] showed that lexical features are most suitable to predict OCR quality, while Alex and Burns [2014] indicated that a simple ratio of recognised words against a lexicon correlates well with human judgement about OCR quality. We thus include lexicon-based methods in our metrics.

Token Ratio The token ratio determines the percentage of tokens recognised by the HTR model that also occur in a reference. Let T be the tokens in the HTR result and $\mathcal{V}$ the corresponding vocabulary. The token ratio is given by a fraction of counts $c(\dots)$

$$\mathcal{R}_{\text{token}} = \frac{c(T \in \mathcal{V})}{c(T)}. \quad (5.1)$$

This measure performs a lexicon lookup, where the lexicon is based on a collection of texts in the same language. For example, if our HTR results is “Christo domino assiduis vestris et ecalesiae vestrae”, we count seven words, of which we do not find “ecalesiae” in the reference vocabulary. Therefore, we have a token ratio of $\frac{6}{7}$.

²We restrict ourselves to HTR models in the following.

Character N -gram Ratio Nguyen et al. [2019] included an n -gram analysis in the OCR error detection process. We also include character n -gram ratios against a lexicon, but without the peculiarity index that Nguyen et al. [2019] introduced.

We consider the ratios of 2- to 7-grams. For each n -gram variant, we build the corresponding n -gram vocabulary from the reference text and the HTR results. Be $\mathcal{G}$ the set of n -grams in the reference text and N the n -grams in the HTR result. The n -gram ratio is given by

$$\mathcal{R}_{n\text{-gram}} = \frac{c(N \in \mathcal{G})}{c(N)}. \quad (5.2)$$

Language Modeling Perplexity

RNN-based HTR architectures learn an implicit character language model (LM) [Sabir et al., 2017]. Other approaches pre-train an LM and incorporate it in an HTR model to boost performance [Kang et al., 2021]. LMs thus form an integral part of the HTR process, so we suggest using external ones to evaluate HTR results. This is the first approach that uses LMs for HTR quality estimation to the best of our knowledge. We differentiate between statistical LMs and Transformer-based LMs.

Statistical Language Modeling Perplexity The task of LMs is to predict a word given a sequence of preceding words. Such probabilistic LMs learn the most probable sequences from massive text corpora. A statistical LM is given by

$$P_{MLE}(w_n | w_1 \dots w_{n-1}) = \frac{c(w_1 \dots w_n)}{c(w_1 \dots w_{n-1})} \quad (5.3)$$

where *MLE* refers to *Maximum Likelihood Estimation* and n to the order of n -grams. For example, a bi-gram LM would predict the next word based on the preceding word.

We can evaluate LMs with the perplexity (PPL) score, a measure from information theory. For example, given an LM P and test set W consisting of words in their sentential contexts, the word-level PPL measures the surprisal of the model when computing the likelihood of the sequence of words:

$$PPL(W) = 2^{H(W)} \quad (5.4)$$

where H is the cross-entropy of the sequence W :

$$H(W) = -\frac{1}{N} \sum_{i=1}^N \log_2 P(w_i) \quad (5.5)$$

The lower the PPL score, the less surprised the model is, and hence, the better the text quality should be for HTR results.

Pseudo-Perplexity Whilst PPL is a suitable metric for traditional LMs, the masked language modeling objective of Transformer-based LMs like BERT (see Section 2.2.4) does not permit a direct calculation of PPL. Instead, Salazar et al. [2019] proposed the use of *pseudo-perplexities* (PPPLs). The PPPL is the exponential of the word-normalised sums of each token’s conditional log probability in all output sentences to evaluate. More concretely, given an input sentence, we mask each token once, add up the conditional log probabilities of all tokens and normalise by the number of words in the sentence. We use the [MLM scorer](#)³ framework proposed by the authors for our experiments.

5.3 Data and Models

5.3.1 HTR Data

In contrast to the evaluation methods in Section 5.1, we work with historical handwritten data, more concretely, with samples from the Bullinger dataset introduced in Section 4.2.2. Recall that many more letters to Bullinger have been preserved than himself, so the author distribution is heavily skewed (see Figure 4.6). The data reveals that over 350 contemporaries wrote to Bullinger only once. This skew presents a considerable hurdle for HTR models since many different hands make it more difficult for them to generalise. Figure 5.2 shows examples of how the handwritings differ among authors. Especially in such a setting, it can be necessary to train multiple models with different author distributions and choose the model that performs best.

Our training data for these models comes from two sources. Firstly, we use the Gwalther dataset introduced in Section 4.2.3. Secondly, we sample around 20,000

³<https://github.com/awslabs/mlm-scoring>

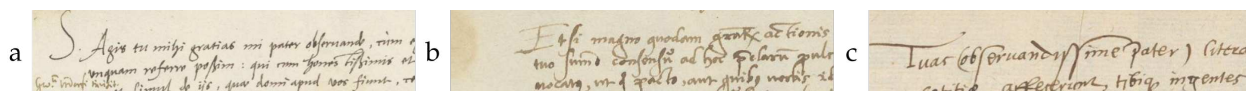


Figure 5.2: Examples of different handwritings in the Bullinger correspondence: a) Rudolf Gwalther (1519–1586), b) Kaspar Wolf (1532–1601), c) Matthieu Coignet (1514–1586).

lines from the Bullinger dataset (see Section 4.2.2) that we call *Misc. Bullinger correspondence* (see box 1a on the left in Figure 5.3). An analysis of the author distribution shows a heavily skewed picture with 69 different authors, where about 4,600 lines stem from Oswald Myconius (1488–1552) and only two, e. g., from Georg Cassander (1513–1566).

We also compiled two separate test sets. On the one hand, we sampled 825 lines from a portion of scans without transcriptions (box 1b in Figure 5.3) and manually transcribed the lines. Again, the author distribution is heavily skewed. Moreover, 35% of the 825 lines are by authors, which we also find in the *Misc. Bullinger correspondence*. On the other hand, we include a separate 57 lines letter from Gwalther that is not part of any training or test data.⁴ Models trained on *Gwalther* data should nevertheless perform better on this test set than other models.

Figure 5.3 presents an overview of the data and models we use in this chapter.

5.3.2 Language Resources

We restrict these experiments to Latin texts only, mostly because Early New High German is not a standardised language, therefore making it difficult to create a representative vocabulary. Generally, if the metrics we test in this chapter work for Latin, they will also work for Early New High German, provided that language resources of similar dimensions are readily available. We will leave it for future research to determine how well the same methods would work for Early New High German. We thus collected Latin reference material. For the experiments in Section 5.4, we downloaded the Latin part of the [CC-100 corpus](https://data.statmt.org/cc-100/)⁵ [Conneau et al., 2020, Wenzek et al., 2019]. The raw file contains 2.5GB of text, i. e., 390 million tokens (white-space tokenised). We removed generic “Lorem ipsum” text, de-duplicated the corpus on the paragraph level and retained only lines consisting of characters in the Latin alpha-

⁴We have already observed that Gwalther’s writing style differs in the poetry volume and the letters.

⁵[http://data.statmt.org/cc-100/](https://data.statmt.org/cc-100/)

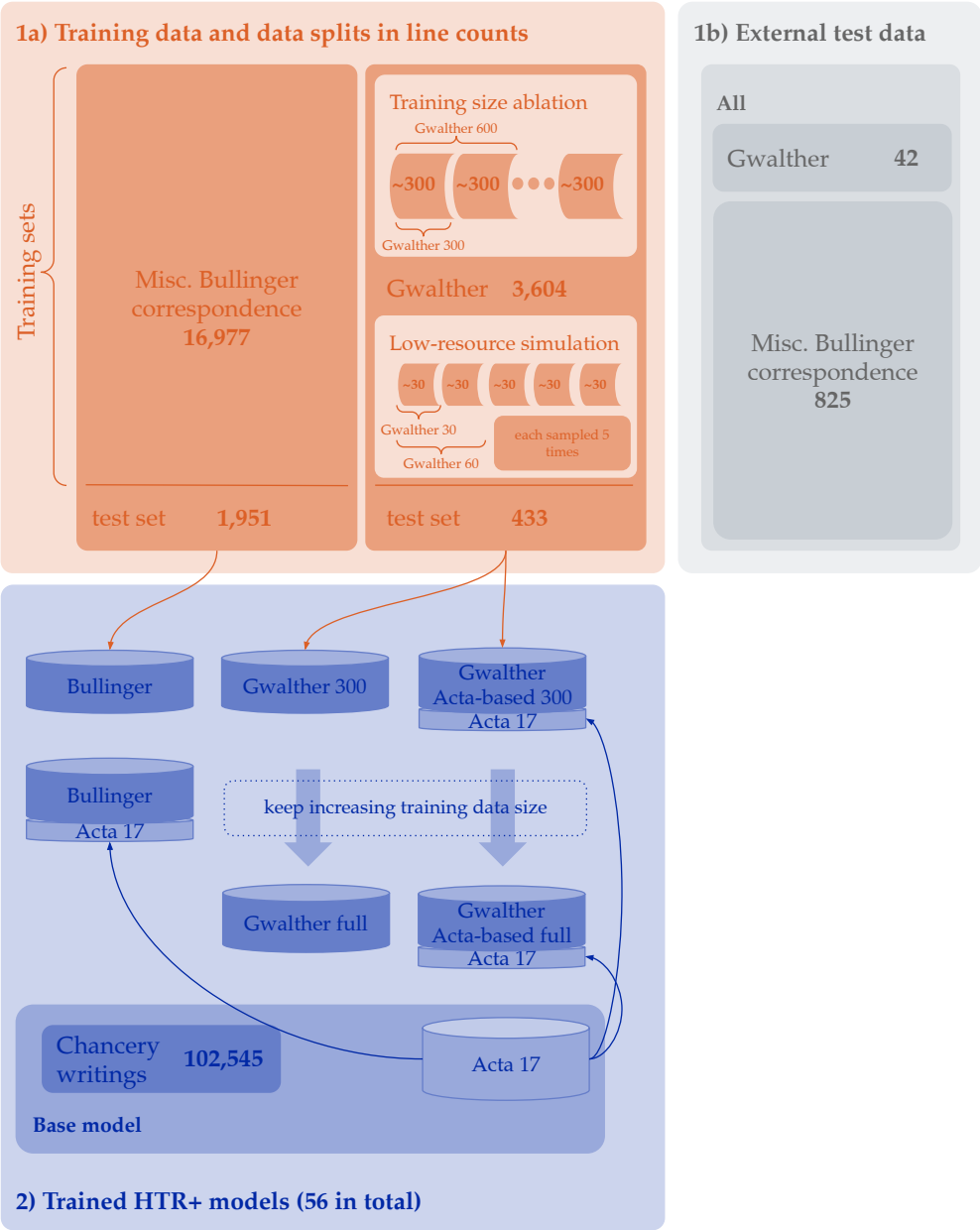


Figure 5.3: Overview of different datasets and models used for the evaluation of different metrics.

bet, numbers, and a selection of punctuation marks. Finally, we tokenised and normalised the corpus with the help of the *Classical Language Toolkit* CLTK [Johnson et al., 2021] and split it into sentences. After pre-processing, ~ 206 million tokens remain.

We trained all our HTR models with Transkribus and the HTR+ model architecture for 50 epochs. For both training datasets in box 1a in Figure 5.3, we set aside 10% of the lines for testing and 90% for training. We used the *Acta_17*HTR+⁶ model as base model.

For the Gwalther dataset, we additionally conducted a size ablation, starting at 300 lines (corresponding to roughly ten pages) up to the usage of the entire training set in increments of 300 lines, following a similar approach as detailed in Ströbel et al. [2020]. Moreover, we simulated an extreme low-resource setting, in which we sampled 30, 90, and 150 lines five times and used those as training data.

We sketched the training procedure in box 2 in Figure 5.3. This training regime results in a total of 56 models. We only considered the best-performing model per training set size for the low-resource simulation setting, thus discarding 24 models. We also evaluated the base model *Acta_17* and the *Spruchakten_M_211*⁷ model on the *Gwalther* test set and the Bullinger test sets. The reason for including the *Spruchakten* model is to check whether the effects of applying a model on paleographically more related data (since it is closer to Bullinger’s time) are more pronounced than just corpus size.

We present the models’ results in Table 5.1. As we have seen in the ablation experiment with the NZZ (see Section 3.6.2), we note a considerable performance improvement the more data we add. At the same time, models trained with a base model consistently perform better than their counterparts without a base model. The *Spruchakten* model, although trained on data closer to Bullinger’s time of living, performs worse than the *Acta_17* model, which indicates that more data helps.

Language Models

We used the Latin corpus presented in Section 5.3.2 to train all LMs. We applied a data split of 90% for training and 10% for testing and kept it constant for all models.

⁶see Section 4.3.1 for details

⁷The model was trained on a subset of the *Acta_17* training set: period: 1583-1653, size: 246k words, epochs: 1k.

	Number of lines for training														
	30	90	150	300	600	900	1200	1500	1800	2100	2400	2700	3000	3300	3600
No base	39.33	14.37	11.39	7.28	5.55	4.96	4.36	4.14	3.9	3.94	3.64	3.59	3.36	3.24	3.29
Acta-based	13.67	8.38	7.08	5.5	4.42	4.03	3.66	3.43	3.27	3.13	3.09	3.2	2.82	2.8	2.74
Other models (evaluated on the <i>Gwalther</i> test set)															
Bullinger	6.99														
Bullinger+Acta	6.56														
Acta_17	14.66														
Spruchakten	15.95														

Table 5.1: CERs for all models trained on different training data and number of lines on their respective test sets.

Statistical LM We used the *KenLM* package [Heafield, 2011, Heafield et al., 2013] to compute a 5-gram Kneser-Ney interpolated LM on the Latin training data. It achieves a PPL score of 23.13 on the Latin corpus test set.

Masked Language Models For the evaluation with the help of MLMs, we used the pre-trained multilingual BERT model based on Devlin et al. [2019]. Its training data consists of Wikipedia articles in 104 languages, among which we also find Latin. This [bert-base-multilingual-cased model](#)⁸ is available via the Hugging Face model hub [Wolf et al., 2019].

Since the multilingual BERT model uses WordPiece tokenisation and a shared vocabulary of 110k subtokens, we hypothesise that a single-language MLM should be superior to multilingual BERT. Thus, we trained a RoBERTa model with standard settings on the training set of our Latin corpus. We make the [model](#)⁹, as well as the [training and test data](#)¹⁰ available on the Hugging Face model hub.

⁸<https://huggingface.co/bert-base-multilingual-cased>

⁹<https://huggingface.co/pstroe/roberta-base-latin-cased>

¹⁰<https://huggingface.co/datasets/pstroe/cc100-latin>

5.4 Experiments

5.4.1 HTR Quality Estimation Suitability

After training the HTR and language models and preparing our language resource, we investigate whether the metrics introduced in Section 5.2.1 are suitable for HTR quality evaluation. We aim to establish a correlation between the CERs and the different metrics. Thus, we applied each model to our test set and computed the scores of our metrics. For evaluating PPPL, PPL, and token ratio, we preprocessed the HTR result like our Latin reference corpus to guarantee a fair comparison.

Our null hypothesis H_0 is that the metrics' scores on the test data and the CERs that the models achieve on the same data do not correlate. We follow Springmann et al. [2016]'s approach in fitting linear models with the CERs as dependent variables and the scores of the metrics as predictors using the R package [Team, 2021]. The adjusted R^2 informs us about the goodness of fit, where a higher value indicates the model can fit the data well. This procedure helps us determine whether a metric reliably estimates HTR quality.

5.4.2 Ranking Ability

The ranking tells us whether our metrics capture differences in HTR models. Our null hypothesis H_0 states that there is no correlation between the rankings of different models based on CERs and the rankings based on the metrics. Rejecting H_0 would prove the existence of such a correlation and validate the proposed metrics. Furthermore, the strength of the correlation between the CER and each of the metrics allows us to determine which metric the ranking should be based on.

We compute Spearman's rank correlation coefficient ρ between the reference and metric rankings to check for correlations. It is given by

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}, \quad (5.6)$$

where d_i denotes the difference between two ranks and n is the number of ranks. A comparison against critical values indicates whether the correlation is significant, thus rejecting H_0 .

Applied to our data, this means that we ran all our models on the Gwalther₅₇ and the Bullinger₈₂₅ test sets in block 1b in Figure 5.3, computed the corresponding CERs, and ranked the models according to the CERs. These are our reference rankings. We added the reference ranking for the Gwalther₄₃₃ test set in block 1a. Table 5.1 shows that many models perform in similar ranges, especially for the Acta-based models. Our metrics should reflect this by returning higher values when we apply the models to the validation set. We then took each model output for the Gwalther₄₃₃, Gwalther₅₇, and the Bullinger₈₂₅ test set and applied our metrics to it. This produces a score for each model and metric, which we used to rank the models. Lastly, we computed Spearman's ρ by comparing each metric ranking to its corresponding reference ranking. For example, we compared the model ranking of the PPL metric on the Gwalther₅₇ test set against the model ranking according to the CERs on the same set.

5.4.3 Hit Top-N

The last investigation we conducted is a check for each metric on how well it identifies a model in the Top-1, Top-3, or Top-5. As we see in Table 5.1, the Top-5 Acta-based models have a difference of at most 0.35 percentage points, which leads us to assume that for practical applications, a method that would identify either of those five models as the best-performing model was suitable for model selection. A single factor ANOVA test on the Gwalther₄₃₃ with a p -value of 0.71 hardens our suspicions that the means between the Top-5 performing models are not significantly different.

5.5 Results and Discussion

Table 5.2 shows the adjusted R^2 values for the models we fitted for each CER vs. metric pair. The superscript behind each score indicates the polynomial degree of the model. The higher the R^2 score, the better the model fits the data, i. e., the higher the correlation between metric and actual CER. We exemplify this with two metrics for the Gwalther₅₇ test set from Table 5.2 in Figure 5.4. Each black dot represents a model. We note the linear relationship (blue line) between metric and CER.

The big picture from HTR quality estimation suitability confirms that our metrics generally find it easier to score the Gwalther₄₃₃ set. This is only logical since we trained most of our models on the Gwalther set.

Metrics		Adjusted R^2		
		Gwalther-433	Bullinger-825	Gwalther-57
PPPL	BERT	0.99 ²	0.80 ²	0.92 ²
	RoBERTa	0.98 ¹	0.69 ²	0.88 ²
PPL	Statistical LM	0.65 ²	0.52 ¹	0.42 ¹
Token ratio		0.99 ¹	0.95 ⁴	0.92 ³
Character n -grams	2-gram	0.42 ²	-0.01 ¹	0.14 ²
	3-gram	-0.03 ¹	0.09 ¹	-0.03 ¹
	4-gram	0.80 ²	0.35 ¹	0.06 ¹
	5-gram	0.96 ²	0.79 ¹	0.81 ¹
	6-gram	0.99 ²	0.96 ³	0.96 ²
	7-gram	0.99 ²	0.99 ²	0.97 ²

Table 5.2: R^2 values for all metrics compared to the performance of the models on the respective validation or test sets. The superscript indicates the polynomial degree of the model we fitted to the data.

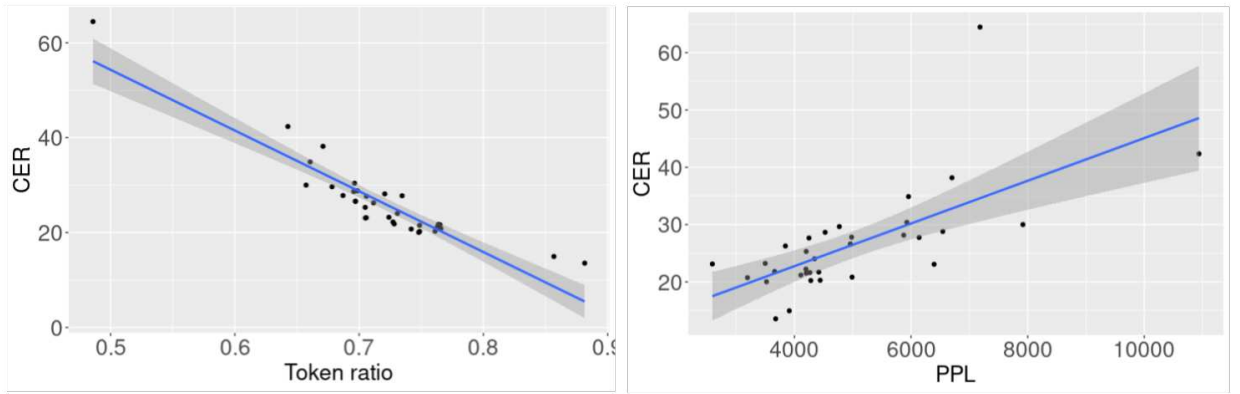


Figure 5.4: Model fit for two measures (token ratio left and PPL right) for the Gwalther₅₇ test set. The regression line in blue, 95% confidence interval in grey.

To put our results into perspective regarding previous work, we would like to point out that some of our R^2 scores are considerably higher than what Springmann et al. [2016] reported for their confidence vs. CER and lexicity vs. CER correlations. However, our results confirm the importance of the lexicity of the output and that its measuring is suitable for HTR quality estimation. As concerns character

n -grams, the general tendency is, the higher the order of the n -gram, the better it correlates with the CER.

As for the (P)PPL metrics, especially the scoring with the multilingual BERT model results in reliable estimates for HTR quality. Surprisingly, a dedicated Latin model does not perform better than multilingual BERT. However, the upside of this finding is that we can use a pre-trained model out of the box for HTR quality estimation without training additional models. Moreover, the solid performance of PPPL makes the collection of language-specific data superfluous. For lesser-resourced languages, like ENHG, this is of course not true.

Moving on to the examination of the ranking capabilities of our metrics, we find that most metrics' rankings exhibit strong correlations to the reference rankings, as Table 5.3 shows. Overall, the ranking correlation shows a similar picture as the R^2 values do. Again, the token ratio metric shows strong correlations. The PPL of the statistical LM cannot keep up with the other measures. At the same time, our newly proposed application of PPPL for ranking purposes performs equally well as the token ratio metric, at least on the Gwalther data.

Metrics		Adjusted R^2		
		Gwalther-433	Bullinger-825	Gwalther-57
PPPL	BERT	0.98	0.90	0.90
	RoBERTa	0.96	0.82	0.85
PPL	Statistical LM	0.78	0.65	0.71
Token ratio		0.98	0.95	0.91
Character n -gram	2-gram	-0.28	0.28	0.13
	3-gram	0.01	0.48	0.36
	4-gram	0.58	0.62	0.15
	5-gram	0.90	0.88	0.70
	6-gram	0.97	0.94	0.92
	7-gram	0.99	0.97	0.94

Table 5.3: The Spearman correlation values for all measures. Significance levels: 0.05, 0.025, **0.01**, **0.005**, **0.001**

Lastly, we take a look at the Top-N evaluation in Table 5.4. We see that the metrics select the best model in 70% of the cases for the Bullinger₈₂₅ test set and in 60% of the cases for the Gwalther₅₇ test set, respectively. This is in stark contrast to the Gwalther₄₃₃ validation set, with only 30% Top-1 hits. This is mainly because the CERs of the models are very close to each other on the validation set, thus making it harder for alternative metrics to pick the best model. All in all, our metrics select one of the five best performing models 90% of the time for the Bullinger₈₂₅ set and 80% for the Gwalther₅₇ set, thus confirming that our metrics are capable of model selection.

Metrics		Adjusted R ²		
		Gwalther-433	Bullinger-825	Gwalther-57
PPPL	BERT	1	1	1
	RoBERTa	1	1	1
PPL	Statistical LM		3	
Token ratio		3	1	1
Character n -gram	2-gram			
	3-gram		3	5
	4-gram		1	3
	5-gram	5	1	1
	6-gram	1	1	1
	7-gram	3	1	1

Table 5.4: Top-N values for each metric.

5.6 Conclusion

In this chapter, we proposed metrics based on lexical features and (pseudo-)perplexity scores for HTR quality estimation and model selection. We confirmed other research that the lexicality of an HTR result is a strong indicator of quality. However, lexical resources are not always readily available, especially for low-resourced languages. While we dealt with Latin in this paper, we would like to recall that the Bullinger correspondence contains Early New High German texts (ENHG). We

would need to assemble a big reference corpus for our or other lexicon-based methods to work. In the end, it is still not guaranteed that our findings for Latin can be transferred to ENHG because ENHG never had a standard. Still, our experiments confirmed that HTR quality estimation metrics based on transformers like multilingual BERT are suitable replacements, therefore responding to **research question 5.1**. In terms of **research question 5.2**, we note that there are strong correlations between the proposed methods from this chapter and the CER.

The pseudo-perplexity score also proved capable of ranking HTR outputs. Applying different models to new data and evaluating the output with ground truth-free metrics also opens up new possibilities to further improve HTR quality. For example, when we apply the metrics to lines of different model outputs, we can puzzle together the lines with the best scores, thus combining the best possible outputs from different models.

Our findings are valuable for digitisation initiatives by GLAM institutions, which are concerned with the automatic transcription of printed and handwritten documents. Especially our suggested approach using Transformers works out of the box and provides quality estimates, based on which rankings of different models are possible. Hence, libraries and archives could quickly produce ATR estimates and identify quality issues. It needs further research to investigate whether such metrics suit mass application.

Chapter 6

Conclusion and Outlook

The main motive of this thesis is the examination of automatic text recognition (ATR) techniques for historical documents. It lays out that ATR is facing great challenges in the digitisation processes of GLAM institutions. However, ATR is an indispensable step in making historical text collections more accessible to researchers and a broader audience. Due to the great variety of these collections, ATR techniques need to be flexible and adapt quickly to new data.

The experiments using the different ATR models showed that especially TrOCR would meet these criteria. The model's pre-training on a vast amount of image and text data allows it to adapt to new scenarios swiftly. For single-authored documents with a regular handwriting style, as we have encountered in the Gwalther dataset, automatically extracted texts with CERs between 2% and 3% are easily legible and suitable for being integrated into search systems. The fine-tuning on the Bullinger dataset showed that, given enough training material, multilingual data does not pose a problem (except in few-shot learning scenarios) and that CERs between 6% and 10% are realistic for data with a highly skewed author distribution.

Finally, the previous chapter introduces metrics which can be used to assess the quality of a text produced by an ATR system without ground truth data. We could confirm that first and foremost simple measures based on lexical features are suitable for the quality estimation of ATRed text. But also pseudo-perplexities based on masked language model scoring show high correlations with CERs. This can prove valuable for large digitisation initiatives which want to indicate an approximation of the texts stemming from ATR models to their users.

In the course of this thesis, we gained insights into applications of text recognition from a multitude of perspectives. We not only discussed and described architectures (such as HTR+, kraken+, and TrOCR) theoretically but also put them to practice in various forms to find answers to questions regarding training data quantities, author distribution, pre-processing, and different forms of error measurement. As such, we want to summarise the main findings of this thesis again:

kraken+ A rebuild of the Transkribus HTR+ architecture with kraken+ almost delivers the same results (see Section 3.6.1). We hypothesise that the same kraken+ model trained on augmented data would perform on par with the HTR+ and PyLaia architecture. All in all, kraken+ is an easy-to-use and open-source system suited for the recognition of both printed and handwritten historical documents.

Traditional approaches For printed documents, traditional approaches, i. e., CNN×RNN architectures, perform on par with Transformer-based techniques. In a bag-of-words evaluation scheme, TrOCR outperforms HTR+ and PyLaia (see Chapter 3, especially Section 3.6.1), while evaluations based on CERs showed slight advantages for CNN×RNN architectures. Because of this, we need to question evaluation methods on the one hand, and further focus on qualitative error analyses of the different systems on the other hand.

Adaptability of TrOCR TrOCR is a highly adaptable system, capable of coping with low-resource scenarios (see Section 4.3.2). For GLAM institutions, this is a valuable insight, since TrOCR enables them to ameliorate their ATR output with only little fine-tuning material.

Language diversity For TrOCR, the quantity of training data is more important than the language distribution (see Figure 4.12). This means, that the knowledge intrinsic to TrOCR’s sub-word tokenisation suffices to reproduce not only English, but also other languages like Latin, (Early New High) German, and possibly many more. It also shows that TrOCR profits more from seeing many different handwriting styles.

Impact of pre-processing TrOCR is indifferent to the pre-processing applied to the line images, at least if trained for more than one epoch (see “Effect of pre-

processing” under Section 4.3.3). This means that in general, no laborious pre-processing after layout recognition is necessary. This is important because detailed polygons, as provided by Transkribus, are not always available, and, with our discoveries, do not need to be available to provide satisfactory results.

Data augmentation Data augmentation benefits even an already powerful model like TrOCR, hypothetically because it has never seen historical data during its pre-training (see Table 4.9). This discovery goes together with an earlier point in this list where we mentioned that it is important for TrOCR to see a great variety of handwriting styles to improve its performance.

Evaluation without GT data The evaluation of ATR results without a GT is possible, even with as simple measures as token and n -gram ratios.

While our investigation of TrOCR showed its suitability to be applied to historical documents, we still have some open issues which we should attend to in future research. The introduction of Transformers into the “vision world” and the use of them for ATR might give the impression that they have replaced CNNs. However, this is not the case. Rather, recent approaches have tried to combine the idea behind CNNs and Transformers, leading to the so-called *Swin Transformers* [Liu et al., 2021]. Swin Transformers compute the attention on smaller windows consisting of a subset of patches first instead of all the patches. Through the merging of the patches in the different windows, Liu et al. [2021] reduce the number of visual tokens and hence enlarge the receptive field of the Transformer until it has seen the whole image. This procedure results in hierarchical Transformers and has shown to be *on par* or outperform classical CNNs. It will be worthwhile to focus further research on these kinds of architectures.

On the side of the language model, the out-of-the-box TrOCR model has seen a lot of English during its pre-training phase. Although it was able to handle other languages like (Early New High) German, Latin, and Italian in our experiments with relatively few problems, future research should try exploiting already trained language models on the decoder side to further improve performance. Approaches like these already exist for sequence-to-sequence ATR models [Kang et al., 2021] and have proven to be beneficial.

What is still an open question is the type of errors TrOCR makes. Since it works differently than the conventional RNN×CNN models, it is reasonable to assume

that it also produces different types of errors. A better understanding of the errors TrOCR commits would help especially in ATR post-correction.

We also leave it for future work whether pseudo-perplexity aligns with CER for historical German, i. e., Early New High German. After all, Early New High German is not part of multilingual BERT's training data, and it is far more irregular and includes many more spelling variants. However, to obtain a comprehensive overview of the ATR results' quality it would be essential to create an ENHG dataset and train a BERT-like model on it.

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